

(12) **United States Patent**
Howe et al.

(10) **Patent No.:** **US 12,408,078 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **5G UDM TO IDP FEDERATION AND IDENTITY FUNCTION**

(58) **Field of Classification Search**

None

See application file for complete search history.

(71) Applicant: **Zscaler, Inc.**, San Jose, CA (US)

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,894,350 B2	2/2011	Kailash et al.
8,010,085 B2	8/2011	Apte et al.
8,259,571 B1	9/2012	Raphel et al.
8,458,786 B1	6/2013	Kailash et al.
8,464,335 B1	6/2013	Sinha et al.
9,350,644 B2	5/2016	Desai et al.

(Continued)

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 389 days.

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(21) Appl. No.: **17/699,498**

(57) **ABSTRACT**

(22) Filed: **Mar. 21, 2022**

(65) **Prior Publication Data**

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Related U.S. Application Data

(63) Continuation-in-part of application No. 17/491,795, filed on Oct. 1, 2021, now Pat. No. 12,010,553, and a continuation-in-part of application No. 17/371,408, filed on Jul. 9, 2021, now Pat. No. 11,765,593, and a continuation-in-part of application No. 17/194,568, filed on Mar. 8, 2021, now Pat. No. 11,596,027.

(51) **Int. Cl.**

H04W 28/08 (2023.01)

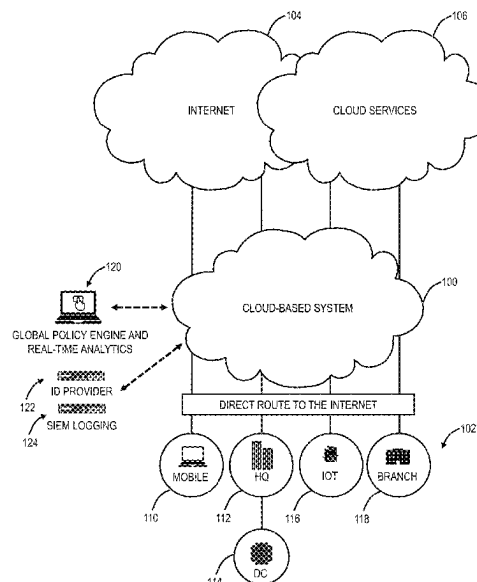
H04W 28/02 (2009.01)

(52) **U.S. Cl.**

CPC ... **H04W 28/0925** (2020.05); **H04W 28/0215** (2013.01); **H04W 28/0268** (2013.01)

The present disclosure relates to systems and methods for cloud-based 5G security network architectures intelligent steering, workload isolation, identity, and secure edge steering. Specifically, various approaches are described to integrate cloud-based security services into Multiaccess Edge Compute servers (MECs). That is, existing cloud-based security services are in line between a UE and the Internet. The present disclosure includes integrating the cloud-based security services and associated cloud-based system within service provider's MECs. In this manner, a cloud-based security service can be integrated with a service provider's 5G network or a 5G network privately operated by the customer. For example, nodes in a cloud-based system can be collocated within a service provider's network, to provide security functions to 5G users or connected by peering from the cloud-based security service into the 5G service provider's regional communications centers.

17 Claims, 37 Drawing Sheets



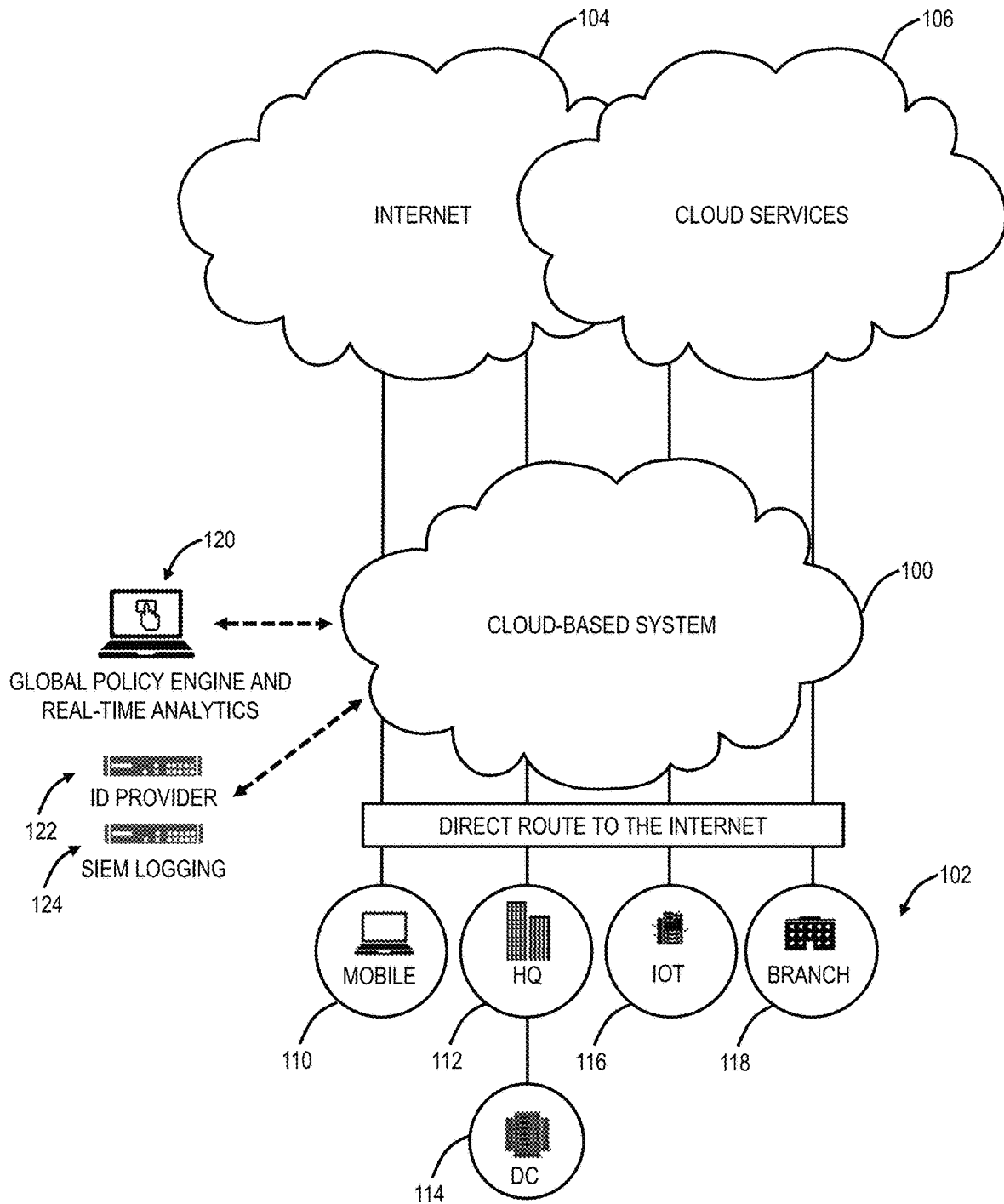
(56)

References Cited

U.S. PATENT DOCUMENTS

9,473,537	B2	10/2016	Sinha et al.
9,531,758	B2	12/2016	Devarajan et al.
10,375,024	B2	8/2019	Foxhoven et al.
10,511,590	B1 *	12/2019	Bosch H04L 63/20
10,609,083	B2	3/2020	Kailash et al.
10,728,246	B2 *	7/2020	Bansal H04L 67/10
11,190,981	B1 *	11/2021	Nelson H04L 65/102
11,343,707	B2 *	5/2022	Briggs H04W 76/15
11,706,705	B2 *	7/2023	Talebi Fard H04W 28/0268
			370/329
11,811,670	B2 *	11/2023	Sun H04M 15/8016
11,832,214	B2 *	11/2023	Stojanovski H04W 76/27
11,985,544	B2 *	5/2024	Narasimha H04W 88/085
2015/0372982	A1	12/2015	Herle et al.
2016/0261564	A1 *	9/2016	Foxhoven H04L 63/20
2020/0077265	A1	3/2020	Singh et al.
2020/0336913	A1	10/2020	Pampati et al.
2021/0037061	A1 *	2/2021	Trost H04L 63/1425

* cited by examiner

**FIG. 1**

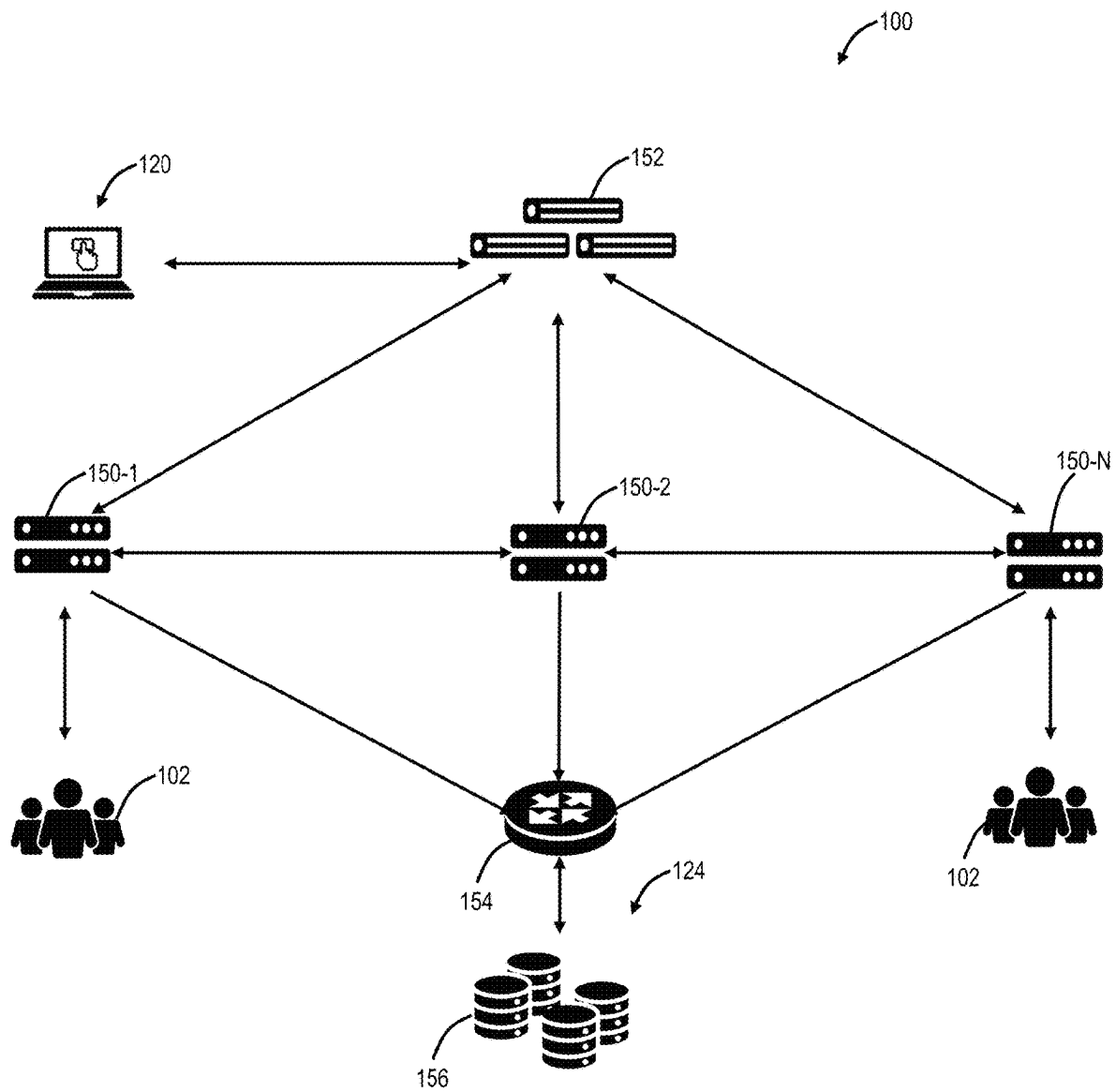
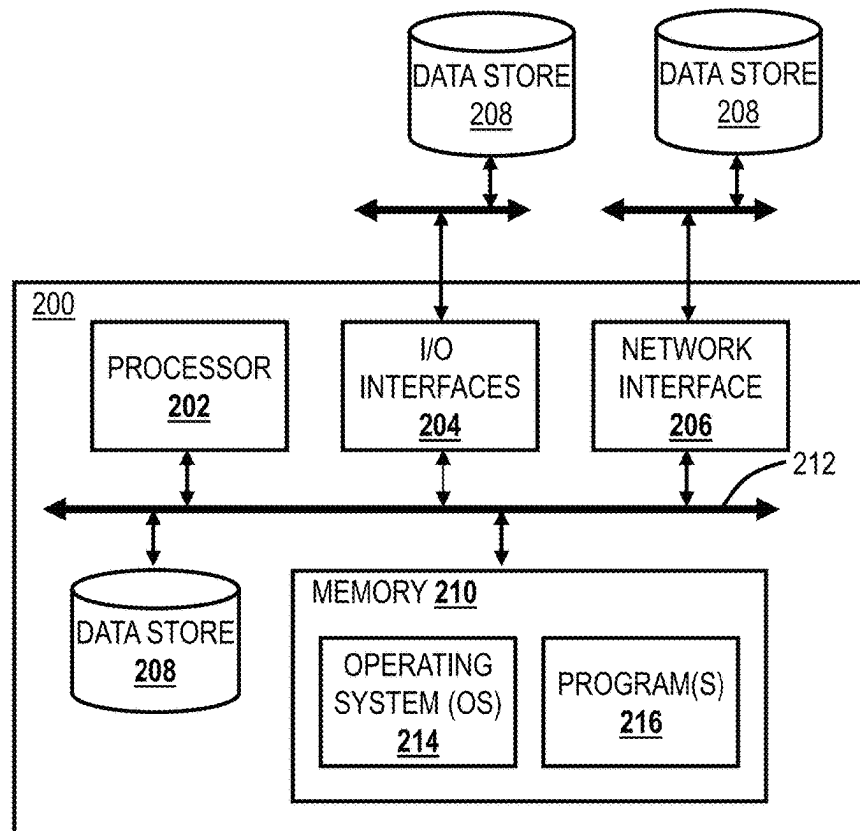
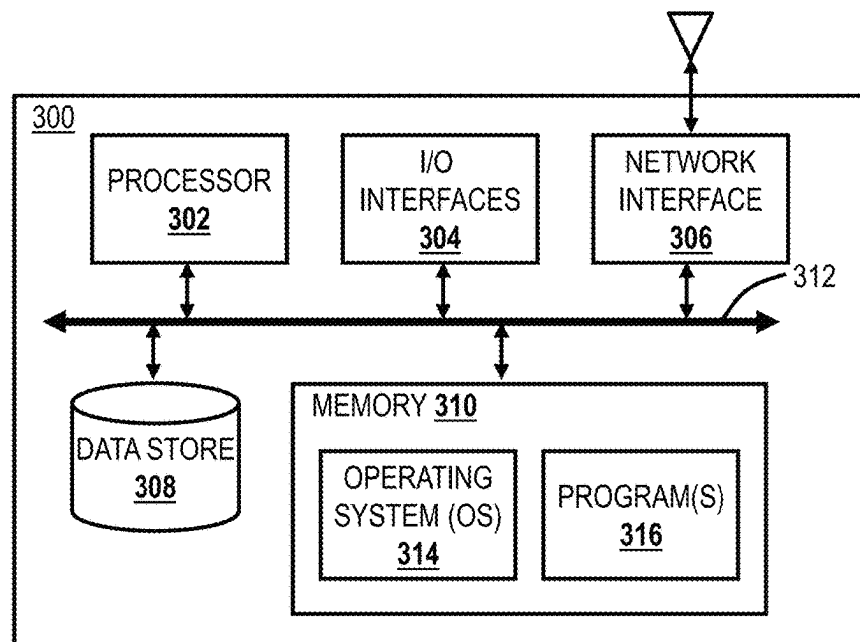
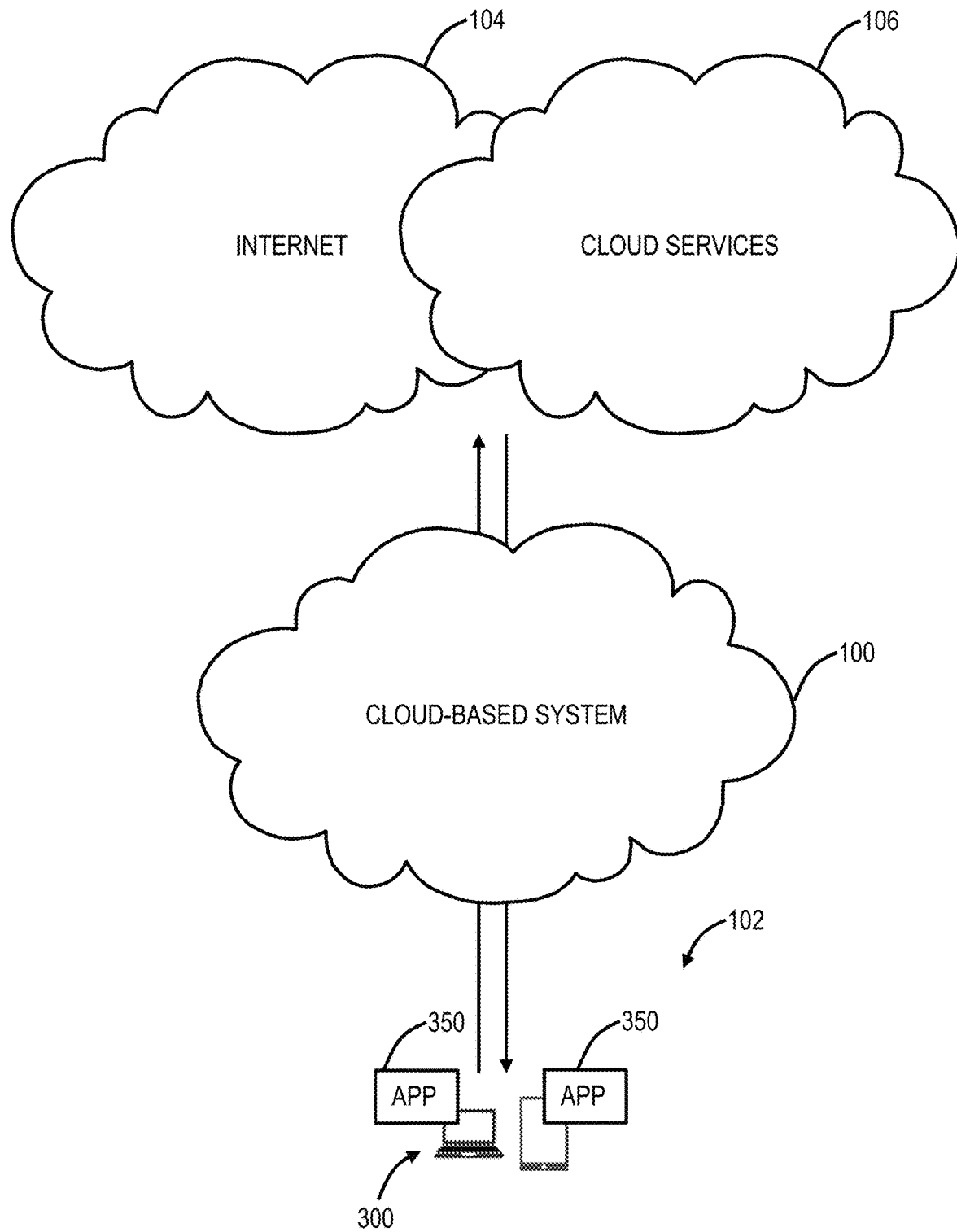


FIG. 2

FIG. 3**FIG. 4**

**FIG. 5**

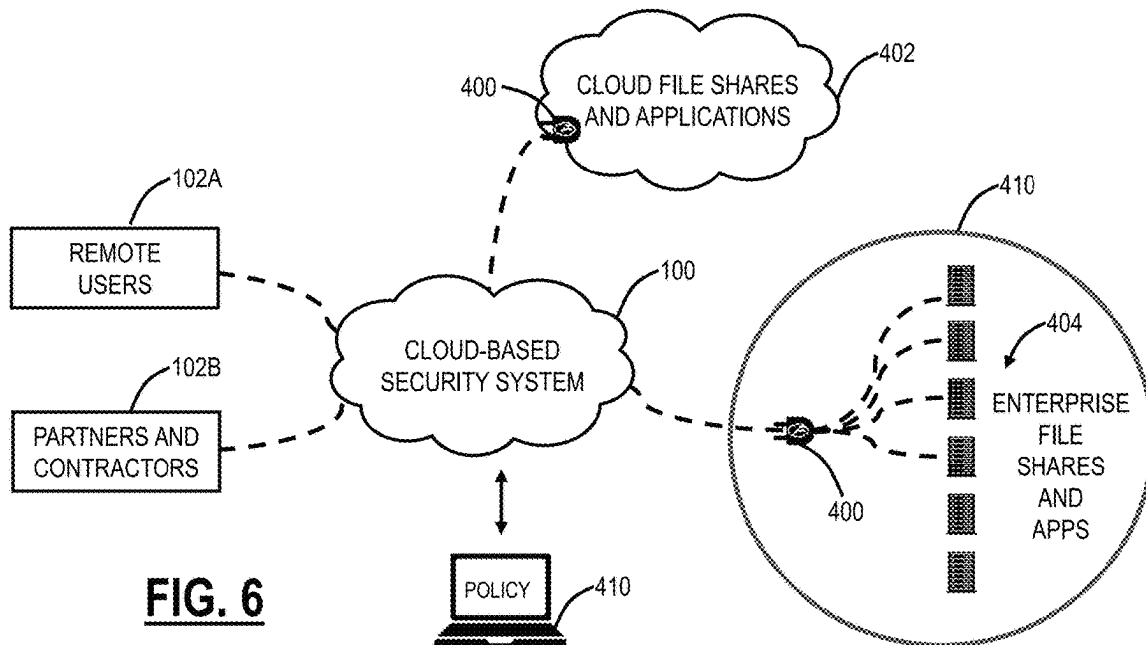


FIG. 6

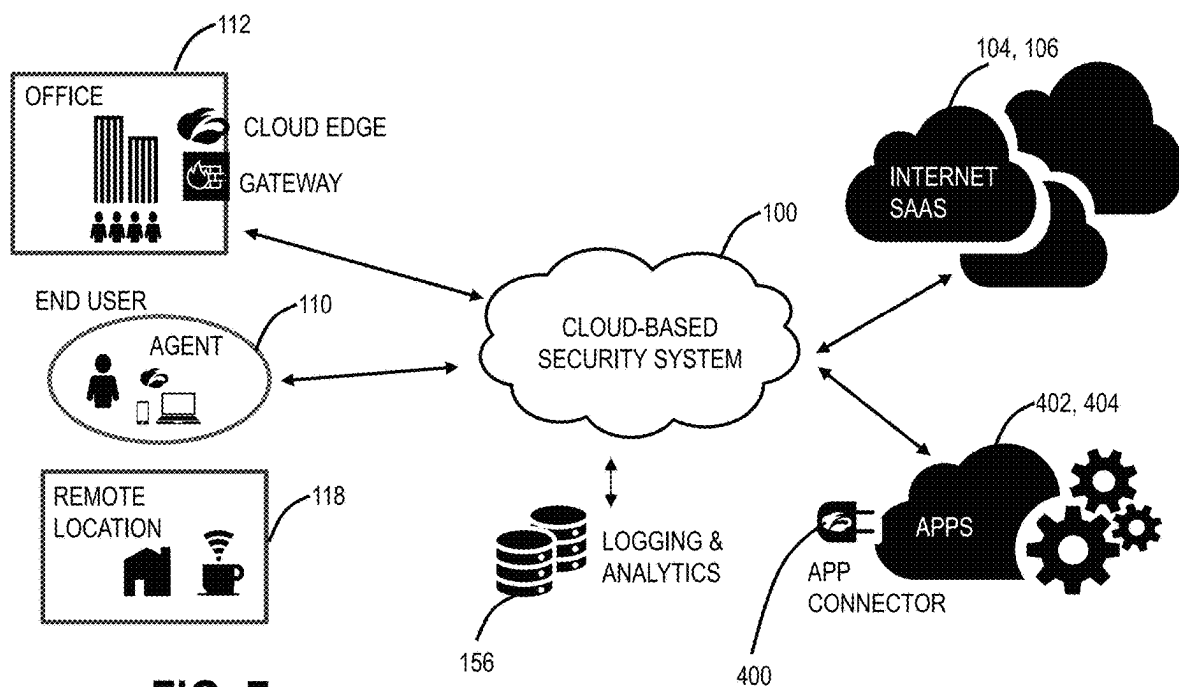
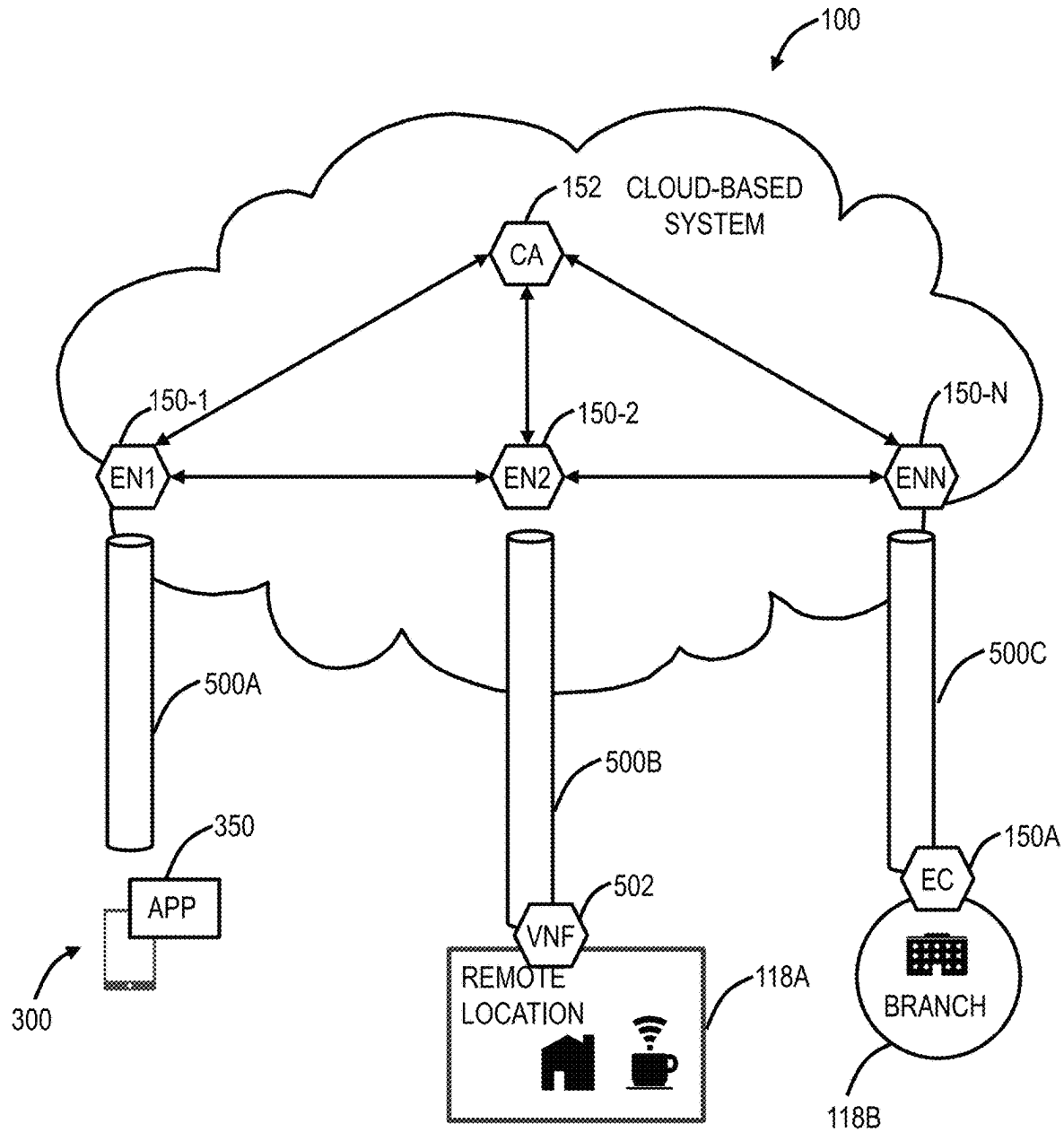
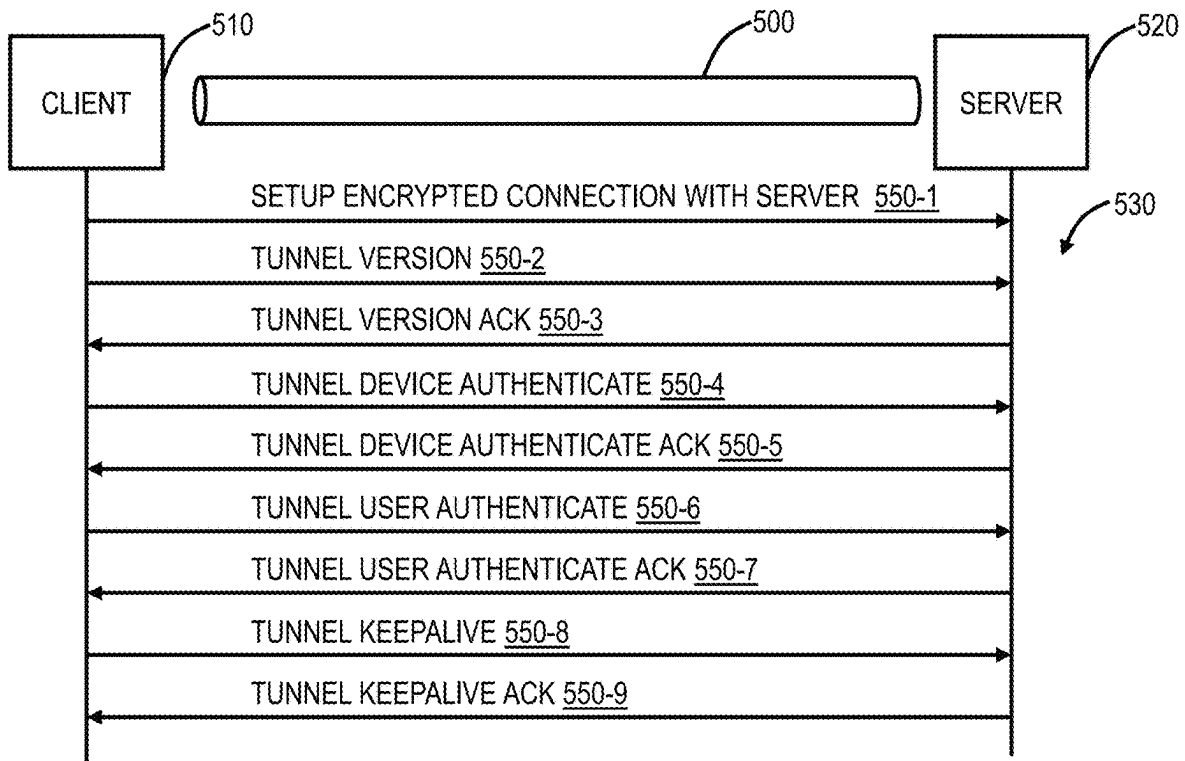
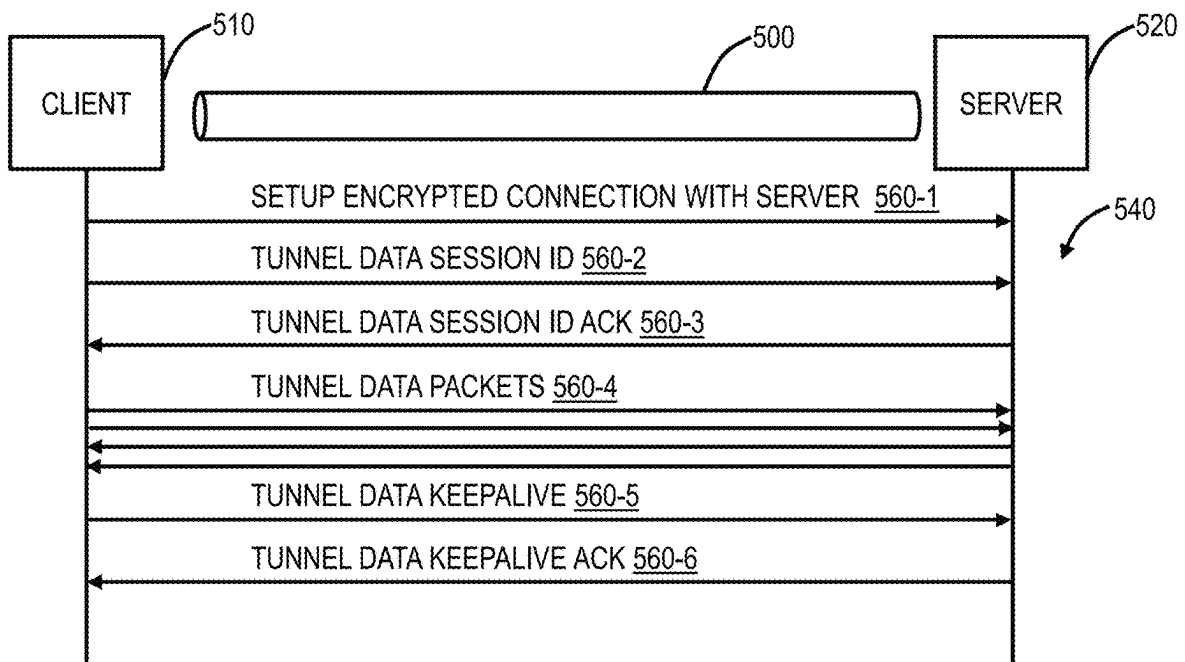
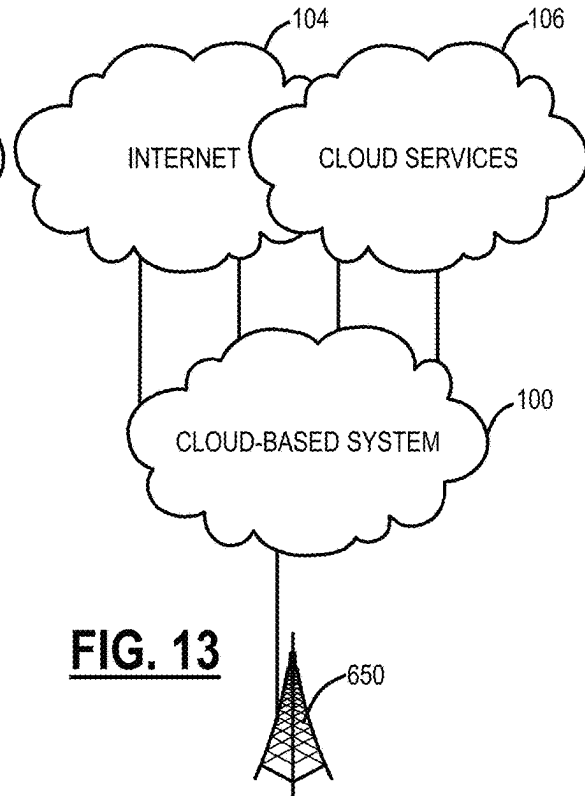
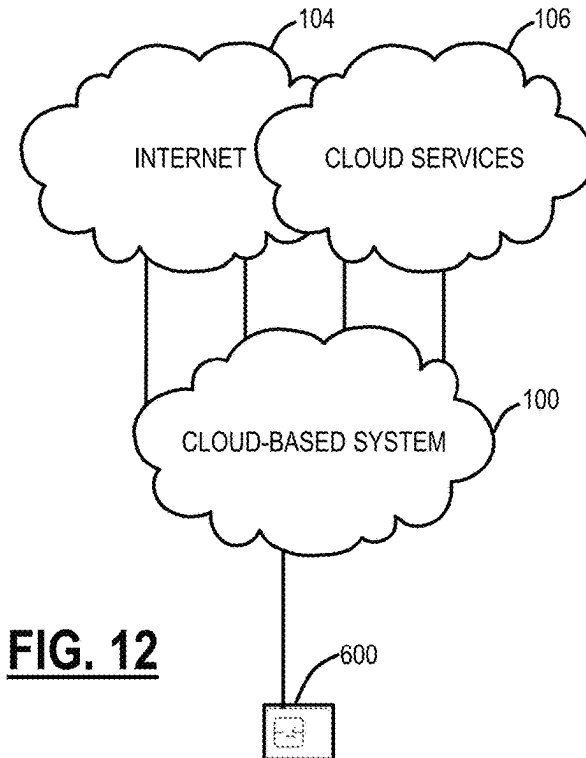
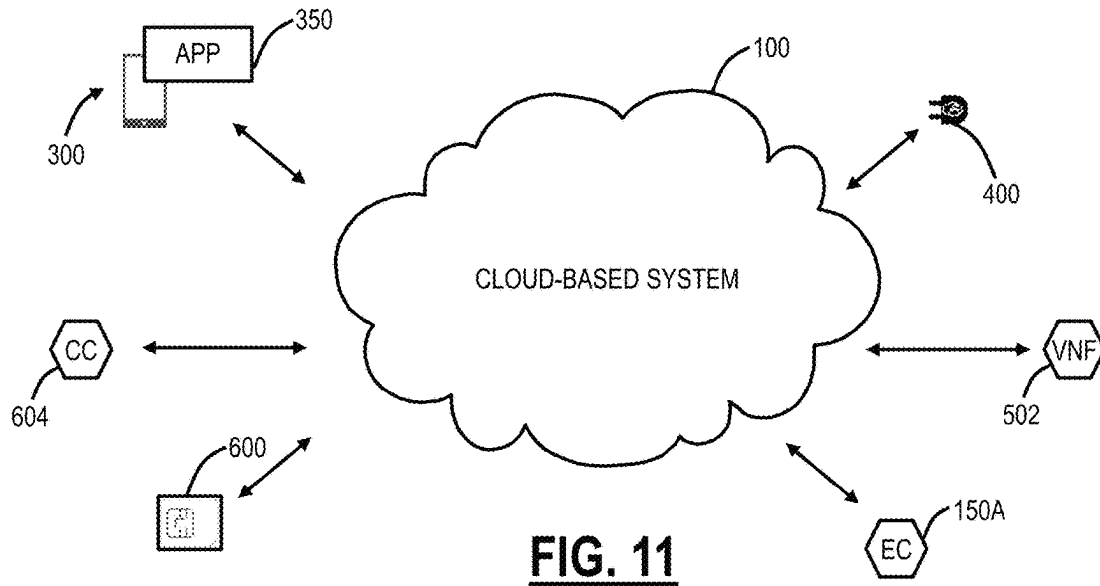


FIG. 7

**FIG. 8**

**FIG. 9****FIG. 10**



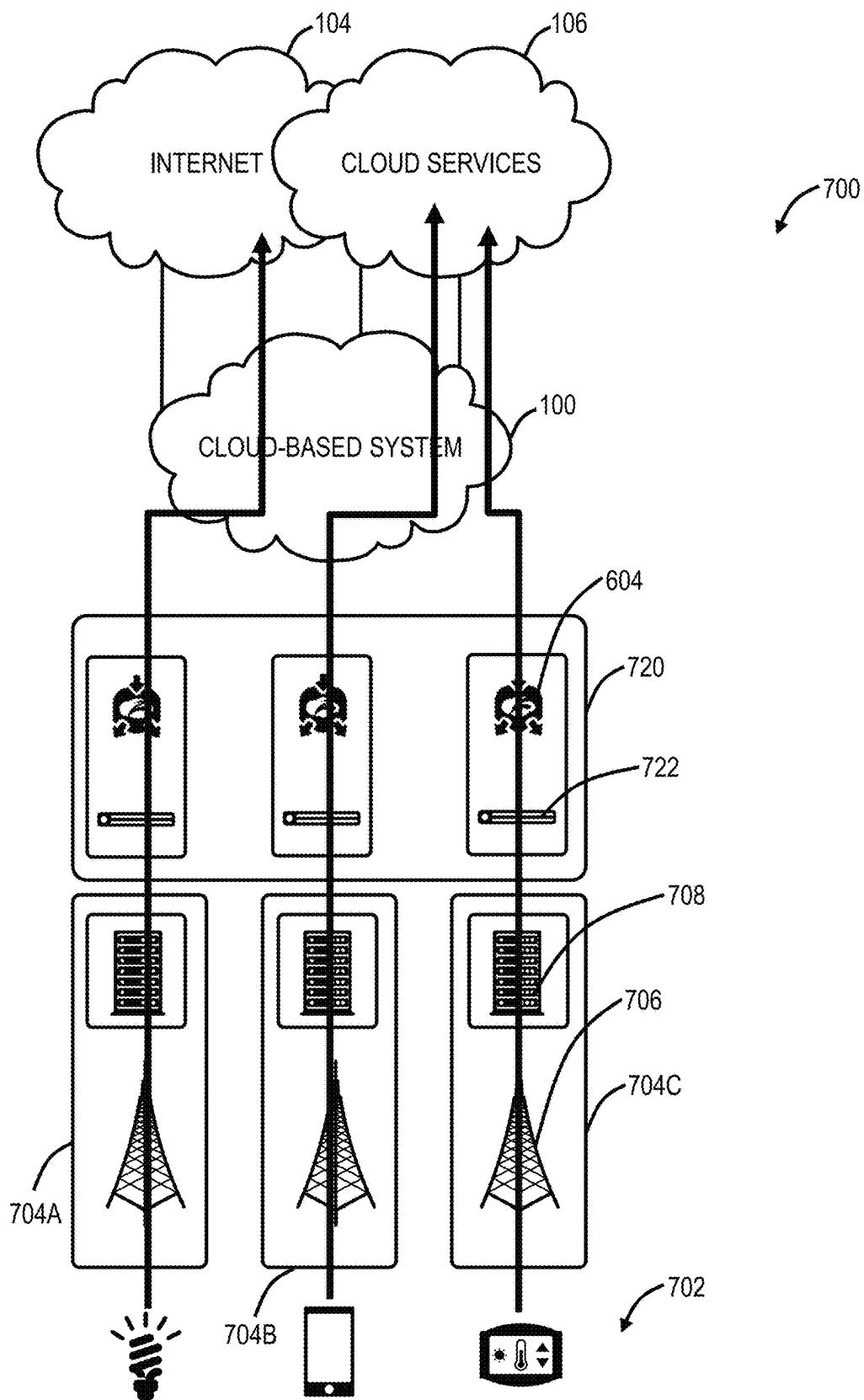


FIG. 14

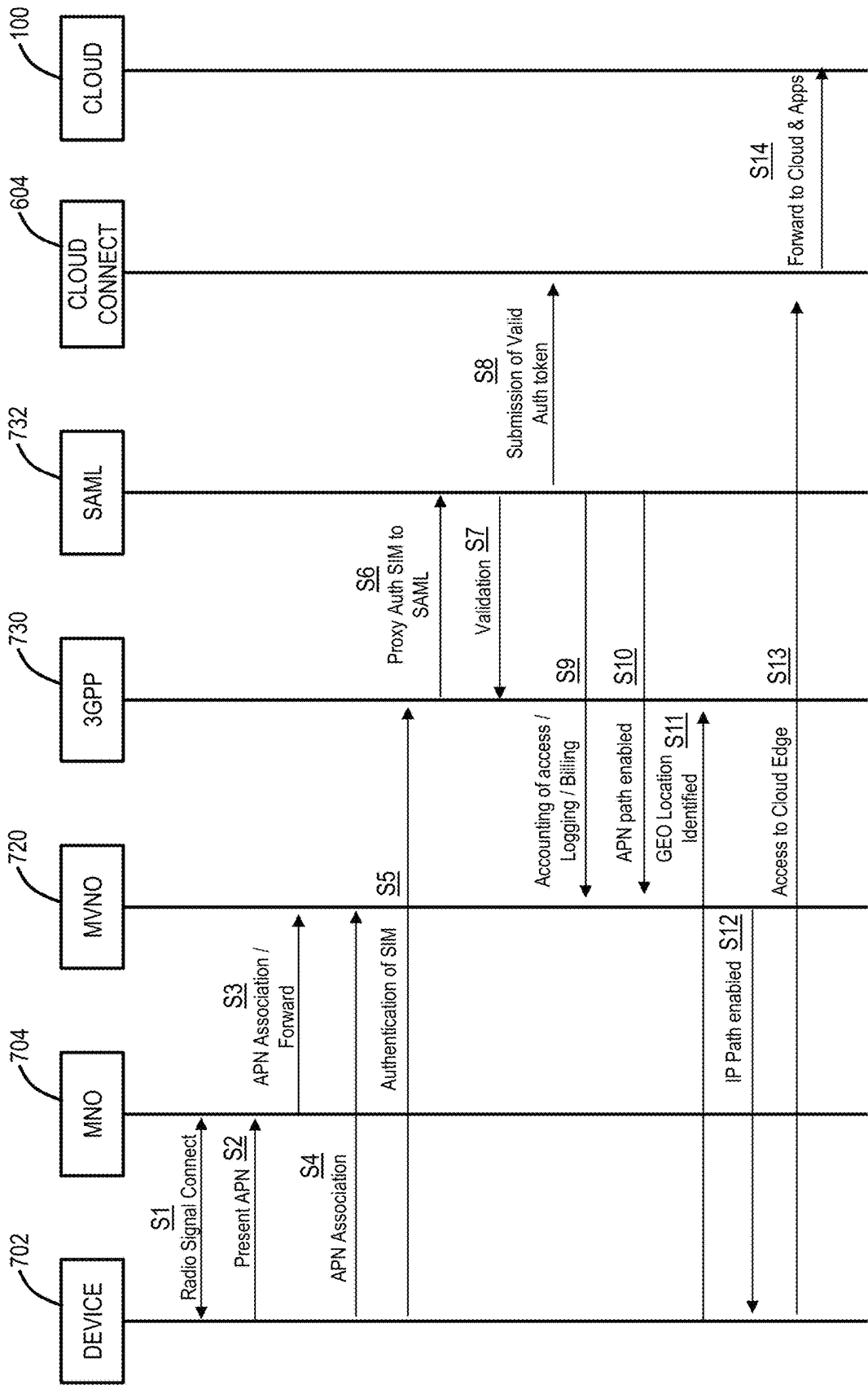


FIG. 15

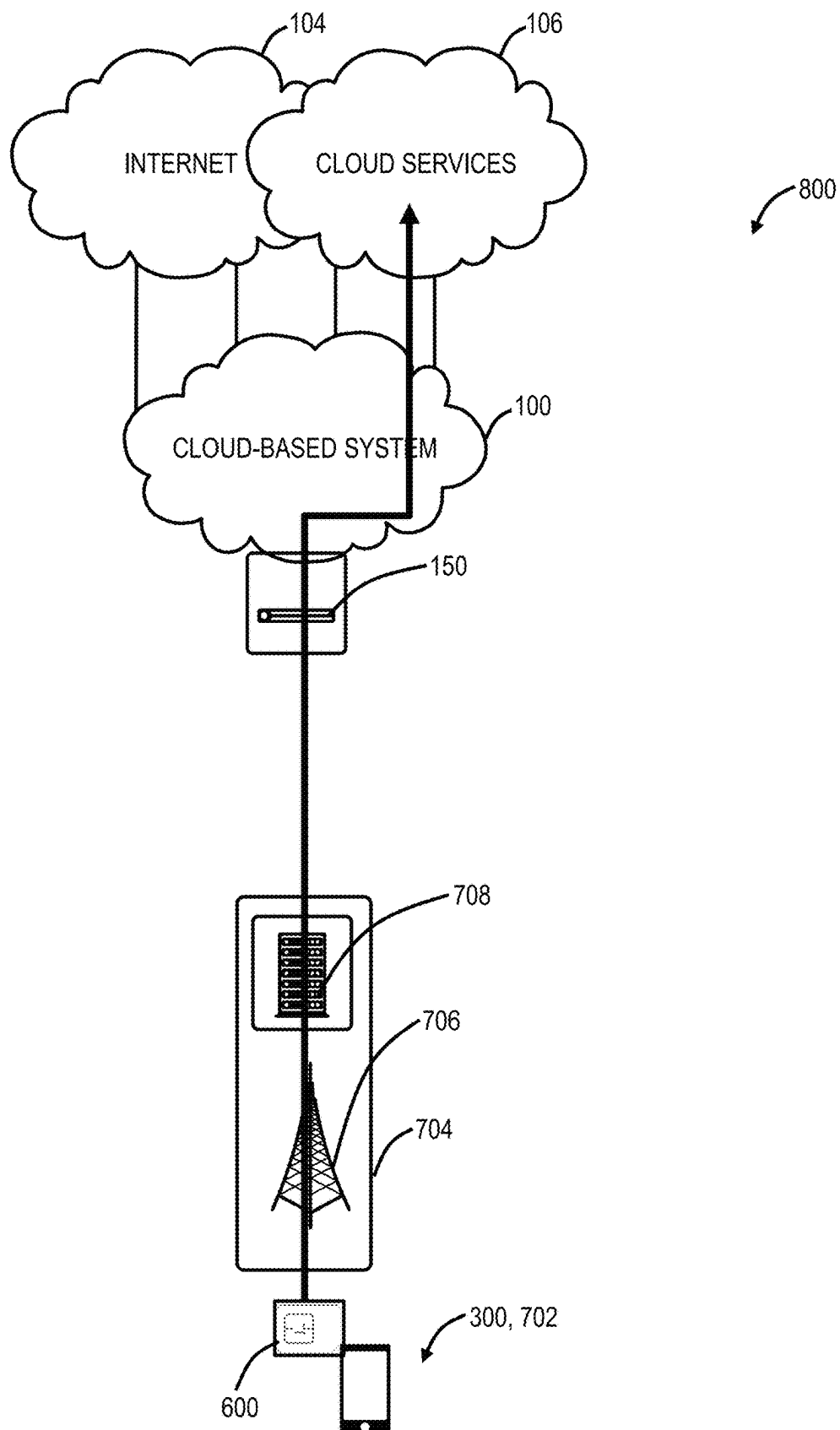


FIG. 16

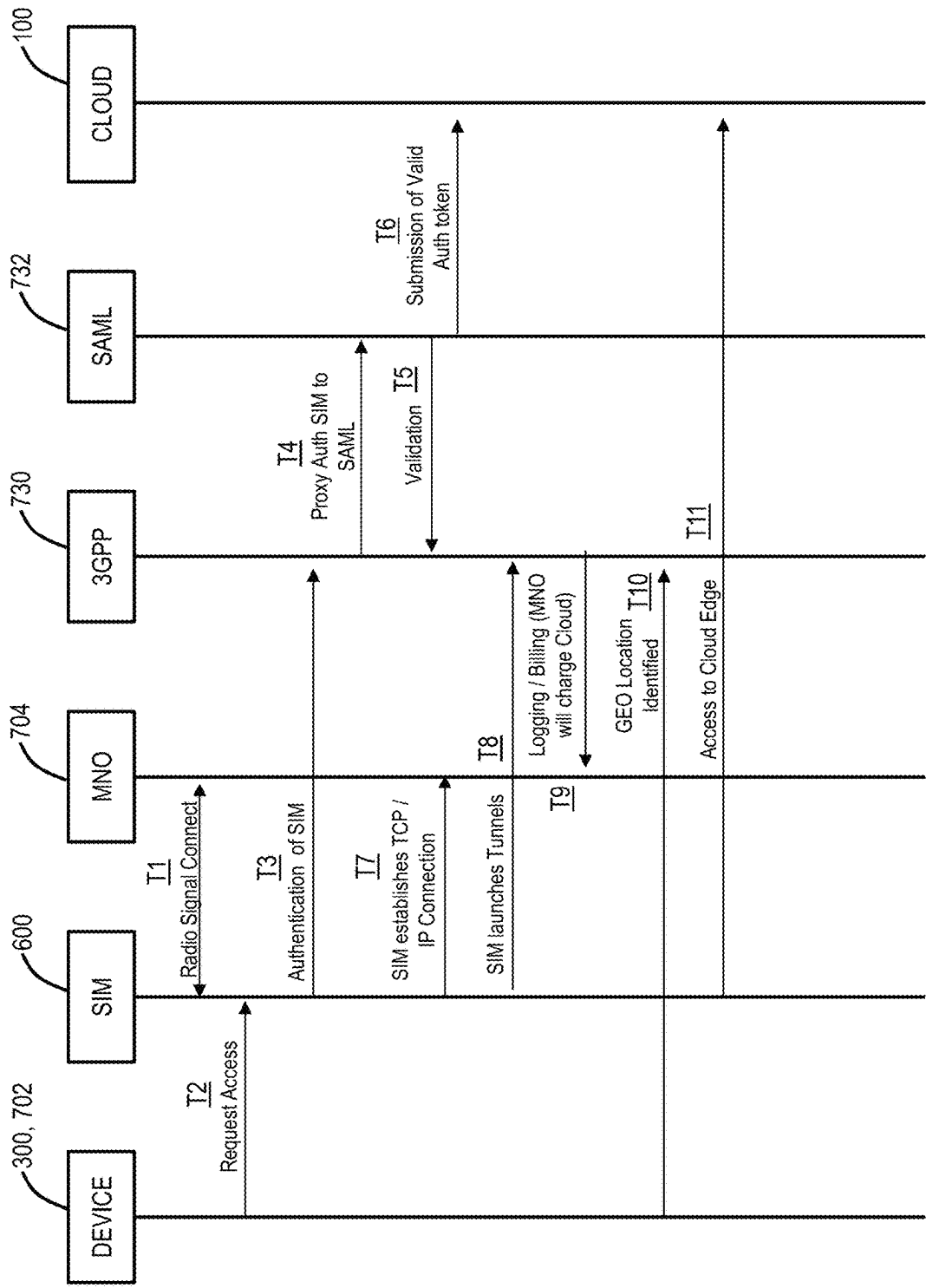
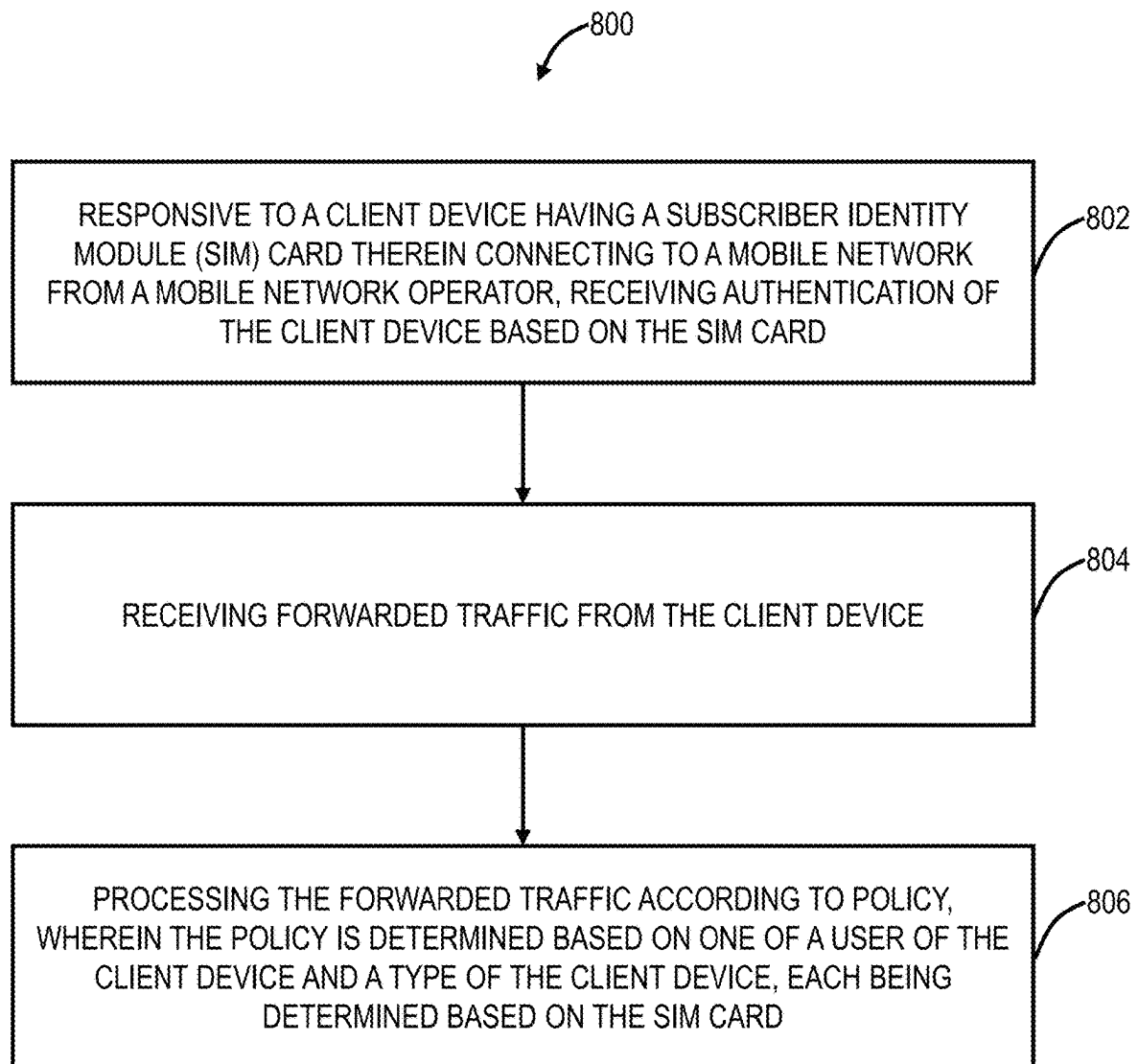


FIG. 17

**FIG. 18**

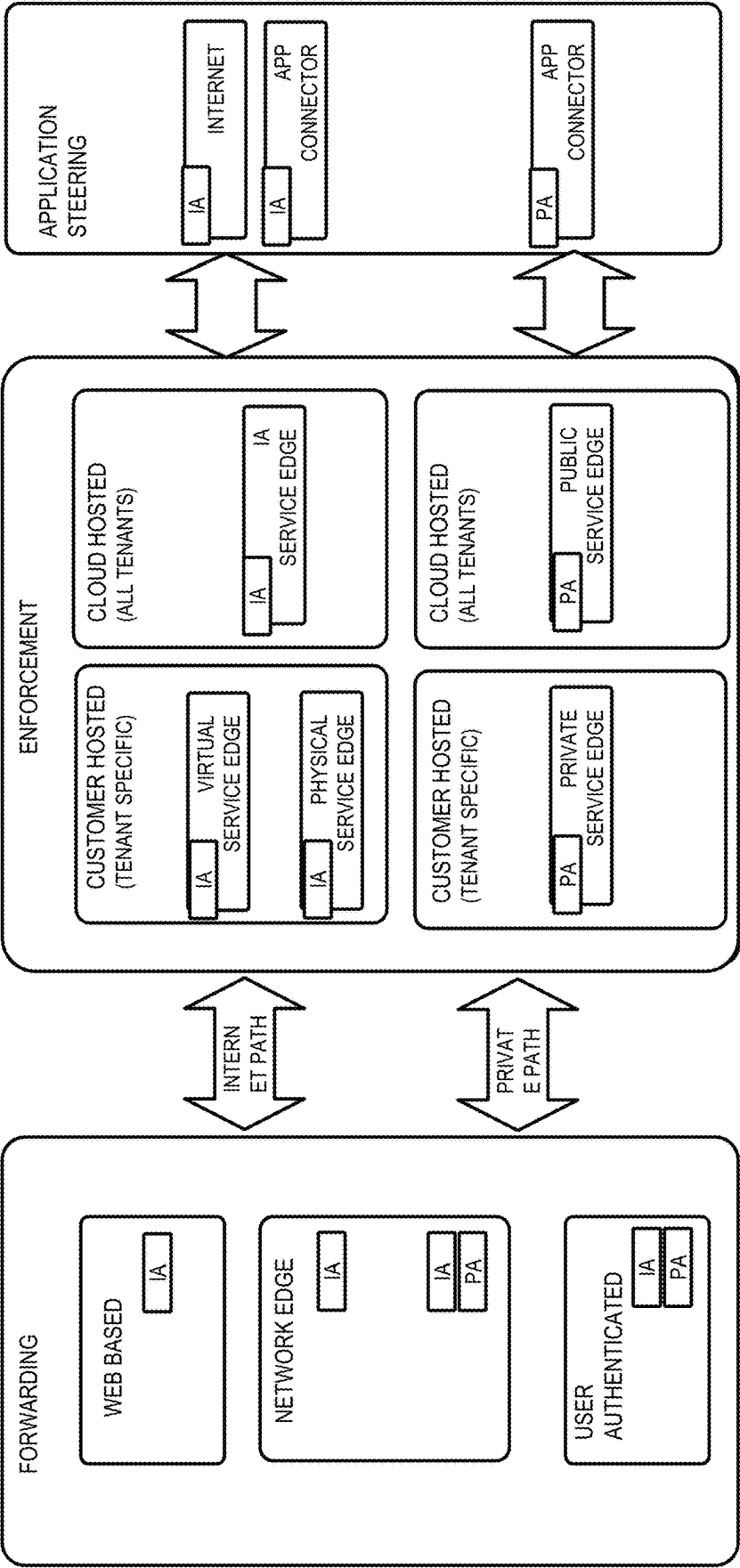


FIG. 19

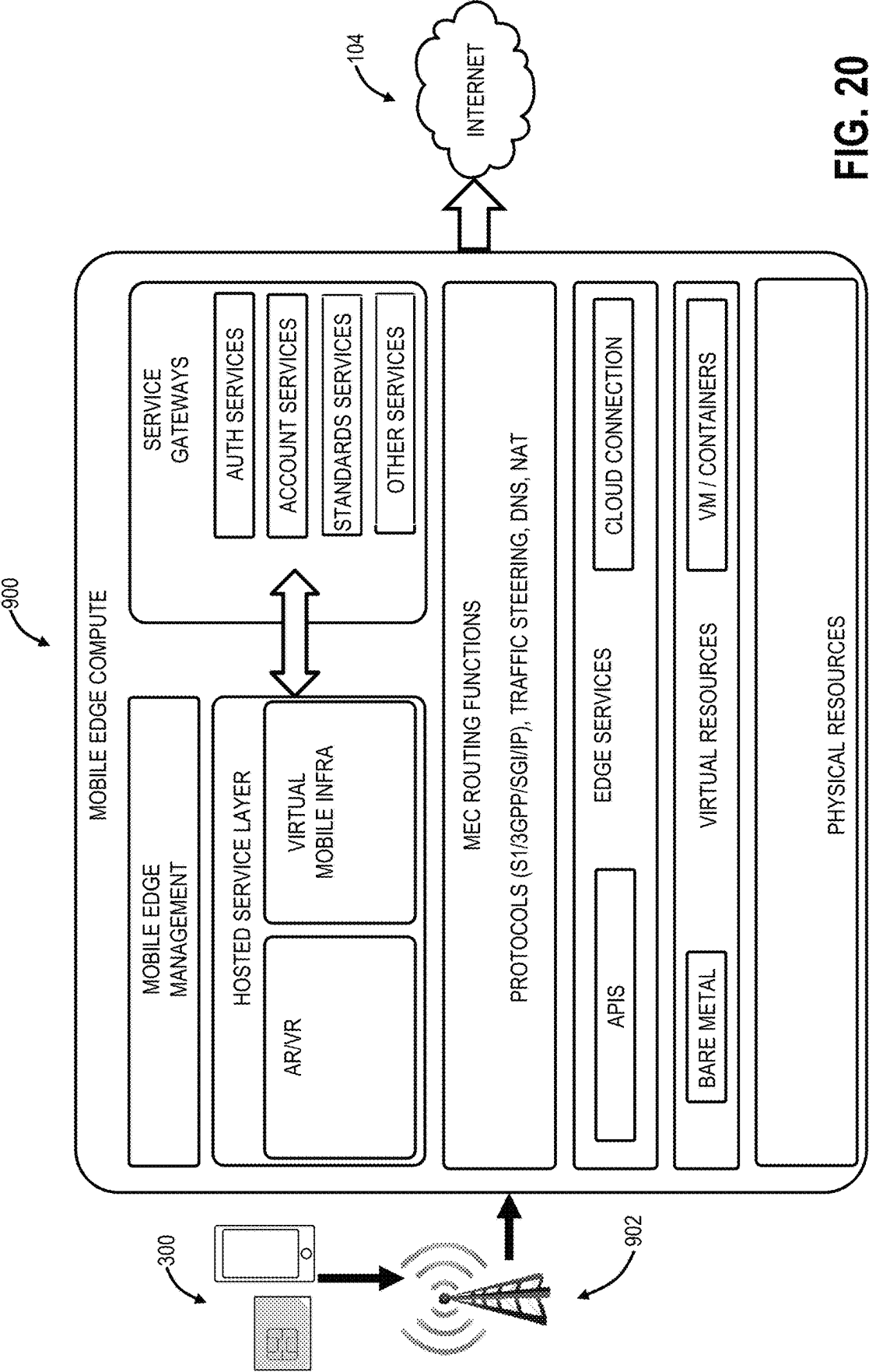


FIG. 20

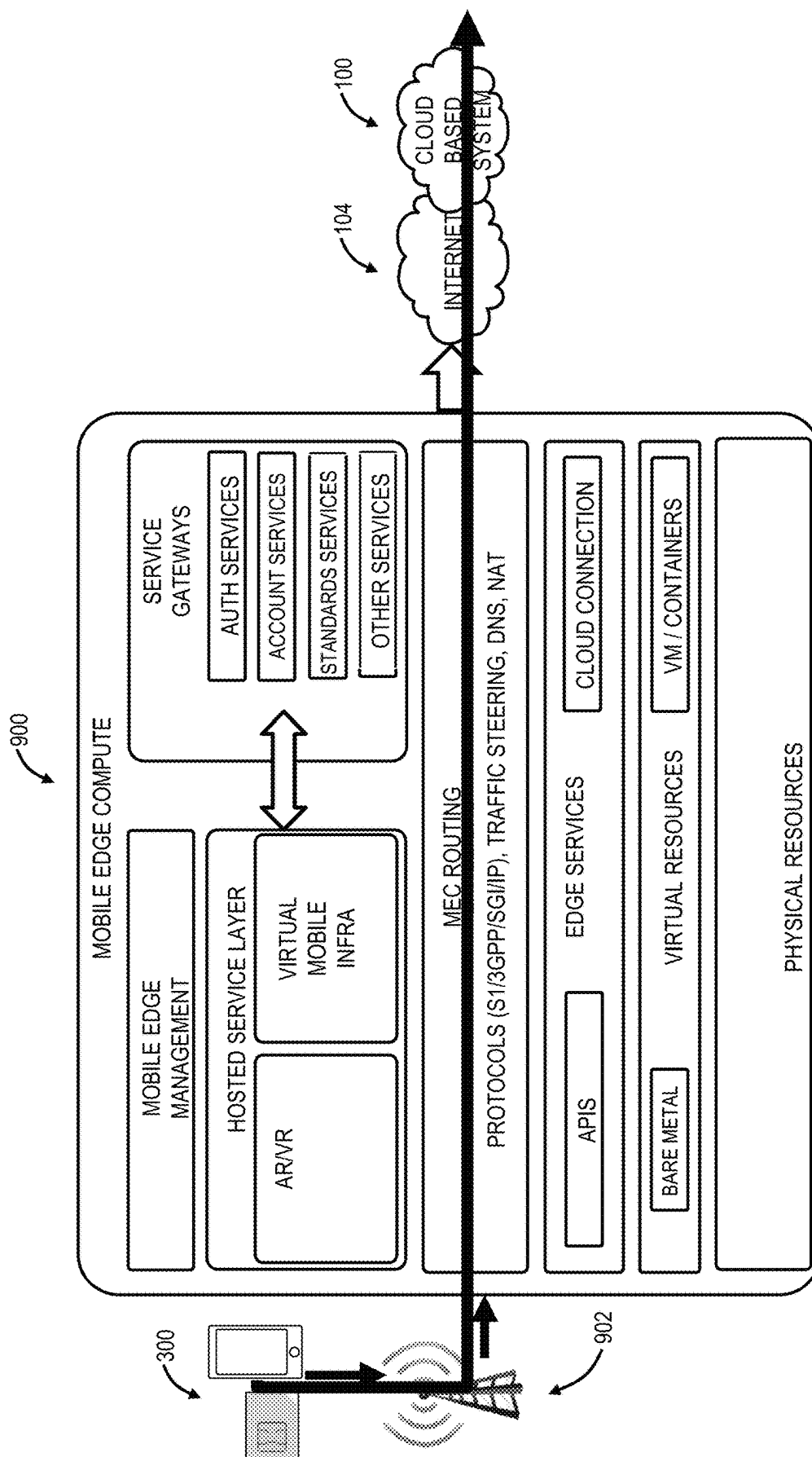


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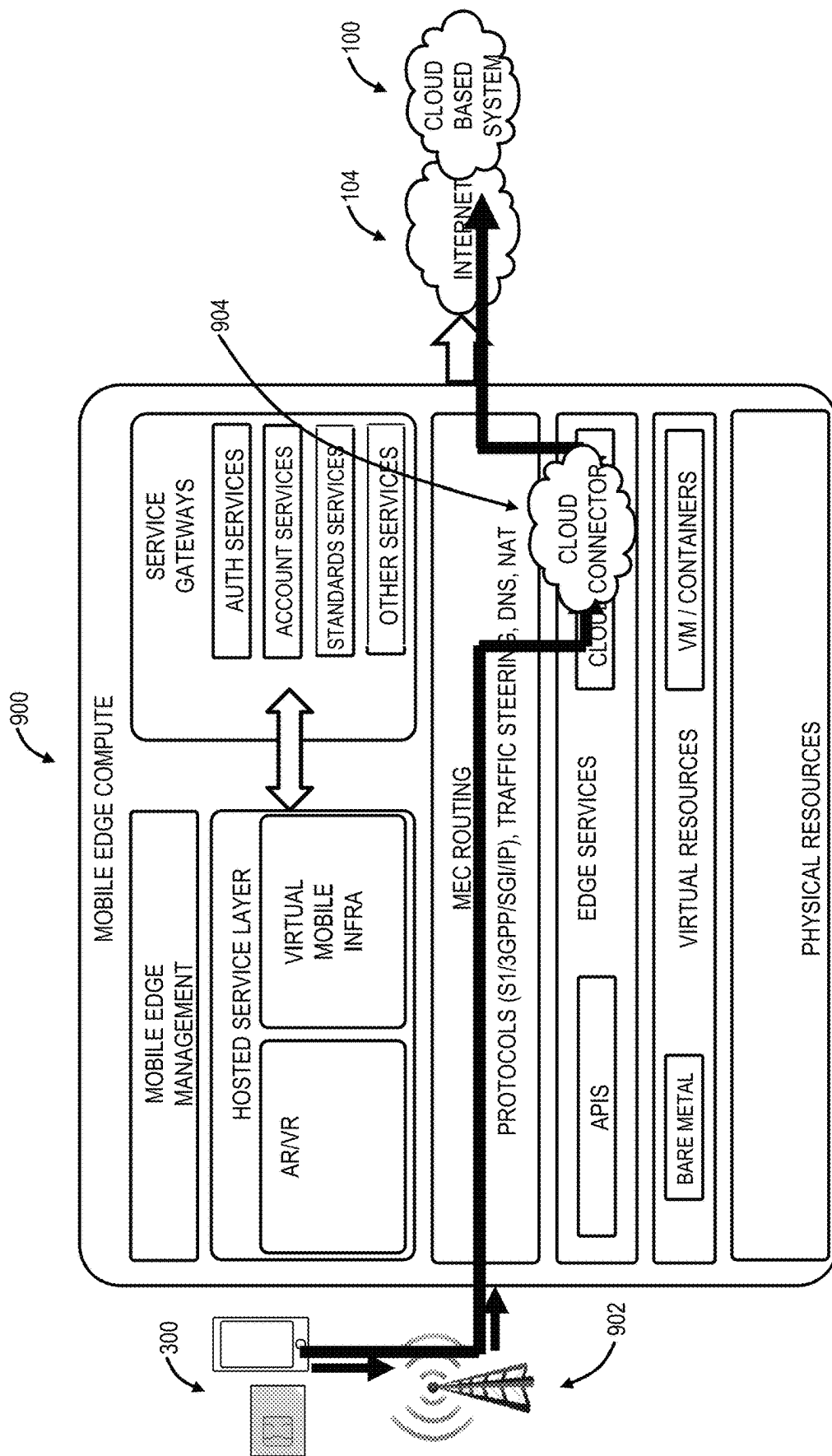


FIG. 22

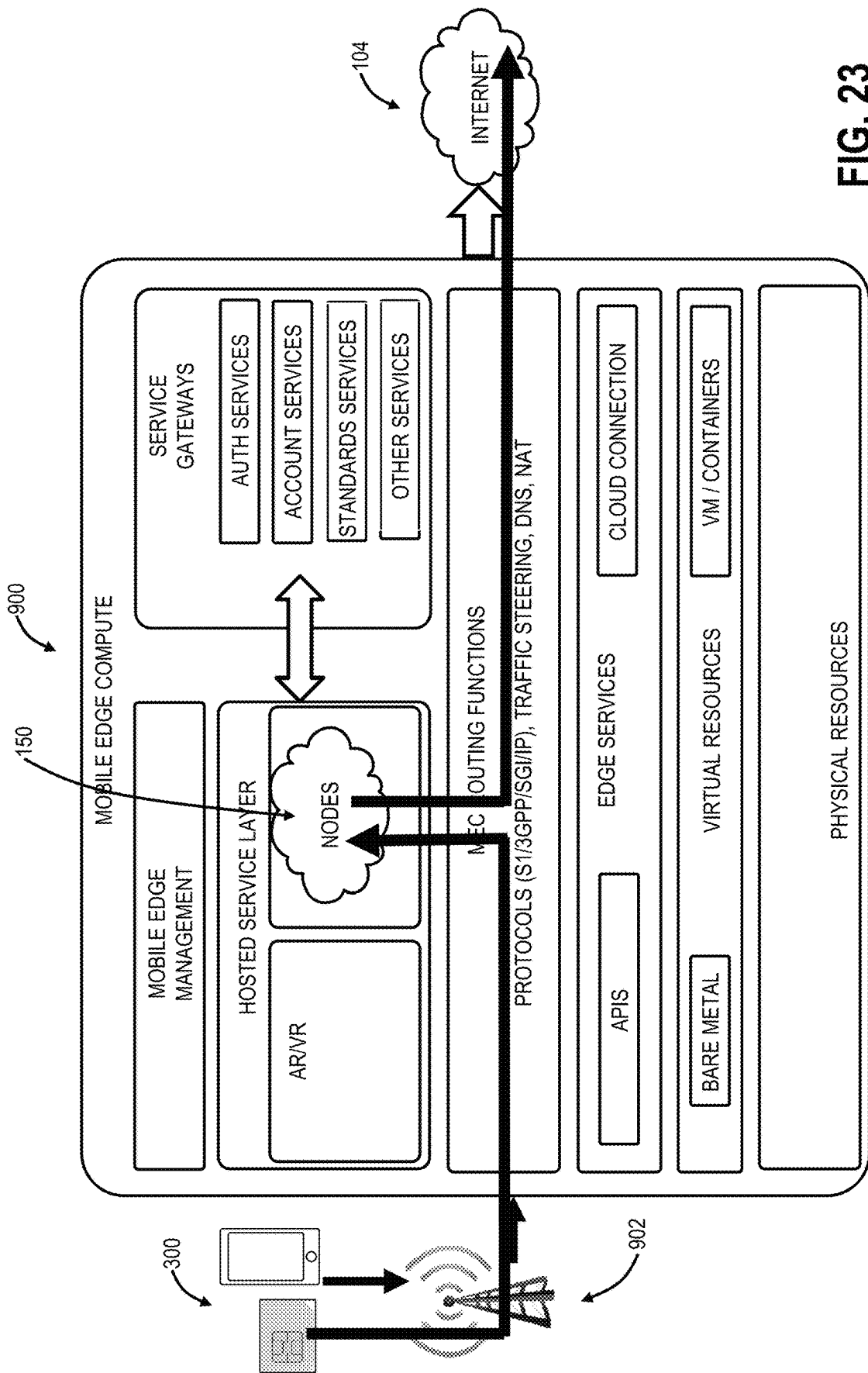
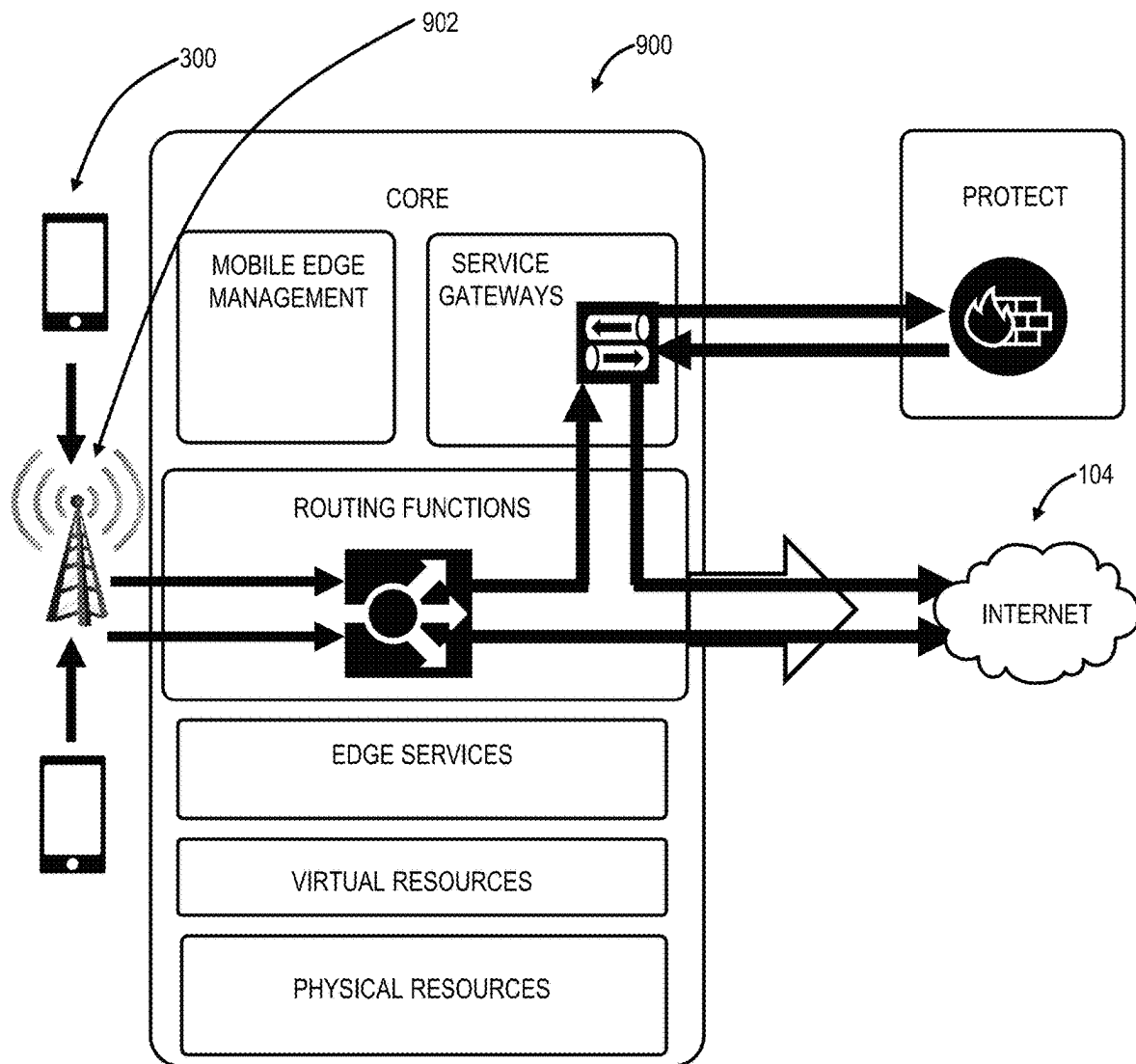
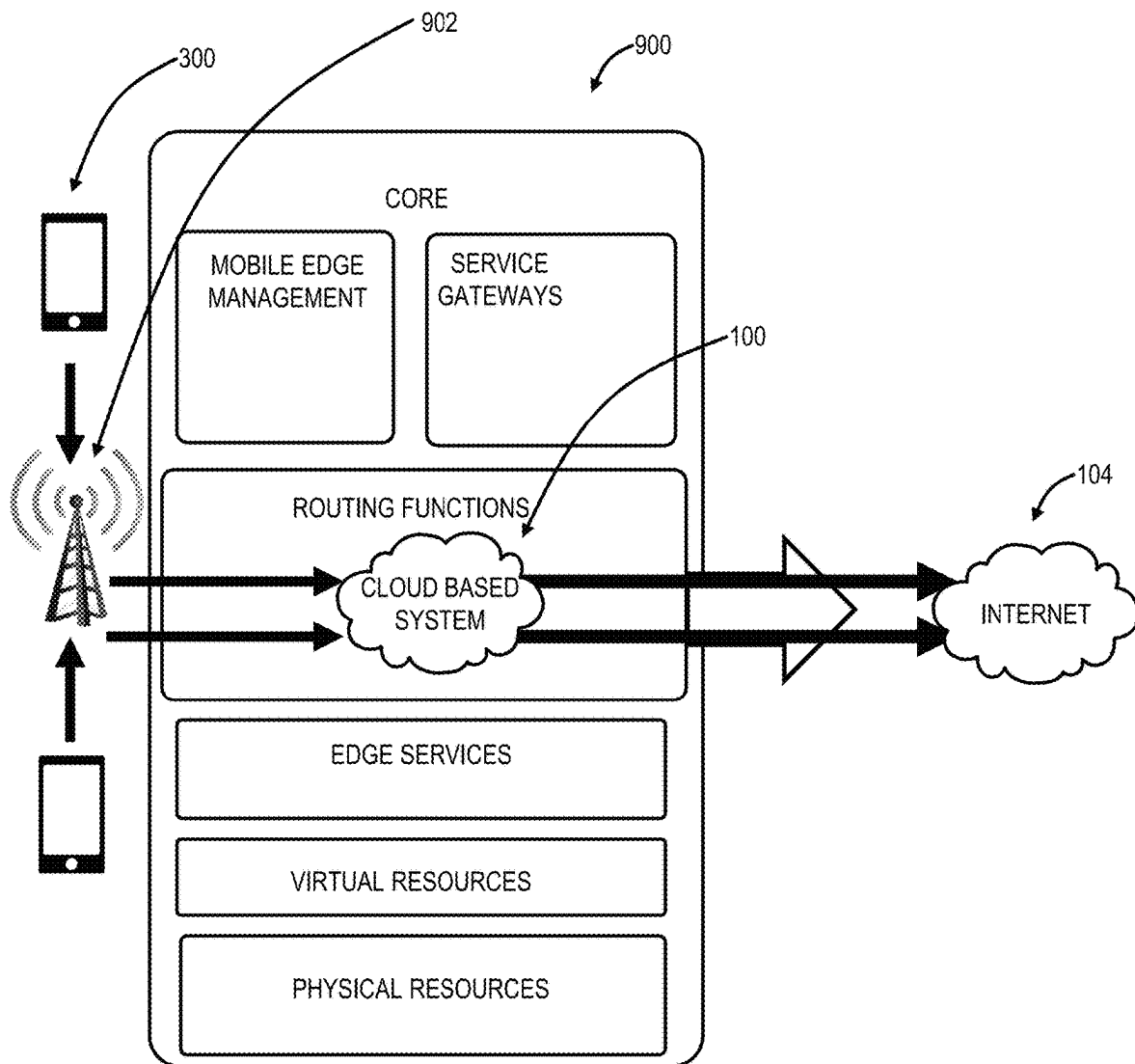


FIG. 23

**FIG. 24**

**FIG. 25**

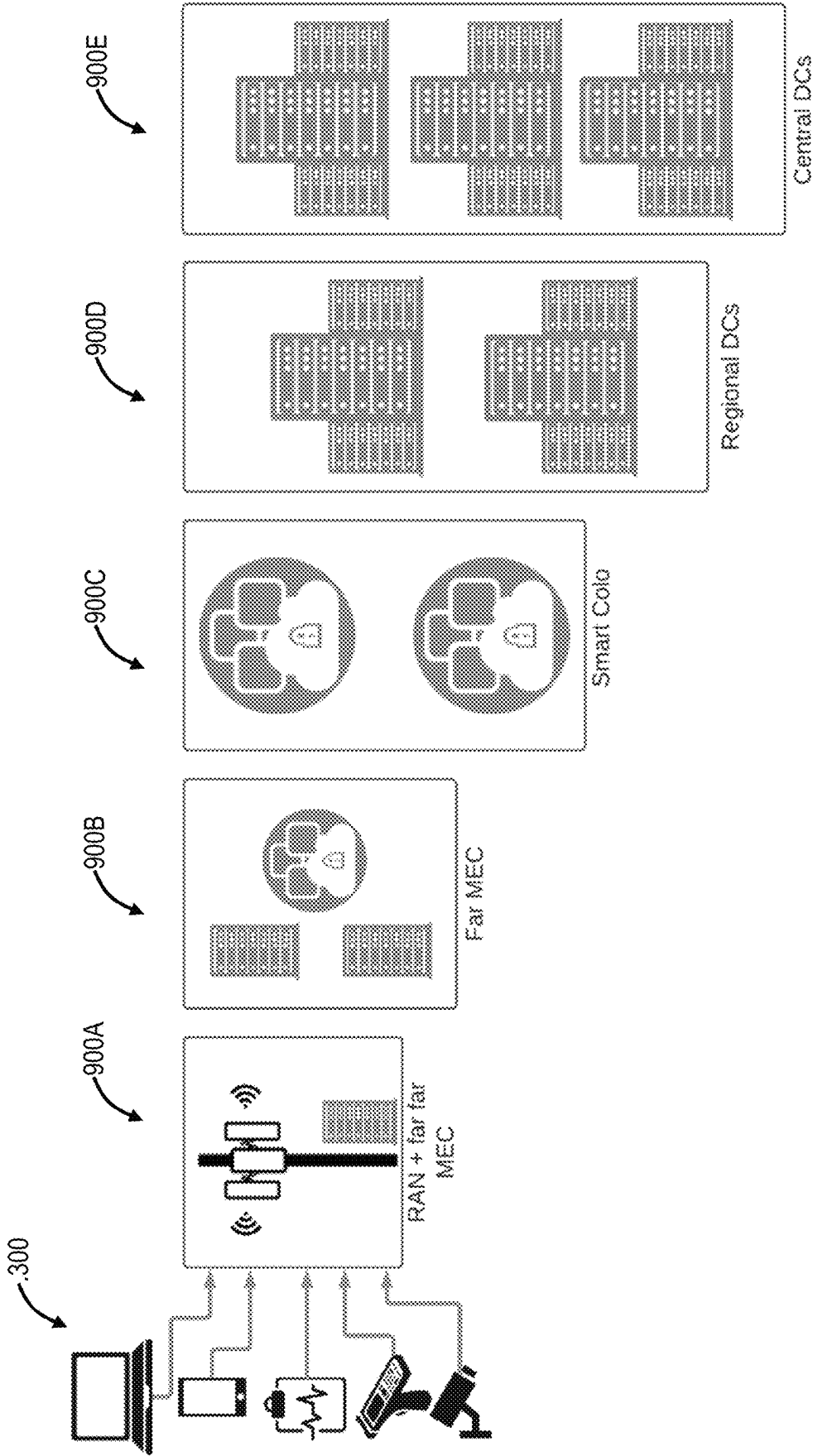


FIG. 26

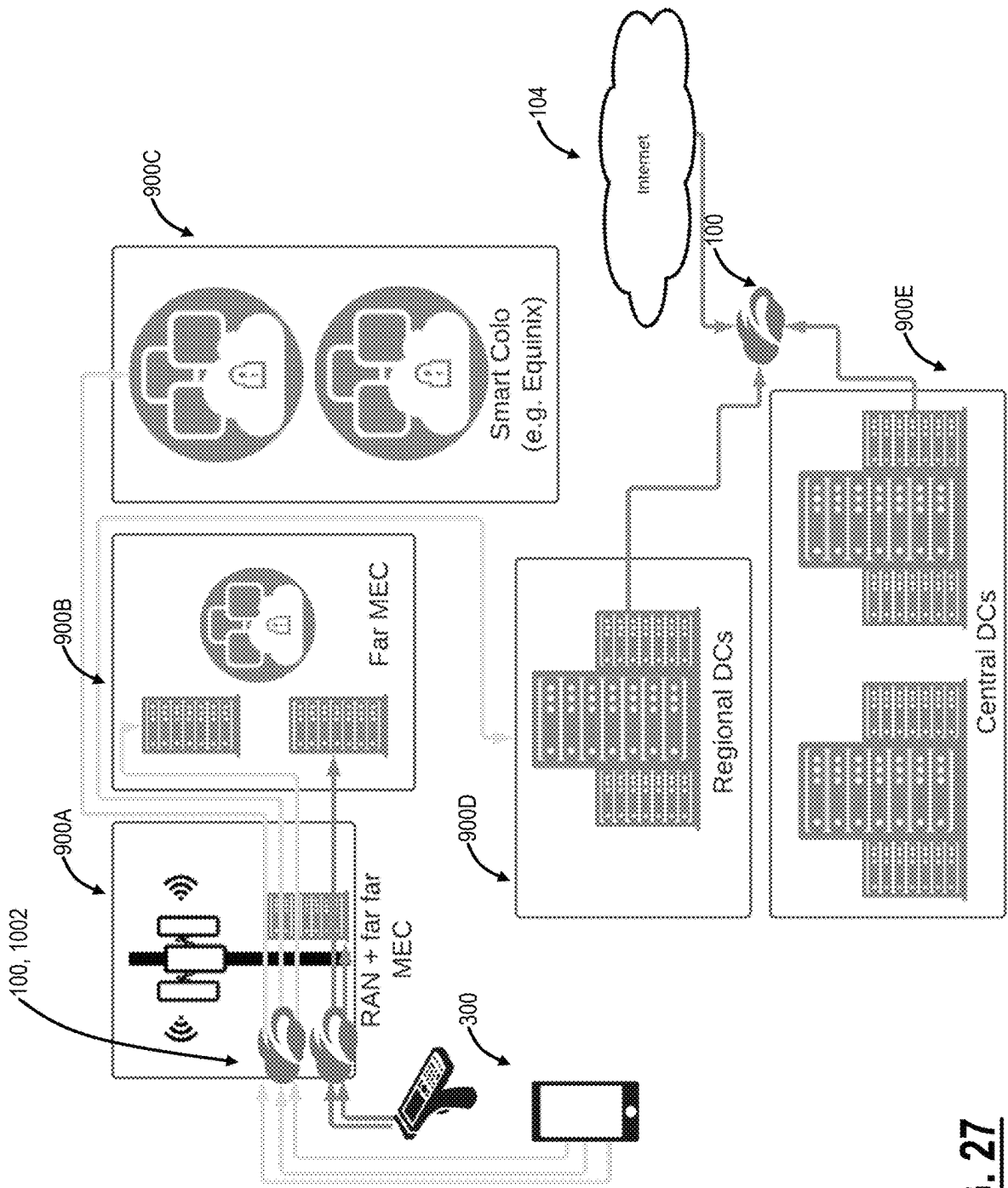


FIG. 27

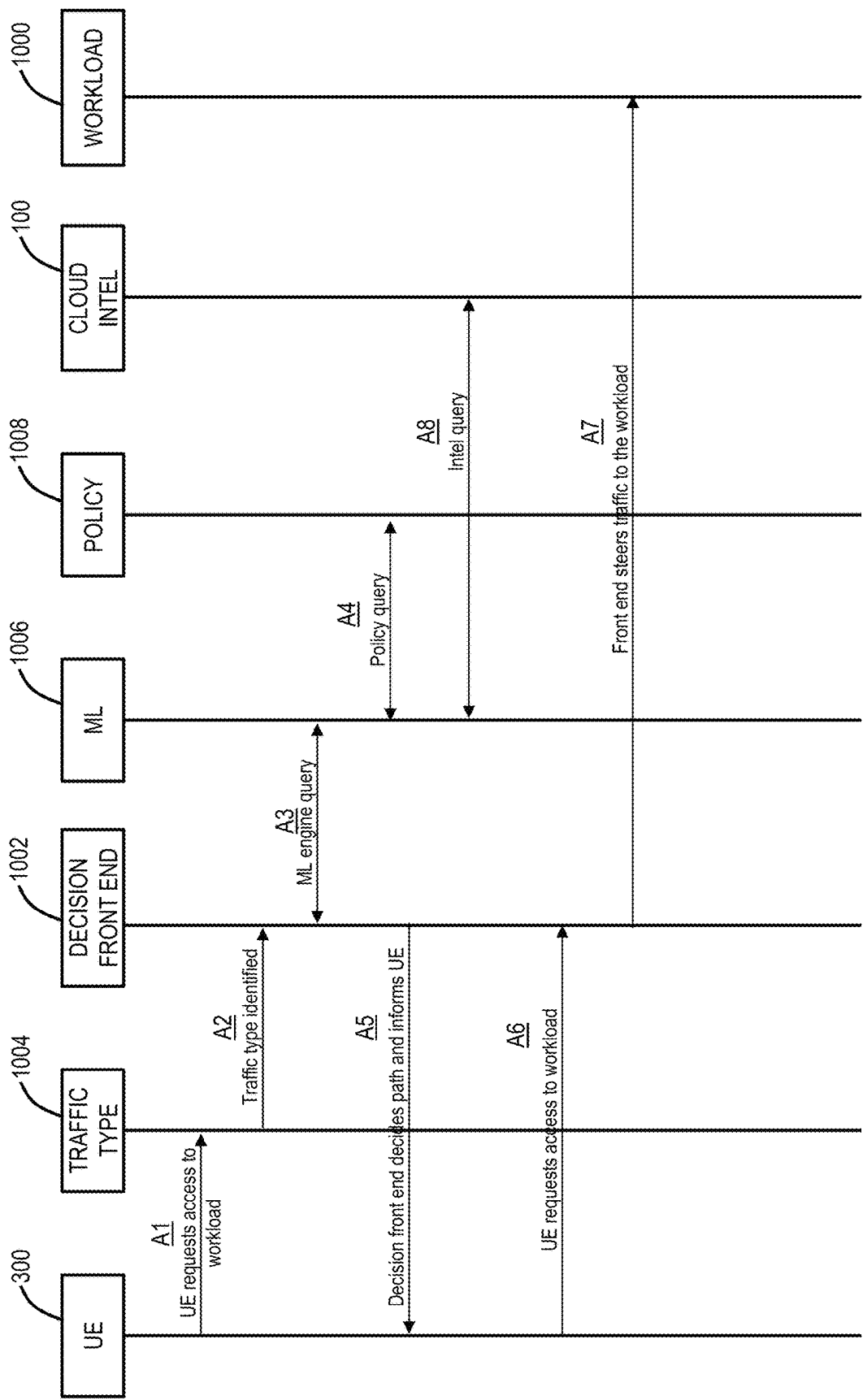
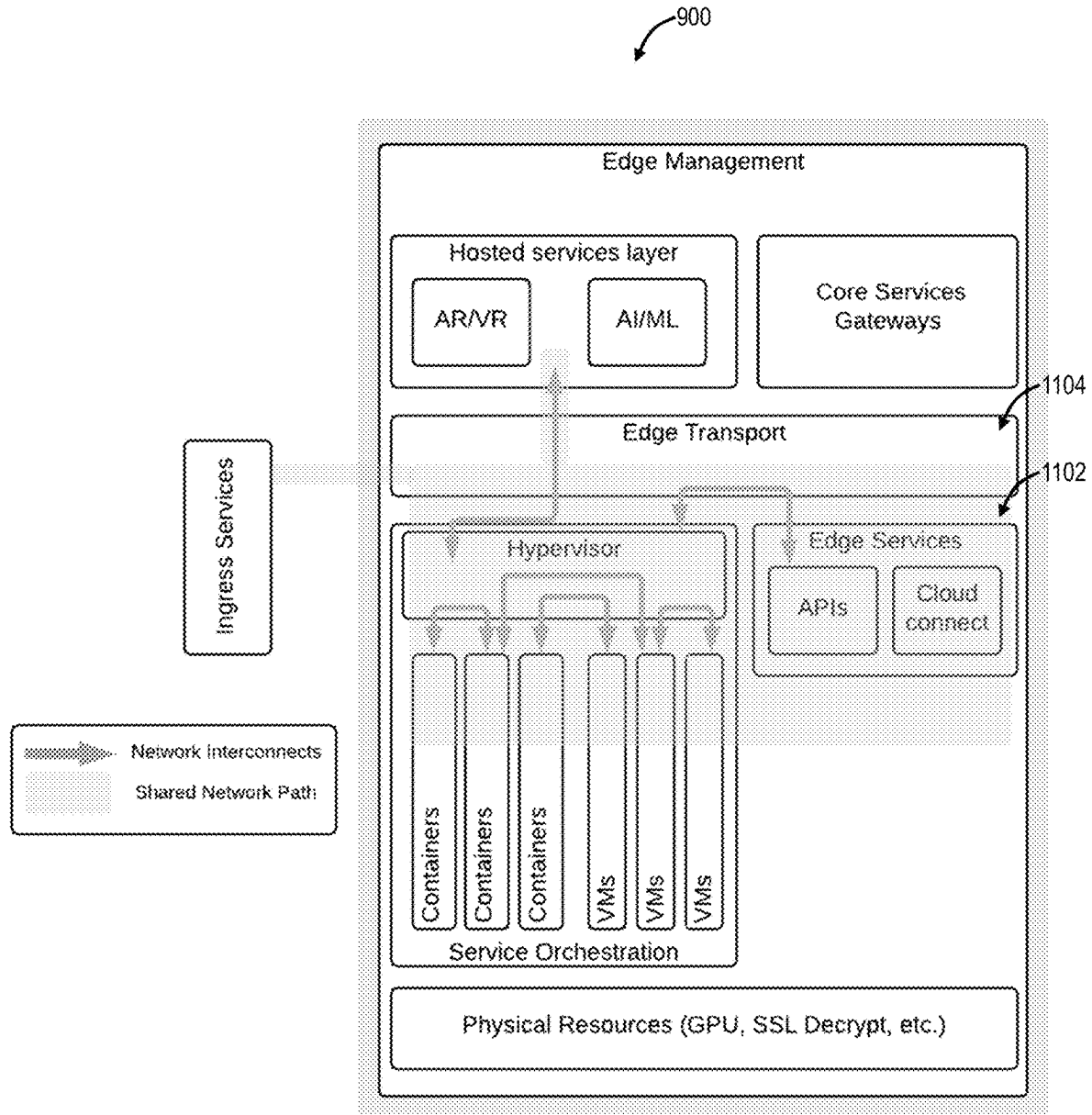
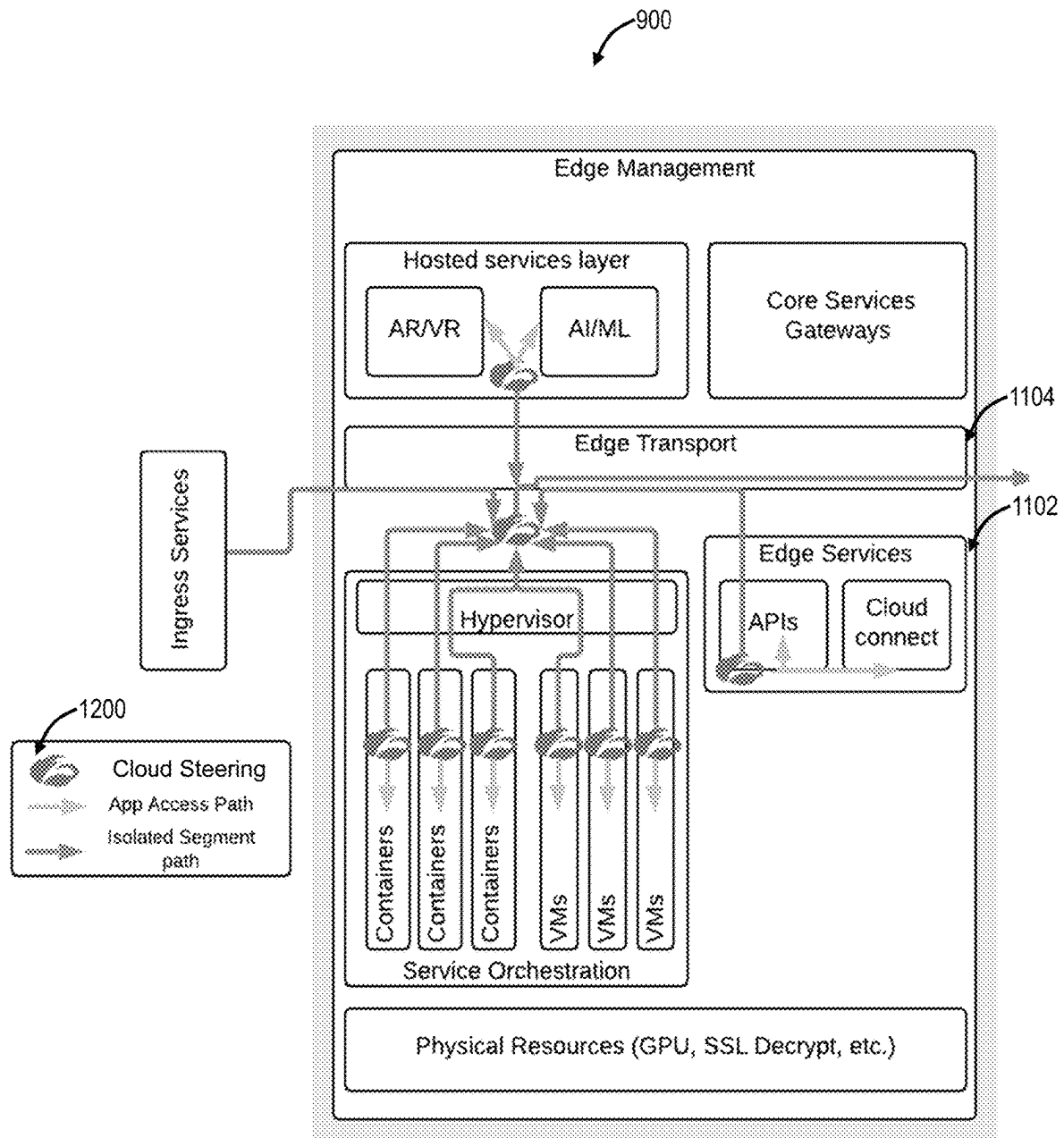


FIG. 28

**FIG. 29**

**FIG. 30**

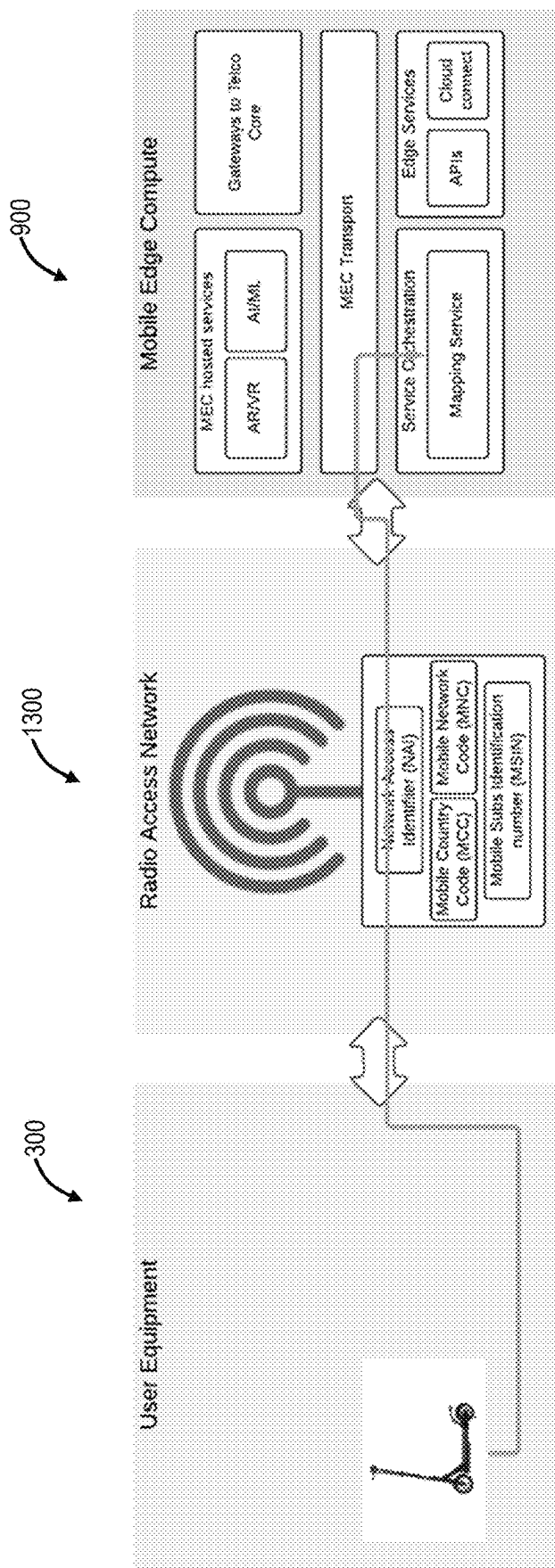
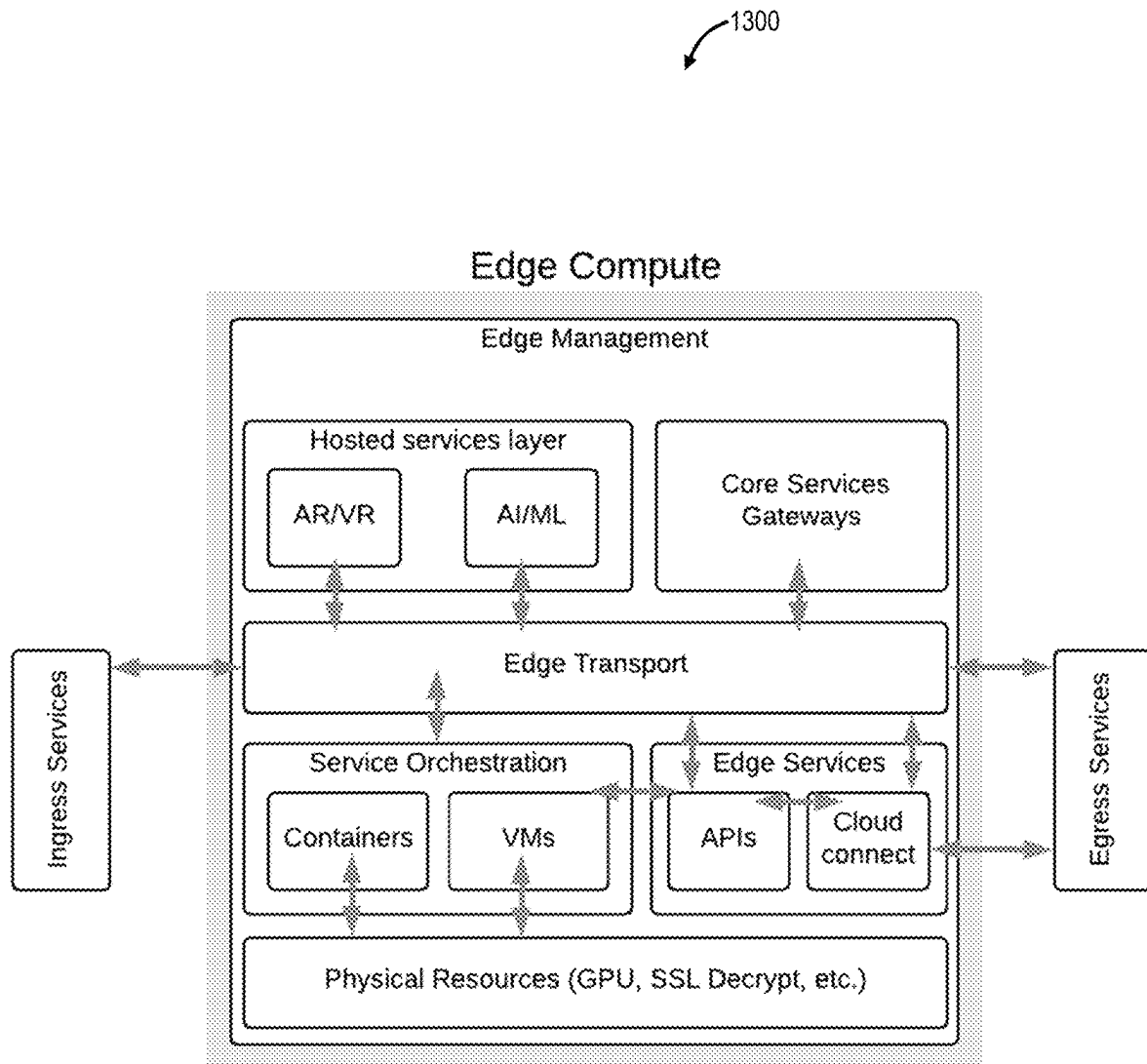


FIG. 31

**FIG. 32**

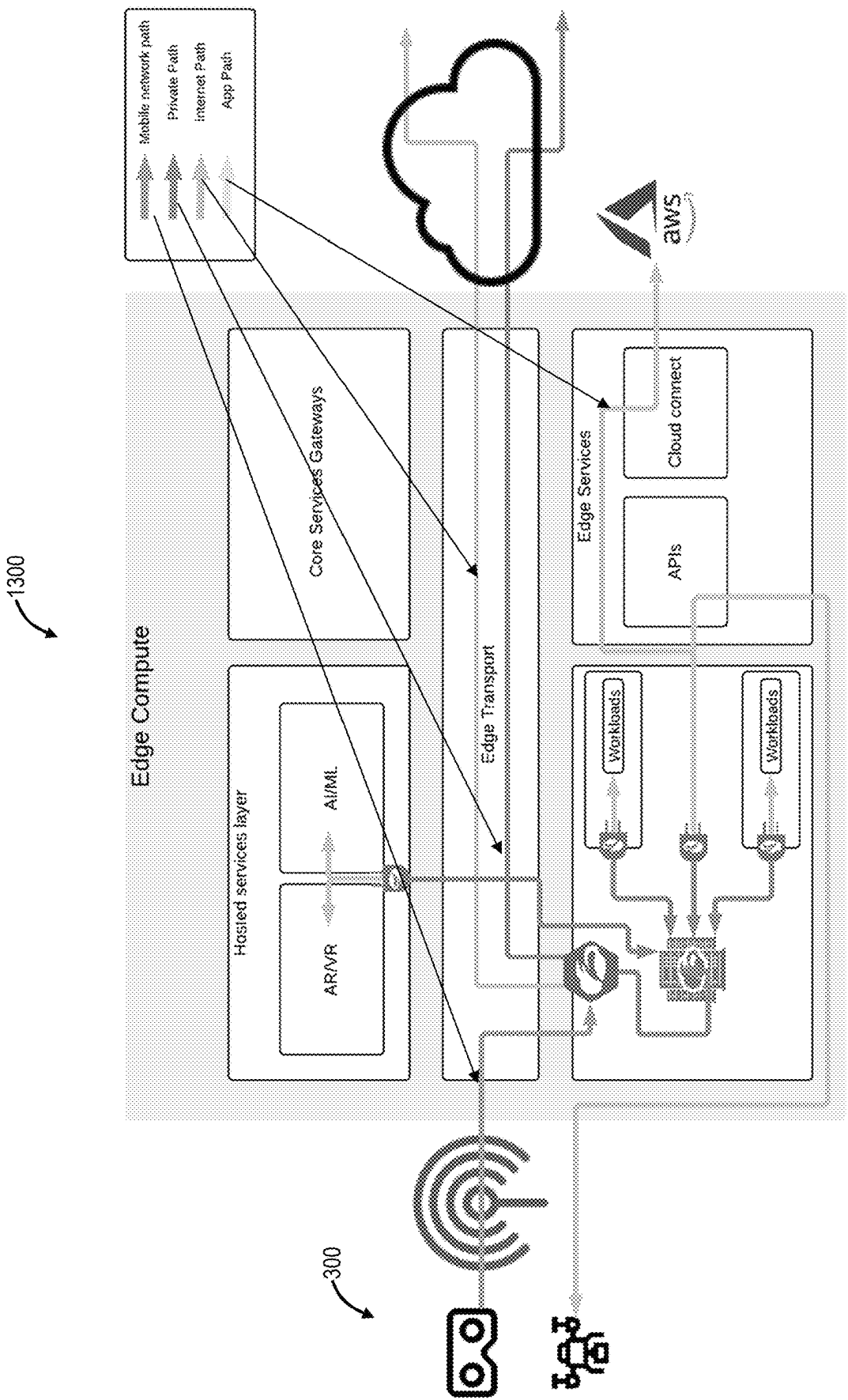
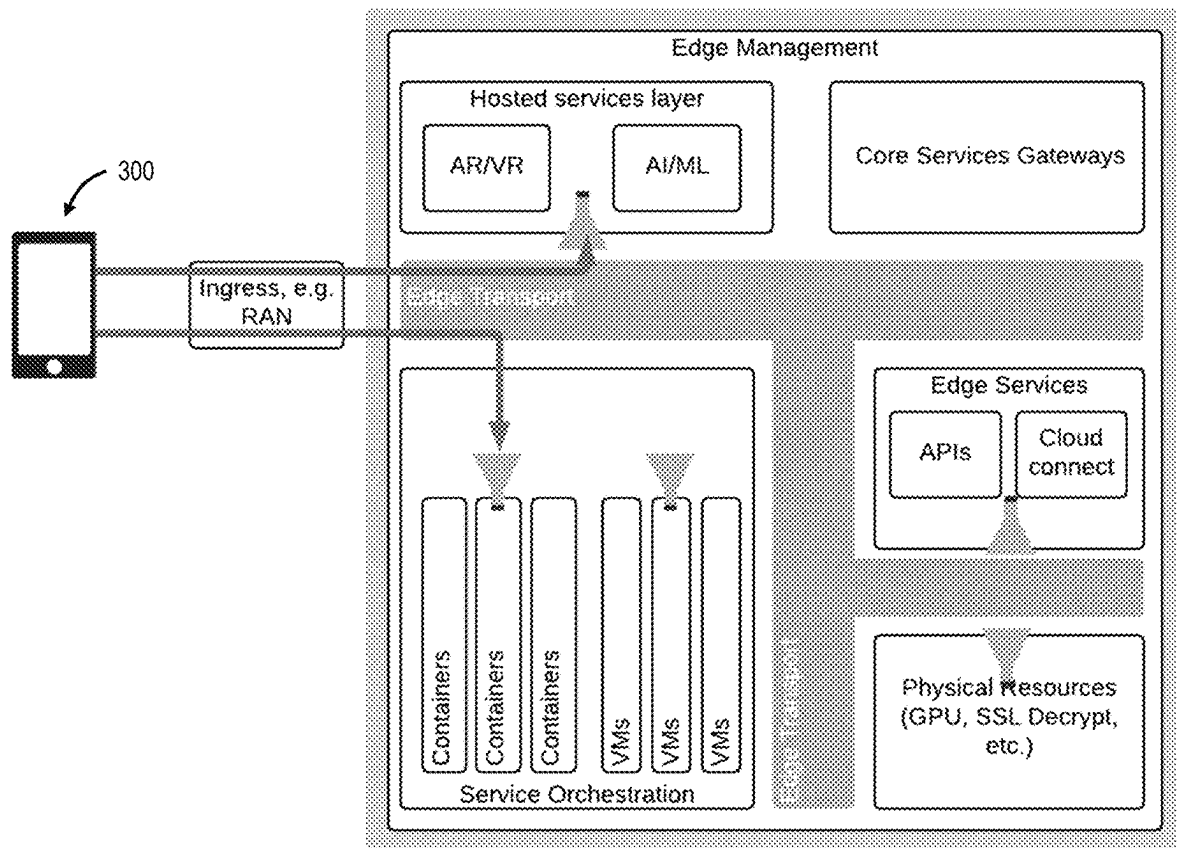
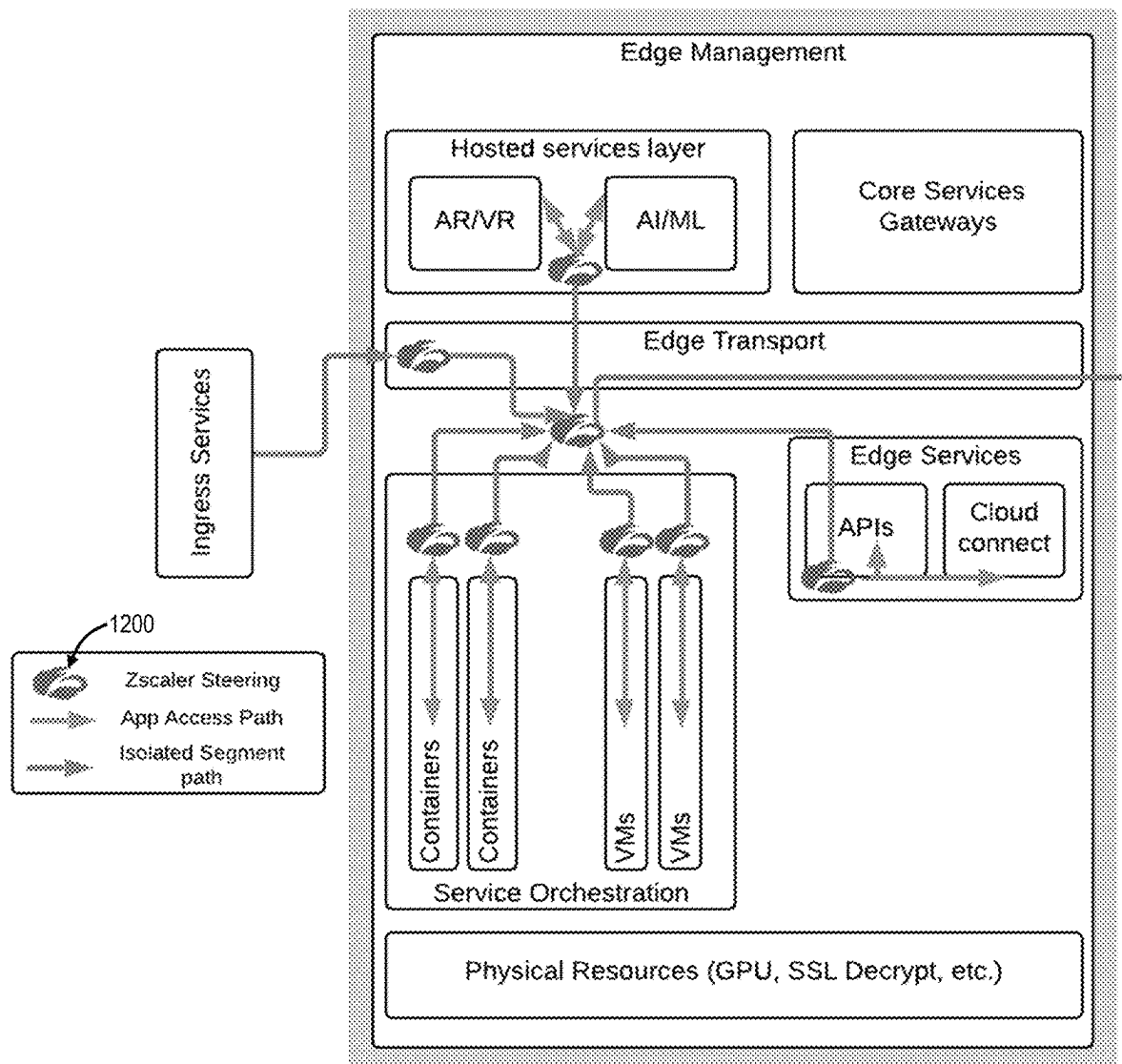


FIG. 33

**FIG. 34**

**FIG. 35**

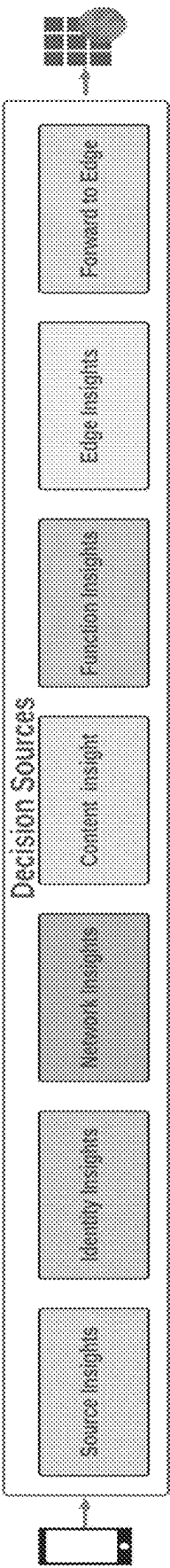


FIG. 36

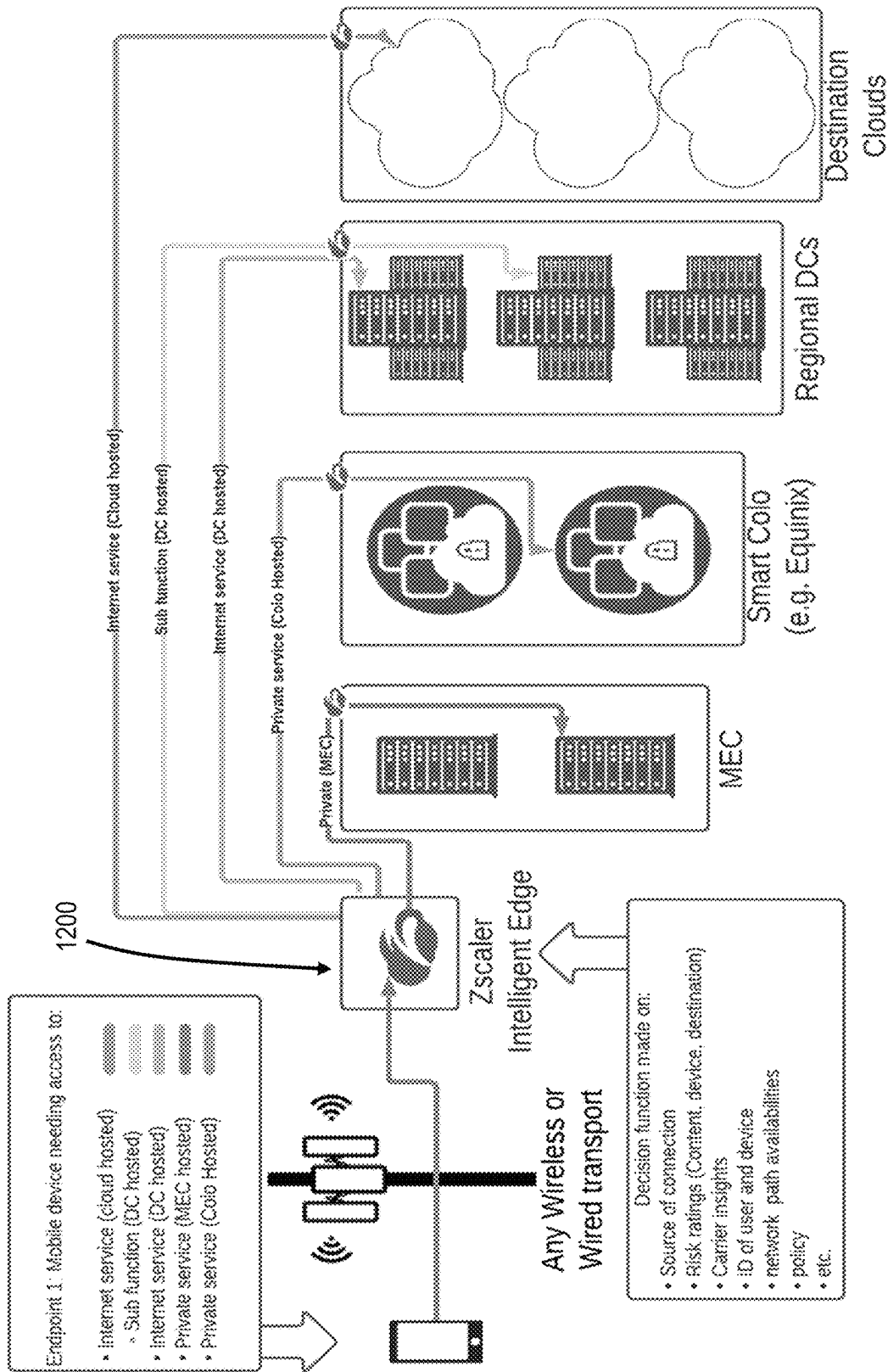


FIG. 37

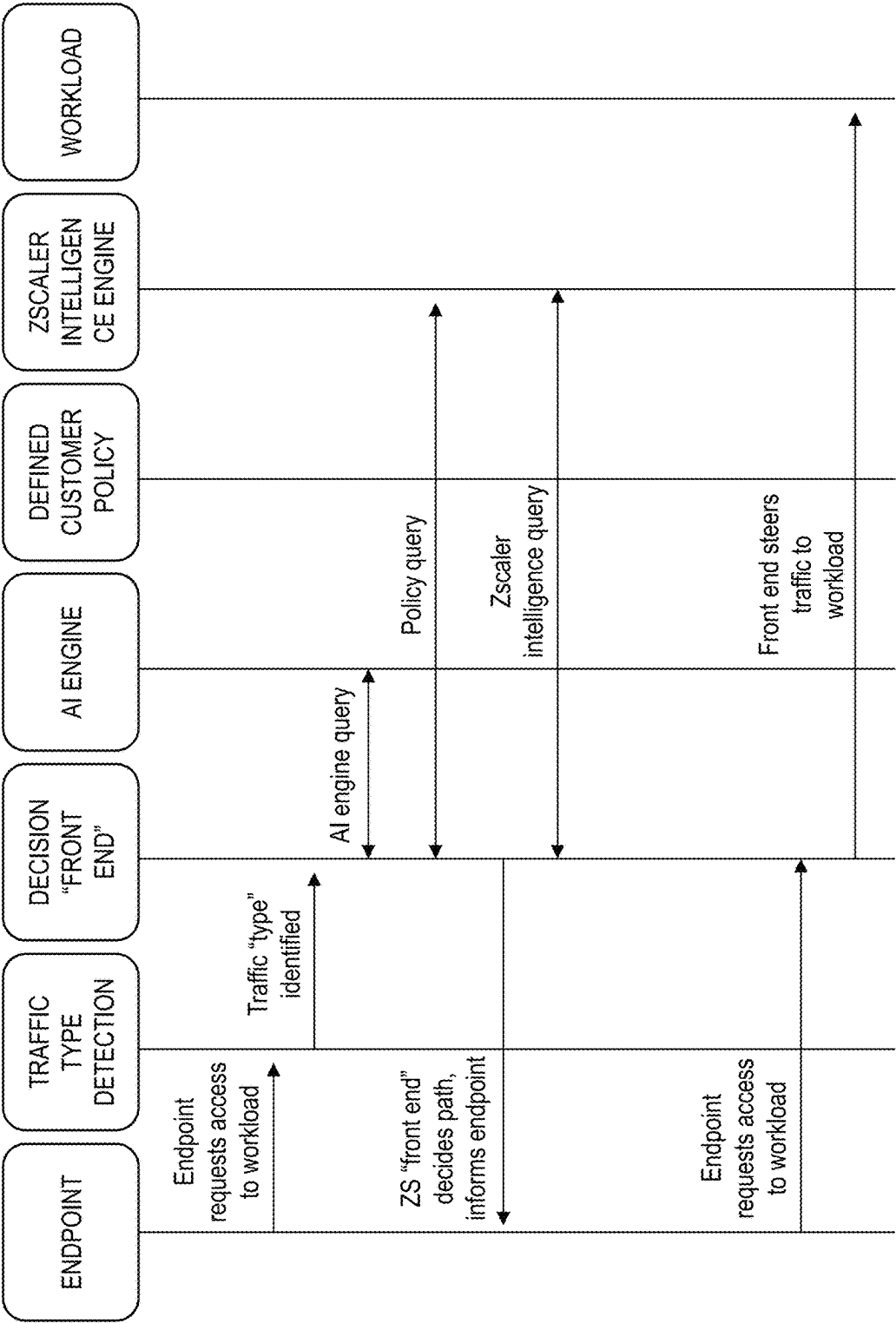


FIG. 38

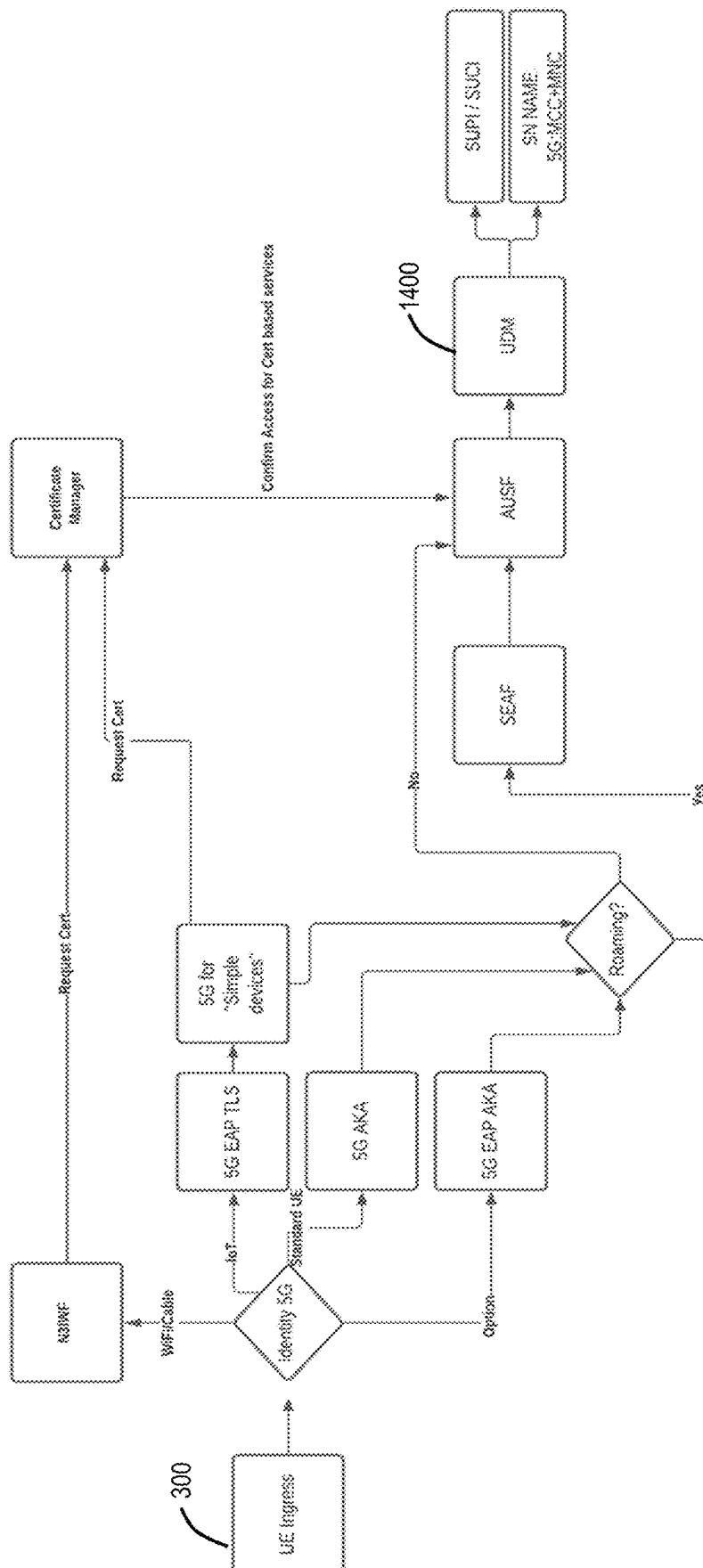
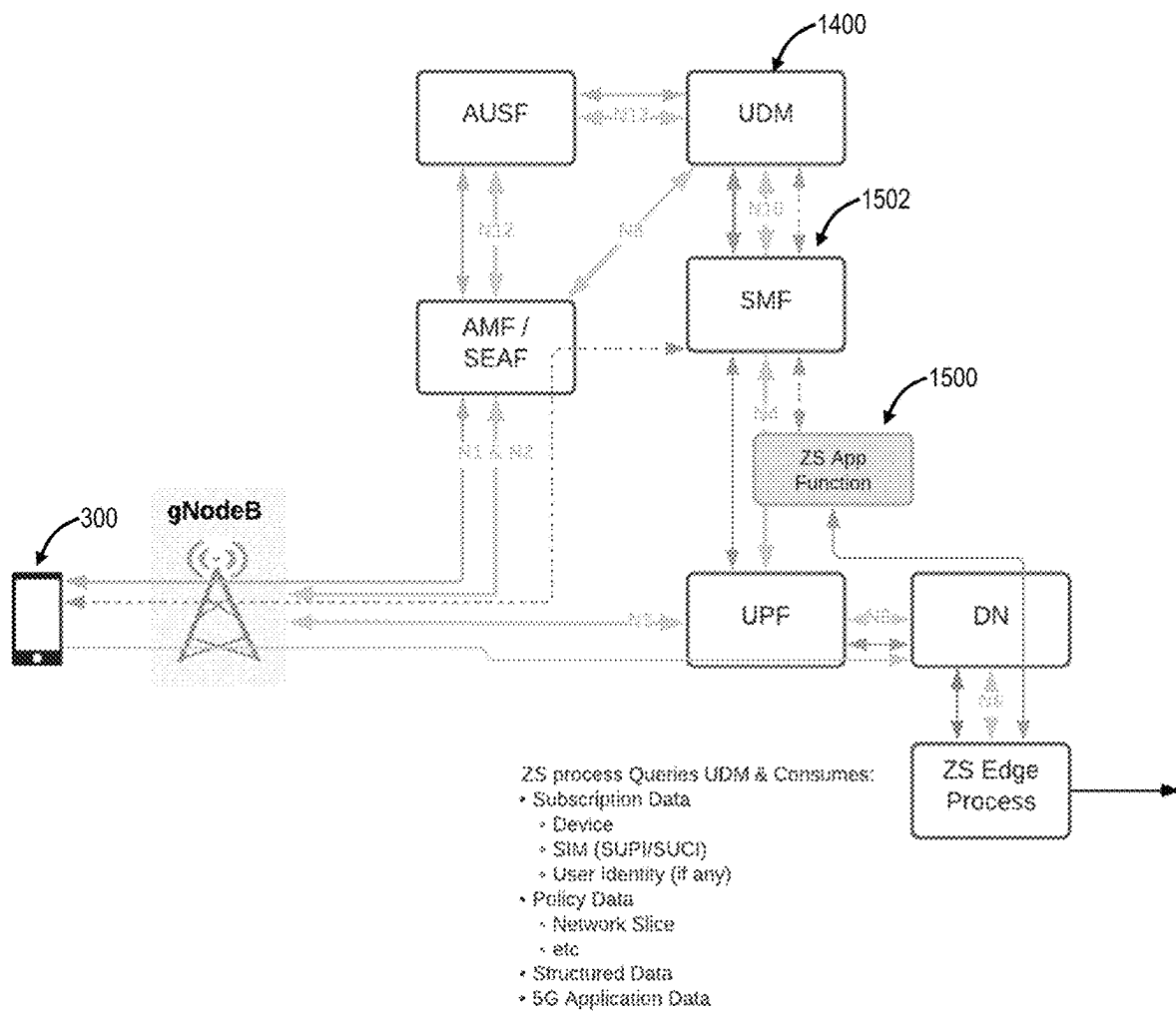


FIG. 39

**FIG. 40**

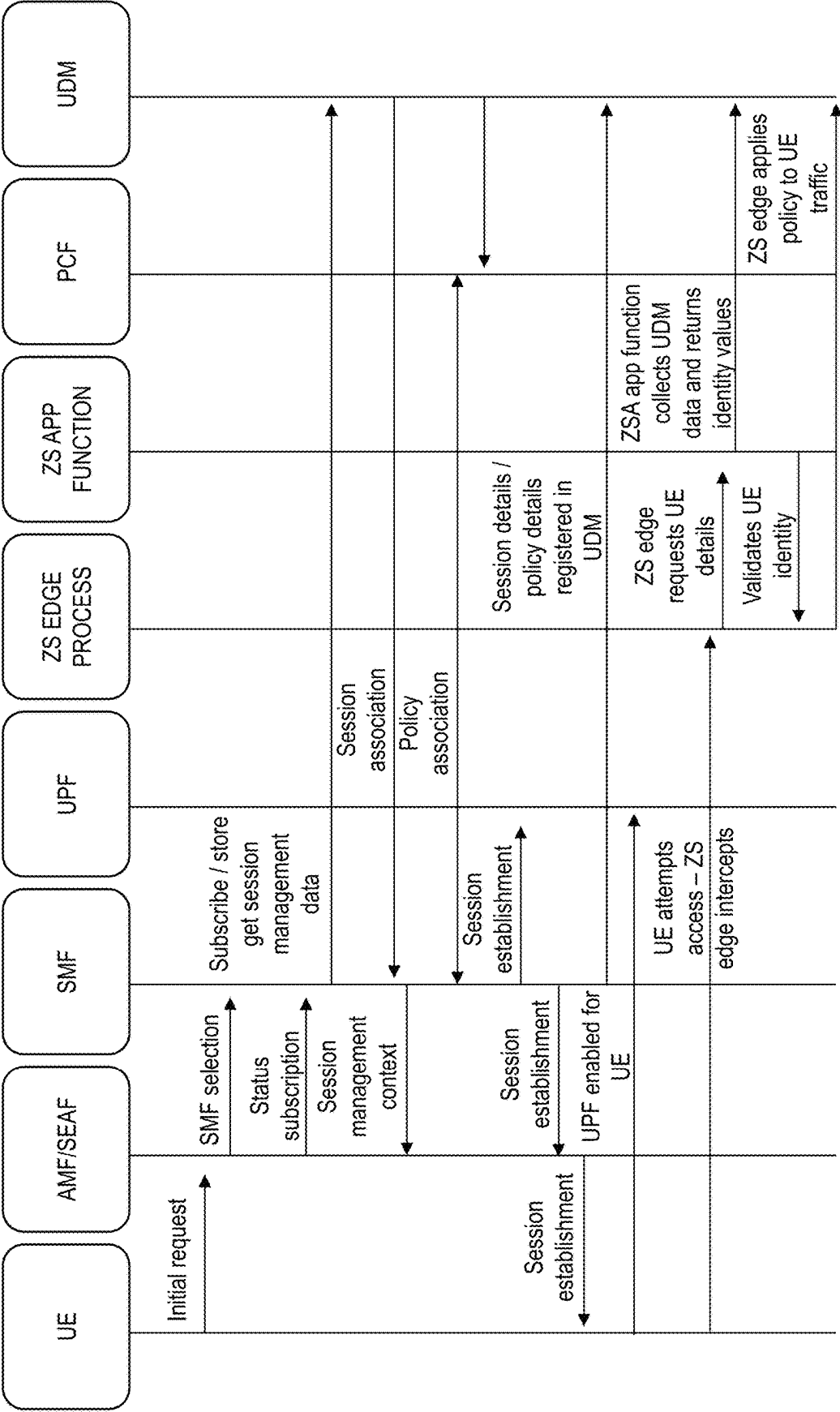


FIG. 41

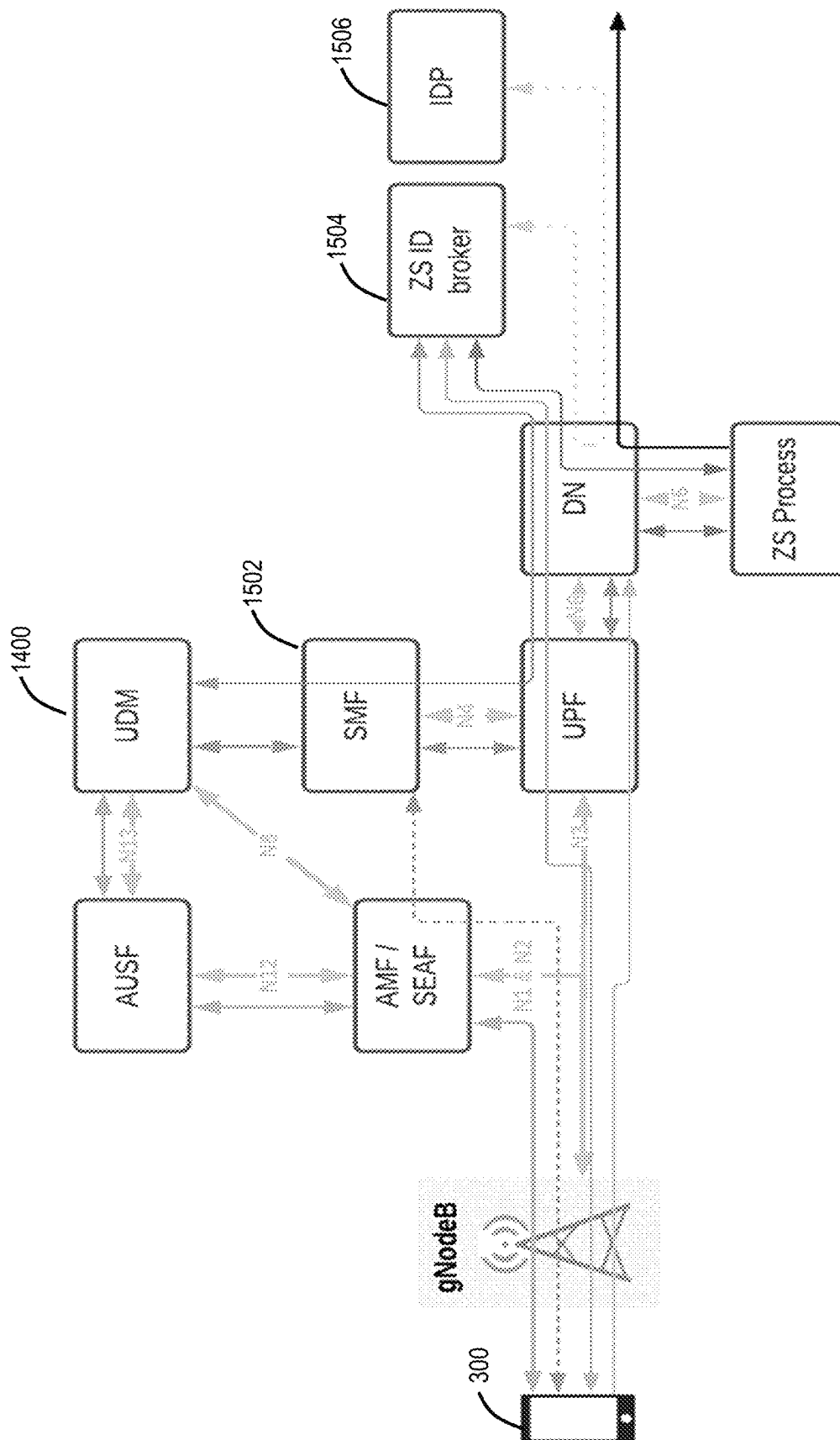


FIG. 42

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**5G UDM TO IDP FEDERATION AND
IDENTITY FUNCTION****CROSS-REFERENCE TO RELATED
APPLICATION(S)**

The present disclosure is a continuation-in-part of U.S. patent application Ser. No. 17/194,568, filed Mar. 8, 2021, entitled "Mobile and IoT device forwarding to the cloud," a continuation-in-part of U.S. patent application Ser. No. 17/371,408, filed Jul. 9, 2021, entitled "Cloud-based 5G security network architectures," and a continuation-in-part of U.S. patent application Ser. No. 17/491,795, filed Oct. 1, 2021, entitled "Cloud-based 5G security network architectures with intelligent steering," the contents of each are incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

The present disclosure relates generally to networking and computing. More particularly, the present disclosure relates to systems and methods for cloud-based 5G security network architectures, intelligent steering, device posture, workload isolation and interaction, identity, and secure edge steering.

BACKGROUND OF THE DISCLOSURE

Fifth generation (5G) wireless deployments are ongoing. 5G means more data, more services, and more devices. The traditional view of an enterprise network (i.e., corporate, private, etc.) included a well-defined perimeter defended by various appliances (e.g., firewalls, intrusion prevention, advanced threat detection, etc.). In this traditional view, mobile users utilize a Virtual Private Network (VPN), etc. and have their traffic backhauled into the well-defined perimeter. This worked when mobile users represented a small fraction of the users, i.e., most users were within the well-defined perimeter. However, this is no longer the case—the definition of the workplace is no longer confined to within the well-defined perimeter. 5G provides connection speeds rivaling wired speeds and drives significant increases in network traffic volumes. This results in an increased risk for enterprise data residing on unsecured and unmanaged devices as well as the security risks in access to the Internet.

Security is a key 5G design principle. 5G is secure where 5G was designed to be secure, but the security is limited to the 5G network itself, not for the workloads running on top of the 5G network. That is, applications and services running on 5G networks that are not associated directly with network traffic management cannot be assumed to be secure. Application providers are responsible for end-to-end service security.

Cloud-based security solutions have emerged, such as Zscaler Internet Access (ZIA) and Zscaler Private Access (ZPA), available from Zscaler, Inc., the applicant and assignee of the present application. These cloud-based services operate inline between User Equipment (UE) and the Internet. However, many 5G use cases do not require moving data between the UE and the cloud, but rather data between the UE and Multiaccess Edge Compute (MECs) devices available physically close to the UE to reduce network latency. When the MEC is owned by the customer, or leased/subscribed by the customer from the 5G network service provider, this arrangement can be used to limit the

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amount of UE data traffic traversing farther into the 5G network for reasons of privacy or cost.

BRIEF SUMMARY OF THE DISCLOSURE

The present disclosure relates to systems and methods for cloud-based 5G security network architectures intelligent steering, device posture, workload isolation and interaction, identity, and secure edge steering. Specifically, various approaches are described to integrate cloud-based security services into Multiaccess Edge Compute servers (MECs). That is, existing cloud-based security services are in line between a UE and the Internet. The present disclosure includes integrating the cloud-based security services and associated cloud-based system within service provider's MECs. In this manner, a cloud-based security service can be integrated with a service provider's 5G network or a 5G network privately operated by the customer. For example, nodes in a cloud-based system can be collocated within a service provider's network, to provide security functions to 5G users or connected by peering from the cloud-based security service into the 5G service provider's regional communications centers.

Intelligent steering includes actively identifying traffic types from User Equipment (UE—a reference to any wireless device equipped with a SIM/eSIM/iSIM/other attached to a telecommunications network) and steering this traffic to the most effective edge for processing, securing, logging, etc. The goal is to provide intelligent use of edge computing resources. Workload isolation includes isolating internal edge workloads from unauthorized exposure, use, or discovery, ensuring that workloads can only be accessed by authorized entities or processes. Secure edge steering includes access from User Equipment (UE) to edge hosted workloads through dynamic, unique, and encrypted tunnels on a per-app, per-session basis, not reliant on underlying mobile network transports (e.g., S1/SGi/F1/IP), 5G network slicing, or TCP/IP networks for security. This will provide protection whether the Radio Access Network (RAN) architecture secures traffic between UE and edge workloads or not. It will also work regardless of the location of the edge or the latency range (e.g., deep edge with <1 ms latency, far edge with <5 ms latency, near edge with 5-20 ms latency, or any other edge and latency combination). It will also work if the transport method is WiFi, wired, fibre, satellite or any transport method other than a radio network.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure is illustrated and described herein with reference to the various drawings, in which like reference numbers are used to denote like system components/method steps, as appropriate, and in which:

FIG. 1 is a network diagram of a cloud-based system offering security as a service.

FIG. 2 is a network diagram of an example implementation of the cloud-based system.

FIG. 3 is a block diagram of a server that may be used in the cloud-based system of FIGS. 1 and 2 or the like.

FIG. 4 is a block diagram of a user device that may be used with the cloud-based system of FIGS. 1 and 2 or the like.

FIG. 5 is a network diagram of the cloud-based system illustrating an application on user devices with users configured to operate through the cloud-based system.

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FIG. 6 is a network diagram of a Zero Trust Network Access (ZTNA) application utilizing the cloud-based system of FIGS. 1 and 2.

FIG. 7 is a network diagram of the cloud-based system of FIGS. 1 and 2 in an application of digital experience monitoring.

FIG. 8 is a network diagram of the cloud-based system of FIGS. 1 and 2 with various cloud tunnels, labeled as cloud tunnels, for forwarding traffic.

FIGS. 9 and 10 are flow diagrams of a cloud tunnel illustrating a control channel (FIG. 9) and a data channel (FIG. 10), with the tunnel illustrated between a client and a server.

FIG. 11 is a diagram illustrating various techniques to forward traffic to the cloud-based system.

FIG. 12 is a diagram of a ESIM/iSIM/SIM-card-based approach for network path and connectivity to the cloud-based system.

FIG. 13 is a diagram of an Application-aware Networking (APN) approach with a mobile network for network path and connectivity to the cloud-based system.

FIG. 14 is a network diagram of an APN network for traffic forwarding of IoT devices to the cloud-based system, such as for secure access to the Internet, to cloud services, etc.

FIG. 15 is a flow diagram of communication in the APN network.

FIG. 16 is a network diagram of a network for traffic forwarding of any device having an ESIM/iSIM/SIM-card, embedded SIM (eSIM), integrated SIM (iSIM) or other SIM functions to the cloud-based system, such as for secure access to the Internet, to cloud services.

FIG. 17 is a flow diagram of communication in the ESIM/iSIM/SIM-card (or eSIM or iSIM) forwarding approach.

FIG. 18 is a flowchart of a process for forwarding traffic to the cloud-based system using an ESIM/iSIM/SIM-card (or eSIM or iSIM).

FIG. 19 is a block diagram illustrating functions of the cloud-based system.

FIG. 20 is a block diagram of a MEC located at a base station of a service provider's network.

FIG. 21 is a block diagram of the MEC and the conventional approach with the cloud-based system.

FIG. 22 is a block diagram of the MEC with a cloud connector located in the MEC edge services.

FIG. 23 is a block diagram of the MEC with nodes located in the MEC hosted service layer.

FIG. 24 is a block diagram of the MEC illustrating a traditional security service from a service provider.

FIG. 25 is a block diagram of the MEC illustrating integration of cloud-security services in the MEC.

FIG. 26 is a network diagram of common edge and sizes encountered when accessing workloads from UE.

FIG. 27 is a diagram of multi-workload access using the cloud-based system providing edge and cloud solutions.

FIG. 28 is a flow diagram of selective edge flow between the UE and a workload with the cloud-based system in-between for selective dynamic steering.

FIG. 29 is a diagram of intra edge connectivity between various services, over a shared transport network.

FIG. 30 is a network diagram of a MEC illustrating cloud service-based edge isolation.

FIG. 31 is a diagram of UEs connected to edge hosted services without intelligent steering.

FIG. 32 is a diagram of possible edge compute components.

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FIG. 33 is a diagram of cloud edge hosted steering deployment for any type of wireless or other means of transport between UE and the edge compute.

FIG. 34 is a diagram of an endpoint device connected to Edge Hosted Services over a traditional network path with service endpoints exposed and reachable by arbitrary endpoints.

FIG. 35 is a flow diagram showing the edge hosted steering allowing endpoint access only to authorized workloads.

FIG. 36 is a diagram outlining high-level category definitions.

FIG. 37 is a flow diagram of multi workload access using the service to steer to the correct enforcement point.

FIG. 38 is a flow diagram showing the selective edge flow of the present disclosure.

FIG. 39 is a flow diagram showing identity values and stores in the 5G landscape.

FIG. 40 is a flow diagram of the edge process to query 5G UDM through the App Function.

FIG. 41 is a flow diagram of the proposed service flow including the Application Function.

FIG. 42 is a flow diagram of the process to query IDP broker in a 5G landscape.

DETAILED DESCRIPTION OF THE DISCLOSURE

Again, the present disclosure relates to systems and methods for cloud-based 5G security network architectures. Specifically, various approaches are described to integrate cloud-based security services in Multiaccess Edge Compute systems (MECs). That is, existing cloud-based security services are in line between a UE and the Internet. The present disclosure includes integrating the cloud-based security services and associated cloud-based system within service provider's MECs. In this manner, a cloud-based security service can be integrated with a service provider's 5G network. For example, nodes in a cloud-based system can be collocated within a service provider's network, to provide security functions to 5G users.

Of note, 5G is emerging amidst a general decline in service provider revenue. There is a need for service providers to provide additional services for new revenue sources, as opposed to being a simple pipe for network connectivity. Integrating existing cloud-based security services in the service provider's MECs provides an opportunity to offer additional, value-added services.

The present disclosure focuses on traffic forwarding techniques that either will not require a user ID, a forwarding gateway, etc., or conversely leverage an intelligent service to direct, control and protect traffic from a UE device, going to the Internet. In an embodiment, the present disclosure includes use of a SIM card, and/or a functionally equivalent digital embedded SIM (eSIM) or integrated SIM (iSIM) that executes code thereon for implementing a tunnel to a cloud-based system. In another embodiment, a secure edge service of a network is configured for forwarding traffic to the cloud-based system. Advantageously, these approaches are ideal for IoT and similar devices.

Intelligent steering includes actively identifying traffic types from User Equipment (UE—a reference to any wireless device equipped with a SIM/eSIM/iSIM/other attached to a telecommunications network) and steering this traffic to the most effective edge for processing, securing, logging, etc. The goal is to provide intelligent use of edge computing resources. Workload isolation includes isolating internal

edge workloads from unauthorized exposure, use, or discovery, ensuring that workloads can only be accessed by authorized entities or processes. Secure edge steering includes access from User Equipment (UE) to edge hosted workloads through dynamic, unique, and encrypted tunnels on a per-app, per-session basis, not reliant on underlying mobile network transports (e.g., S1/SGi/F1/IP), 5G network slicing, or TCP/IP networks for security. This will provide protection whether the Radio Access Network (RAN) architecture secures traffic between UE and edge workloads or not. It will also work regardless of the location of the edge or the latency range (e.g., deep edge with <1 ms latency, far edge with <5 ms latency, near edge with 5-20 ms latency, or any other edge and latency combination). As described herein, a workload can be an application or any online resources being requested by a UE 300.

§ 1.0 EXAMPLE CLOUD-BASED SYSTEM ARCHITECTURE

FIG. 1 is a network diagram of a cloud-based system 100 offering security as a service. Specifically, the cloud-based system 100 can offer a Secure Internet and Web Gateway as a service to various users 102, as well as other cloud services. In this manner, the cloud-based system 100 is located between the users 102 and the Internet as well as any cloud services 106 (or applications) accessed by the users 102. As such, the cloud-based system 100 provides inline monitoring inspecting traffic between the users 102, the Internet 104, and the cloud services 106, including Secure Sockets Layer (SSL) traffic. The cloud-based system 100 can offer access control, threat prevention, data protection, etc. The access control can include a cloud-based firewall, cloud-based intrusion detection, Uniform Resource Locator (URL) filtering, bandwidth control, Domain Name System (DNS) filtering, etc. Threat prevention can include cloud-based intrusion prevention, protection against advanced threats (malware, spam, Cross-Site Scripting (XSS), phishing, etc.), cloud-based sandbox, antivirus, DNS security, etc. The data protection can include Data Loss Prevention (DLP), cloud application security such as via a Cloud Access Security Broker (CASB), file type control, etc.

The cloud-based firewall can provide Deep Packet Inspection (DPI) and access controls across various ports and protocols as well as being application and user aware. The URL filtering can block, allow, or limit website access based on policy for a user, group of users, or entire organization, including specific destinations or categories of URLs (e.g., gambling, social media, etc.). The bandwidth control can enforce bandwidth policies and prioritize critical applications such as relative to recreational traffic. DNS filtering can control and block DNS requests against known and malicious destinations. DNS filtering can also provide visibility into resources accessed by the 5G infrastructure, radio networks, and UE.

The cloud-based intrusion prevention and advanced threat protection can deliver full threat protection against malicious content such as browser exploits, scripts, identified botnets and malware callbacks, etc. The cloud-based sandbox can block zero-day exploits (just identified) by analyzing unknown files for malicious behavior. Advantageously, the cloud-based system 100 is multi-tenant and can service a large volume of the users 102. As such, newly discovered threats can be promulgated throughout the cloud-based system 100 for all tenants practically instantaneously. The antivirus protection can include antivirus, antispayware, anti-malware, etc. protection for the users 102, using signatures

sourced and constantly updated. The DNS security can identify and route command-and-control connections to threat detection engines for full content inspection.

The DLP can use standard and/or custom dictionaries to continuously monitor the users 102, including compressed and/or SSL-encrypted traffic. Again, being in a cloud implementation, the cloud-based system 100 can scale this monitoring with near-zero latency on the users 102. The cloud application security can include CASB functionality to discover and control user access to known and unknown cloud services 106. The file type controls enable true file type control by the user, location, destination, etc. to determine which files are allowed or not.

The cloud-based system 100 can provide other security functions, including, for example, micro-segmentation, workload segmentation, API security, Cloud Security Posture Management (CSPM), user identity management, and the like. That is, the cloud-based system 100 provides a network architecture that enables delivery of any cloud-based security service, including emerging frameworks.

For illustration purposes, the users 102 of the cloud-based system 100 can include a mobile device 110, a headquarters (HQ) 112 which can include or connect to a data center (DC) 114, Internet of Things (IoT) devices 116, a branch office/remote location 118, etc., and each includes one or more user devices (an example user device 300 (User Equipment (UE)) is illustrated in FIG. 5). The devices 110, 116, and the locations 112, 114, 118 are shown for illustrative purposes, and those skilled in the art will recognize there are various access scenarios and other users 102 for the cloud-based system 100, all of which are contemplated herein. The users 102 can be associated with a tenant, which may include an enterprise, a corporation, an organization, etc. That is, a tenant is a group of users who share a common access with specific privileges to the cloud-based system 100, a cloud service, etc. In an embodiment, the headquarters 112 can include an enterprise's network with resources in the data center 114. The mobile device 110 can be a so-called road warrior, i.e., users that are off-site, on-the-road, etc. Those skilled in the art will recognize a user 102 has to use a corresponding user device 300 for accessing the cloud-based system 100 and the like, and the description herein may use the user 102 and/or the user device 300 interchangeably.

Further, the cloud-based system 100 can be multi-tenant, with each tenant having its own users 102 and configuration, policy, rules, etc. One advantage of the multi-tenancy and a large volume of users is the zero-day/zero-hour protection in that a new vulnerability can be detected and then instantly remediated across the entire cloud-based system 100. The same applies to policy, rule, configuration, etc. changes—they are instantly remediated across the entire cloud-based system 100. As well, new features in the cloud-based system 100 can also be rolled up simultaneously across the user base, as opposed to selective and time-consuming upgrades on every device at the locations 112, 114, 118, and the devices 110, 116.

Logically, the cloud-based system 100 can be viewed as an overlay network between users (at the locations 112, 114, 118, and the devices 110, 116) and the Internet 104 and the cloud services 106. Previously, the IT deployment model included enterprise resources and applications stored within the data center 114 (i.e., physical devices) behind a firewall (perimeter), accessible by employees, partners, contractors, etc. on-site or remote via Virtual Private Networks (VPNs), etc. The cloud-based system 100 is replacing the conventional deployment model. The cloud-based system 100 can be used to implement these services in the cloud without

requiring the physical devices and management thereof by enterprise IT administrators. As an ever-present overlay network, the cloud-based system **100** can provide the same functions as the physical devices and/or appliances regardless of geography or location of the users **102**, as well as independent of platform, operating system, network access technique, network access provider, etc.

There are various techniques to forward traffic between the users **102** at the locations **112**, **114**, **118**, and via the devices **110**, **116**, and the cloud-based system **100**. Typically, the locations **112**, **114**, **118** can use tunneling where all traffic is forward through the cloud-based system **100**. For example, various tunneling protocols are contemplated, such as GRE, L2TP, IPsec, customized tunneling protocols, etc. The devices **110**, **116**, when not at one of the locations **112**, **114**, **118** can use a local application that forwards traffic, a proxy such as via a Proxy Auto-Config (PAC) file, and the like. An application of the local application is the application **350** described in detail herein as a connector application. A key aspect of the cloud-based system **100** is all traffic between the users **102** and the Internet **104** or the cloud services **106** is via the cloud-based system **100**. As such, the cloud-based system **100** has visibility to enable various functions, all of which are performed off the user device in the cloud.

The cloud-based system **100** can also include a management system **120** for tenant access to provide global policy and configuration as well as real-time analytics. This enables IT administrators to have a unified view of user activity, threat intelligence, application usage, etc. For example, IT administrators can drill-down to a per-user level to understand events and correlate threats, to identify compromised devices, to have application visibility, and the like. The cloud-based system **100** can further include connectivity to an Identity Provider (IDP) **122** for authentication of the users **102** and to a Security Information and Event Management (SIEM) system **124** for event logging. The system **124** can provide alert and activity logs on a per-user **102** basis.

FIG. 2 is a network diagram of an example implementation of the cloud-based system **100**. In an embodiment, the cloud-based system **100** includes a plurality of enforcement nodes (EN) **150**, labeled as enforcement nodes **150-1**, **150-2**, **150-N**, interconnected to one another and interconnected to a central authority (CA) **152**. Note, the nodes **150** are called “enforcement” nodes **150** but they can be simply referred to as nodes **150** in the cloud-based system **100**. Also, the nodes **150** can be referred to as service edges. The nodes **150** and the central authority **152**, while described as nodes, can include one or more servers, including physical servers, virtual machines (VM) executed on physical hardware, etc. An example of a server is illustrated in FIG. 4. The cloud-based system **100** further includes a log router **154** that connects to a storage cluster **156** for supporting log maintenance from the enforcement nodes **150**. The central authority **152** provide centralized policy, real-time threat updates, etc. and coordinates the distribution of this data between the enforcement nodes **150**. The enforcement nodes **150** provide an onramp to the users **102** and are configured to execute policy, based on the central authority **152**, for each user **102**. The enforcement nodes **150** can be geographically distributed, and the policy for each user **102** follows that user **102** as he or she connects to the nearest (or other criteria) enforcement node **150**. Of note, the cloud-based system is an external system meaning it is separate from the tenant’s private networks (enterprise networks) as well as from networks associated with the devices **110**, **116**, and locations **112**, **118**.

The enforcement nodes **150** are full-featured secure internet gateways that provide integrated internet security. They inspect all web traffic bi-directionally for malware and enforce security, compliance, and firewall policies, as described herein, as well as various additional functionality. In an embodiment, each enforcement node **150** has two main modules for inspecting traffic and applying policies: a web module and a firewall module. The enforcement nodes **150** are deployed around the world and can handle hundreds of thousands of concurrent users with millions of concurrent sessions. Because of this, regardless of where the users **102** are, they can access the Internet **104** from any device, and the enforcement nodes **150** protect the traffic and apply corporate policies. The enforcement nodes **150** can implement various inspection engines therein, and optionally, send sandboxing to another system. The enforcement nodes **150** include significant fault tolerance capabilities, such as deployment in active-active mode to ensure availability and redundancy as well as continuous monitoring.

In an embodiment, customer traffic is not passed to any other component within the cloud-based system **100**, and the enforcement nodes **150** can be configured never to store any data to disk. Packet data is held in memory for inspection and then, based on policy, is either forwarded or dropped. Log data generated for every transaction is compressed, tokenized, and exported over secure Transport Layer Security (TLS) connections to the log routers **154** that direct the logs to the storage cluster **156**, hosted in the appropriate geographical region, for each organization. In an embodiment, all data destined for or received from the Internet is processed through one of the enforcement nodes **150**. In another embodiment, specific data specified by each tenant, e.g., only email, only executable files, etc., is processed through one of the enforcement nodes **150**.

Each of the enforcement nodes **150** may generate a decision vector $D=[d_1, d_2, \dots, d_n]$ for a content item of one or more parts $C=[c_1, c_2, \dots, c_m]$. Each decision vector may identify a threat classification, e.g., clean, spyware, malware, undesirable content, innocuous, spam email, unknown, etc. For example, the output of each element of the decision vector D may be based on the output of one or more data inspection engines. In an embodiment, the threat classification may be reduced to a subset of categories, e.g., violating, non-violating, neutral, unknown. Based on the subset classification, the enforcement node **150** may allow the distribution of the content item, preclude distribution of the content item, allow distribution of the content item after a cleaning process, or perform threat detection on the content item. In an embodiment, the actions taken by one of the enforcement nodes **150** may be determinative on the threat classification of the content item and on a security policy of the tenant to which the content item is being sent from or from which the content item is being requested by. A content item is violating if, for any part $C=[c_1, c_2, \dots, c_m]$ of the content item, at any of the enforcement nodes **150**, any one of the data inspection engines generates an output that results in a classification of “violating.”

The central authority **152** hosts all customer (tenant) policy and configuration settings. It monitors the cloud and provides a central location for software and database updates and threat intelligence. Given the multi-tenant architecture, the central authority **152** is redundant and backed up in multiple different data centers. The enforcement nodes **150** establish persistent connections to the central authority **152** to download all policy configurations. When a new user connects to an enforcement node **150**, a policy request is sent to the central authority **152** through this connection. The

central authority **152** then calculates the policies that apply to that user **102** and sends the policy to the enforcement node **150** as a highly compressed bitmap.

The policy can be tenant-specific and can include access privileges for users, websites and/or content that is disallowed, restricted domains, DLP dictionaries, etc. Once downloaded, a tenant's policy is cached until a policy change is made in the management system **120**. The policy can be tenant-specific and can include access privileges for users, websites and/or content that is disallowed, restricted domains, DLP dictionaries, etc. When this happens, all of the cached policies are purged, and the enforcement nodes **150** request the new policy when the user **102** next makes a request. In an embodiment, the enforcement nodes **150** exchange "heartbeats" periodically, so all enforcement nodes **150** are informed when there is a policy change. Any enforcement node **150** can then pull the change in policy when it sees a new request.

The cloud-based system **100** can be a private cloud, a public cloud, a combination of a private cloud and a public cloud (hybrid cloud), or the like. Cloud computing systems and methods abstract away physical servers, storage, networking, etc., and instead offer these as on-demand and elastic resources. The National Institute of Standards and Technology (NIST) provides a concise and specific definition which states cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. Cloud computing differs from the classic client-server model by providing applications from a server that are executed and managed by a client's web browser or the like, with no installed client version of an application required. Centralization gives cloud service providers complete control over the versions of the browser-based and other applications provided to clients, which removes the need for version upgrades or license management on individual client computing devices. The phrase "Software as a Service" (SaaS) is sometimes used to describe application programs offered through cloud computing. A common shorthand for a provided cloud computing service (or even an aggregation of all existing cloud services) is "the cloud." The cloud-based system **100** is illustrated herein as an example embodiment of a cloud-based system, and other implementations are also contemplated.

As described herein, the terms cloud services and cloud applications may be used interchangeably. The cloud service **106** is any service made available to users on-demand via the Internet, as opposed to being provided from a company's on-premises servers. A cloud application, or cloud app, is a software program where cloud-based and local components work together. The cloud-based system **100** can be utilized to provide example cloud services, including Zscaler Internet Access (ZIA), Zscaler Private Access (ZPA), and Zscaler Digital Experience (ZDX), all from Zscaler, Inc. (the assignee and applicant of the present application). Also, there can be multiple different cloud-based systems **100**, including ones with different architectures and multiple cloud services. The ZIA service can provide the access control, threat prevention, and data protection described above with reference to the cloud-based system **100**. ZPA can include access control, microservice segmentation, etc. The ZDX service can provide monitoring of user experience, e.g., Quality of Experience (QoE), Quality of Service (QoS), etc., in a manner that can gain insights based on continuous,

inline monitoring. For example, the ZIA service can provide a user with Internet Access, and the ZPA service can provide a user with access to enterprise resources instead of traditional Virtual Private Networks (VPNs), namely ZPA provides Zero Trust Network Access (ZTNA). Those of ordinary skill in the art will recognize various other types of cloud services **106** are also contemplated. Also, other types of cloud architectures are also contemplated, with the cloud-based system **100** presented for illustration purposes.

§ 1.1 Private Nodes Hosted by Tenants or Service Providers

The nodes **150** that service multi-tenant users **102** may be located in data centers. These nodes **150** can be referred to as public nodes **150** or public service edges. In embodiment, the nodes **150** can be located on-premises with tenants (enterprise) as well as service providers. These nodes can be referred to as private nodes **150** or private service edges. In operation, these private nodes **150** can perform the same functions as the public nodes **150**, can communicate with the central authority **152**, and the like. In fact, the private nodes **150** can be considered in the same cloud-based system **100** as the public nodes **150**, except located on-premises. When a private node **150** is located in an enterprise network, the private node **150** can have a single tenant corresponding to the enterprise; of course, the cloud-based system **100** is still multi-tenant, but these particular nodes are serving only a single tenant. When a private node **150** is located in a service provider's network, the private node **150** can be multi-tenant for customers of the service provider. Those skilled in the art will recognize various architectural approaches are contemplated. The cloud-based system **100** is a logical construct providing a security service.

§ 2.0 USER DEVICE APPLICATION FOR TRAFFIC FORWARDING AND MONITORING

FIG. 3 is a network diagram of the cloud-based system **100** illustrating an application **350** on user devices **300** with users **102** configured to operate through the cloud-based system **100**. Different types of user devices **300** are proliferating, including Bring Your Own Device (BYOD) as well as IT-managed devices. The conventional approach for a user device **300** to operate with the cloud-based system **100** as well as for accessing enterprise resources includes complex policies, VPNs, poor user experience, etc. The application **350** can automatically forward user traffic with the cloud-based system **100** as well as ensuring that security and access policies are enforced, regardless of device, location, operating system, or application. The application **350** automatically determines if a user **102** is looking to access the open Internet **104**, a SaaS app, or an internal app running in public, private, or the datacenter and routes mobile traffic through the cloud-based system **100**. The application **350** can support various cloud services, including ZIA, ZPA, ZDX, etc., allowing the best-in-class security with zero trust access to internal apps. As described herein, the application **350** can also be referred to as a connector application.

The application **350** is configured to auto-route traffic for seamless user experience. This can be protocol as well as application-specific, and the application **350** can route traffic with a nearest or best fit enforcement node **150**. Further, the application **350** can detect trusted networks, allowed applications, etc. and support secure network access. The application **350** can also support the enrollment of the user device **300** prior to accessing applications. The application **350** can uniquely detect the users **102** based on fingerprinting the user device **300**, using criteria like device model, platform, operating system, etc. The application **350** can support

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Mobile Device Management (MDM) functions, allowing IT personnel to deploy and manage the user devices **300** seamlessly. This can also include the automatic installation of client and SSL certificates during enrollment. Finally, the application **350** provides visibility into device and app usage of the user **102** of the user device **300**.

The application **350** supports a secure, lightweight tunnel between the user device **300** and the cloud-based system **100**. For example, the lightweight tunnel can be HTTP-based. With the application **350**, there is no requirement for PAC files, an IPsec VPN, authentication cookies, or user **102** setup.

§ 3.0 EXAMPLE SERVER ARCHITECTURE

FIG. **4** is a block diagram of a server **200**, which may be used in the cloud-based system **100**, in other systems, or standalone. For example, the enforcement nodes **150** and the central authority **152** may be formed as one or more of the servers **200**. The server **200** may be a digital computer that, in terms of hardware architecture, generally includes a processor **202**, input/output (I/O) interfaces **204**, a network interface **206**, a data store **208**, and memory **210**. It should be appreciated by those of ordinary skill in the art that FIG. **4** depicts the server **200** in an oversimplified manner, and a practical embodiment may include additional components and suitably configured processing logic to support known or conventional operating features that are not described in detail herein. The components (**202**, **204**, **206**, **208**, and **210**) are communicatively coupled via a local interface **212**. The local interface **212** may be, for example, but not limited to, one or more buses or other wired or wireless connections, as is known in the art. The local interface **212** may have additional elements, which are omitted for simplicity, such as controllers, buffers (caches), drivers, repeaters, and receivers, among many others, to enable communications. Further, the local interface **212** may include address, control, and/or data connections to enable appropriate communications among the aforementioned components.

The processor **202** is a hardware device for executing software instructions. The processor **202** may be any custom made or commercially available processor, a Central Processing Unit (CPU), an auxiliary processor among several processors associated with the server **200**, a semiconductor-based microprocessor (in the form of a microchip or chip-set), or generally any device for executing software instructions. When the server **200** is in operation, the processor **202** is configured to execute software stored within the memory **210**, to communicate data to and from the memory **210**, and to generally control operations of the server **200** pursuant to the software instructions. The I/O interfaces **204** may be used to receive user input from and/or for providing system output to one or more devices or components.

The network interface **206** may be used to enable the server **200** to communicate on a network, such as the Internet **104**. The network interface **206** may include, for example, an Ethernet card or adapter or a Wireless Local Area Network (WLAN) card or adapter. The network interface **206** may include address, control, and/or data connections to enable appropriate communications on the network. A data store **208** may be used to store data. The data store **208** may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, and the like)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, and the like), and combinations thereof.

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Moreover, the data store **208** may incorporate electronic, magnetic, optical, and/or other types of storage media. In one example, the data store **208** may be located internal to the server **200**, such as, for example, an internal hard drive connected to the local interface **212** in the server **200**. Additionally, in another embodiment, the data store **208** may be located external to the server **200** such as, for example, an external hard drive connected to the I/O interfaces **204** (e.g., SCSI or USB connection). In a further embodiment, the data store **208** may be connected to the server **200** through a network, such as, for example, a network-attached file server.

The memory **210** may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, etc.)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, etc.), and combinations thereof. Moreover, the memory **210** may incorporate electronic, magnetic, optical, and/or other types of storage media. Note that the memory **210** may have a distributed architecture, where various components are situated remotely from one another but can be accessed by the processor **202**. The software in memory **210** may include one or more software programs, each of which includes an ordered listing of executable instructions for implementing logical functions. The software in the memory **210** includes a suitable Operating System (O/S) **214** and one or more programs **216**. The operating system **214** essentially controls the execution of other computer programs, such as the one or more programs **216**, and provides scheduling, input-output control, file and data management, memory management, and communication control and related services. The one or more programs **216** may be configured to implement the various processes, algorithms, methods, techniques, etc. described herein.

§ 4.0 EXAMPLE USER DEVICE ARCHITECTURE

FIG. **5** is a block diagram of a user device **300**, which may be used with the cloud-based system **100** or the like. Specifically, the user device **300** can form a device used by one of the users **102**, and this may include common devices such as laptops, smartphones, tablets, netbooks, personal digital assistants, MP3 players, cell phones, e-book readers, IoT devices, servers, desktops, printers, televisions, streaming media devices, and the like. The user device **300** can be a digital device that, in terms of hardware architecture, generally includes a processor **302**, I/O interfaces **304**, a network interface **306**, a data store **308**, and memory **310**. It should be appreciated by those of ordinary skill in the art that FIG. **5** depicts the user device **300** in an oversimplified manner, and a practical embodiment may include additional components and suitably configured processing logic to support known or conventional operating features that are not described in detail herein. The components (**302**, **304**, **306**, **308**, and **302**) are communicatively coupled via a local interface **312**. The local interface **312** can be, for example, but not limited to, one or more buses or other wired or wireless connections, as is known in the art. The local interface **312** can have additional elements, which are omitted for simplicity, such as controllers, buffers (caches), drivers, repeaters, and receivers, among many others, to enable communications. Further, the local interface **312** may include address, control, and/or data connections to enable appropriate communications among the aforementioned components.

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The processor 302 is a hardware device for executing software instructions. The processor 302 can be any custom made or commercially available processor, a CPU, an auxiliary processor among several processors associated with the user device 300, a semiconductor-based microprocessor (in the form of a microchip or chipset), or generally any device for executing software instructions. When the user device 300 is in operation, the processor 302 is configured to execute software stored within the memory 310, to communicate data to and from the memory 310, and to generally control operations of the user device 300 pursuant to the software instructions. In an embodiment, the processor 302 may include a mobile optimized processor such as optimized for power consumption and mobile applications. The I/O interfaces 304 can be used to receive user input from and/or for providing system output. User input can be provided via, for example, a keypad, a touch screen, a scroll ball, a scroll bar, buttons, a barcode scanner, and the like. System output can be provided via a display device such as a Liquid Crystal Display (LCD), touch screen, and the like.

The network interface 306 enables wireless communication to an external access device or network. Any number of suitable wireless data communication protocols, techniques, or methodologies can be supported by the network interface 306, including any protocols for wireless communication. The data store 308 may be used to store data. The data store 308 may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, and the like)), nonvolatile memory elements (e.g., ROM, hard drive, tape, CDROM, and the like), and combinations thereof. Moreover, the data store 308 may incorporate electronic, magnetic, optical, and/or other types of storage media.

The memory 310 may include any of volatile memory elements (e.g., random access memory (RAM, such as DRAM, SRAM, SDRAM, etc.)), nonvolatile memory elements (e.g., ROM, hard drive, etc.), and combinations thereof. Moreover, the memory 310 may incorporate electronic, magnetic, optical, and/or other types of storage media. Note that the memory 310 may have a distributed architecture, where various components are situated remotely from one another but can be accessed by the processor 302. The software in memory 310 can include one or more software programs, each of which includes an ordered listing of executable instructions for implementing logical functions. In the example of FIG. 3, the software in the memory 310 includes a suitable operating system 314 and programs 316. The operating system 314 essentially controls the execution of other computer programs and provides scheduling, input-output control, file and data management, memory management, and communication control and related services. The programs 316 may include various applications, add-ons, etc. configured to provide end user functionality with the user device 300. For example, example programs 316 may include, but not limited to, a web browser, social networking applications, streaming media applications, games, mapping and location applications, electronic mail applications, financial applications, and the like. In a typical example, the end-user typically uses one or more of the programs 316 along with a network such as the cloud-based system 100.

§ 5.0 ZERO TRUST NETWORK ACCESS USING THE CLOUD-BASED SYSTEM

FIG. 6 is a network diagram of a Zero Trust Network Access (ZTNA) application utilizing the cloud-based system

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100. For ZTNA, the cloud-based system 100 can dynamically create a connection through a secure tunnel between an endpoint (e.g., users 102A, 102B) that are remote and an on-premises connector 400 that is either located in cloud file shares and applications 402 and/or in an enterprise network 410 that includes enterprise file shares and applications 404. The connection between the cloud-based system 100 and on-premises connector 400 is dynamic, on-demand, and orchestrated by the cloud-based system 100. A key feature is its security at the edge—there is no need to punch any holes in the existing on-premises firewall. The connector 400 inside the enterprise (on-premises) “dials out” and connects to the cloud-based system 100 as if too were an endpoint. This on-demand dial-out capability and tunneling authenticated traffic back to the enterprise is a key differentiator for ZTNA. Also, this functionality can be implemented in part by the application 350 on the user device 300. Also, the applications 402, 404 can include B2B applications. Note, the difference between the applications 402, 404 is the applications 402 are hosted in the cloud, whereas the applications 404 are hosted on the enterprise network 410. The B2B service described herein contemplates use with either or both of the applications 402, 404.

The paradigm of virtual private access systems and methods is to give users network access to get to an application and/or file share, not to the entire network. If a user is not authorized to get the application, the user should not be able even to see that it exists, much less access it. The virtual private access systems and methods provide an approach to deliver secure access by decoupling applications 402, 404 from the network, instead of providing access with a connector 400, in front of the applications 402, 404, an application on the user device 300, a central authority 152 to push policy, and the cloud-based system 100 to stitch the applications 402, 404 and the software connectors 400 together, on a per-user, per-application basis.

With the virtual private access, users can only see the specific applications 402, 404 allowed by the central authority 152. Everything else is “invisible” or “dark” to them. Because the virtual private access separates the application from the network, the physical location of the application 402, 404 becomes irrelevant—if applications 402, 404 are located in more than one place, the user is automatically directed to the instance that will give them the best performance. The virtual private access also dramatically reduces configuration complexity, such as policies/firewalls in the data centers. Enterprises can, for example, move applications to Amazon Web Services or Microsoft Azure, and take advantage of the elasticity of the cloud, making private, internal applications behave just like the marketing leading enterprise applications. Advantageously, there is no hardware to buy or deploy because the virtual private access is a service offering to end-users and enterprises.

§ 6.0 DIGITAL EXPERIENCE MONITORING

FIG. 7 is a network diagram of the cloud-based system 100 in an application of digital experience monitoring. Here, the cloud-based system 100 providing security as a service as well as ZTNA, can also be used to provide real-time, continuous digital experience monitoring, as opposed to conventional approaches (synthetic probes). A key aspect of the architecture of the cloud-based system 100 is the inline monitoring. This means data is accessible in real-time for individual users from end-to-end. As described herein, digital experience monitoring can include monitoring, analyzing, and improving the digital user experience.

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The cloud-based system **100** connects users **102** at the locations **110**, **112**, **118** to the applications **402**, **404**, the Internet **104**, the cloud services **106**, etc. The inline, end-to-end visibility of all users enables digital experience monitoring. The cloud-based system **100** can monitor, diagnose, generate alerts, and perform remedial actions with respect to network endpoints, network components, network links, etc. The network endpoints can include servers, virtual machines, containers, storage systems, or anything with an IP address, including the Internet of Things (IoT), cloud, and wireless endpoints. With these components, these network endpoints can be monitored directly in combination with a network perspective. Thus, the cloud-based system **100** provides a unique architecture that can enable digital experience monitoring, network application monitoring, infrastructure component interactions, etc. Of note, these various monitoring aspects require no additional components—the cloud-based system **100** leverages the existing infrastructure to provide this service.

Again, digital experience monitoring includes the capture of data about how end-to-end application availability, latency, and quality appear to the end user from a network perspective. This is limited to the network traffic visibility and not within components, such as what application performance monitoring can accomplish. Networked application monitoring provides the speed and overall quality of networked application delivery to the user in support of key business activities. Infrastructure component interactions include a focus on infrastructure components as they interact via the network, as well as the network delivery of services or applications. This includes the ability to provide network path analytics.

The cloud-based system **100** can enable real-time performance and behaviors for troubleshooting in the current state of the environment, historical performance and behaviors to understand what occurred or what is trending over time, predictive behaviors by leveraging analytics technologies to distill and create actionable items from the large dataset collected across the various data sources, and the like. The cloud-based system **100** includes the ability to directly ingest any of the following data sources network device-generated health data, network device-generated traffic data, including flow-based data sources inclusive of NetFlow and IPFIX, raw network packet analysis to identify application types and performance characteristics, HTTP request metrics, etc. The cloud-based system **100** can operate at 10 gigabits (10G) Ethernet and higher at full line rate and support a rate of 100,000 or more flows per second or higher.

The applications **402**, **404** can include enterprise applications, Office **365**, Salesforce, Skype, Google apps, internal applications, etc. These are critical business applications where user experience is important. The objective here is to collect various data points so that user experience can be quantified for a particular user, at a particular time, for purposes of analyzing the experience as well as improving the experience. In an embodiment, the monitored data can be from different categories, including application-related, network-related, device-related (also can be referred to as endpoint-related), protocol-related, etc. Data can be collected at the application **350** or the cloud edge to quantify user experience for specific applications, i.e., the application-related and device-related data. The cloud-based system **100** can further collect the network-related and the protocol-related data (e.g., Domain Name System (DNS) response time).

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Application-Related Data

Page Load Time	Redirect count (#)
Page Response Time	Throughput (bps)
Document Object Model (DOM) Load Time	Total size (bytes)
Total Downloaded bytes	Page error count (#)
App availability (%)	Page element count by category (#)

Network-Related Data

HTTP Request metrics	Bandwidth
Server response time	Jitter
Ping packet loss (%)	Trace Route
Ping round trip	DNS lookup trace
Packet loss (%)	GRE/IPSec tunnel monitoring
Latency	MTU and bandwidth measurements

Device-Related Data (Endpoint-Related Data)

System details	Network (config)
Central Processing Unit (CPU)	Disk
Memory (RAM)	Processes
Network (interfaces)	Applications

Metrics could be combined. For example, device health can be based on a combination of CPU, memory, etc. Network health could be a combination of Wi-Fi/LAN connection health, latency, etc. Application health could be a combination of response time, page loads, etc. The cloud-based system **100** can generate service health as a combination of CPU, memory, and the load time of the service while processing a user's request. The network health could be based on the number of network path(s), latency, packet loss, etc.

The lightweight connector **400** can also generate similar metrics for the applications **402**, **404**. In an embodiment, the metrics can be collected while a user is accessing specific applications that user experience is desired for monitoring. In another embodiment, the metrics can be enriched by triggering synthetic measurements in the context of an in-line transaction by the application **350** or cloud edge. The metrics can be tagged with metadata (user, time, app, etc.) and sent to a logging and analytics service for aggregation, analysis, and reporting. Further, network administrators can get UEX reports from the cloud-based system **100**. Due to the inline nature and the fact the cloud-based system **100** is an overlay (in-between users and services/applications), the cloud-based system **100** enables the ability to capture user experience metric data continuously and to log such data historically. As such, a network administrator can have a long-term detailed view of the network and associated user experience.

§ 7.0 CLOUD TUNNEL

FIG. **8** is a network diagram of the cloud-based system **100** with various cloud tunnels **500**, labeled as cloud tunnels **500A**, **500B**, **500C**, for forwarding traffic. FIGS. **9** and **10** are flow diagrams of a cloud tunnel **500** illustrating a control channel (FIG. **9**) and a data channel (FIG. **10**), with the tunnel illustrated between a client **510** and a server **520**. The cloud tunnel **500** is a lightweight tunnel that is configured to forward traffic between the client **510** and the server **520**. The present disclosure focuses on the specific mechanisms used in the cloud tunnel **500** between two points, namely the

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client **510** and the server **520**. Those skilled in the art will recognize the cloud tunnel **500** can be used with the cloud-based system **100** as an example use case, and other uses are contemplated. That is, the client **510** and the server **520** are just endpoint devices that support the exchange of data traffic and control traffic for tunnel **500**. For description, the server **520** can be referred to as a local node and the client **510** as a remote node, where the tunnel operates between the local and remote nodes.

In an embodiment, the cloud-based system **100** can use the cloud tunnel **500** to forward traffic to the enforcement nodes **150**, such as from a user device **300** with the application **350**, from a branch office/remote location **118**, etc. FIG. **8** illustrates three example use cases for the cloud tunnel **500** with the cloud-based system **100**, and other uses are also contemplated. In a first use case, a cloud tunnel **500A** is formed between a user device **300**, such as with the application **350**, and an enforcement node **150-1**. For example, when a user **102** associated with the user device **300** connects to a network, the application **350** can establish the cloud tunnel **500A** to the closest or best enforcement node **150-1** and forward the traffic through the cloud tunnel **500A** so that the enforcement node **150-1** can apply the appropriate security and access policies. Here, the cloud tunnel **500A** supports a single user **102**, associated with the user device **300**.

In a second use case, a cloud tunnel **500B** is formed between a Virtual Network Function (VNF) **502** or some other device at a remote location **118A** and an enforcement node **150-2**. Here, the VNF **502** is used to forward traffic from any user **102** at the remote location **118A** to the enforcement node **150-2**. In a third use case, a cloud tunnel **110C** is formed between an on-premises enforcement node, referred to as an Edge Connector (EC) **150A**, and an enforcement node **150-N**. The edge connector **150A** can be located at a branch office **118A** or the like. In some embodiments, the edge connector **150A** can be an enforcement node **150** in the cloud-based system **100** but located on-premises with a tenant. Here, in the second and third use cases, the cloud tunnels **500B**, **500C** support multiple users **102**.

There can be two versions of the cloud tunnel **500**, referred to as tunnel 1 and tunnel 2. The tunnel 1 can only support Web protocols as an HTTP connect tunnel operating on a Transmission Control Protocol (TCP) stream. That is, tunnel 1 can send all proxy-aware traffic or port 80/443 traffic to the enforcement node **150**, depending on the forwarding profile configuration. This can be performed via CONNECT requests, similar to a traditional proxy.

The tunnel 2 can support multiple ports and protocols, extending beyond only web protocols. As described herein, the cloud tunnels **500** are tunnel 2. In all of the use cases, the cloud tunnel **500** enables each user device **300** to redirect traffic destined to all ports and protocols to a corresponding enforcement node **150**. Note, the cloud-based system **100** can include load balancing functionality to spread the cloud tunnels **500** from a single source IP address. The cloud tunnel **500** supports device logging for all traffic, firewall, etc., such as in the storage cluster **156**. The cloud tunnel **500** utilizes encryption, such as via TLS or Datagram TLS (DTLS), to tunnel packets between the two points, namely the client **510** and the server **520**. As described herein, the client **510** can be the user device **300**, the VNF **502**, and/or the edge connector **150A**, and the server **520** can be the enforcement node **150**. Again, other devices are contemplated with the cloud tunnel **500**.

The cloud tunnel **500** can use a Network Address Translation (NAT) device that does not require a different egress

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IP for each device's **300** separate sessions. Again, the cloud tunnel **500** has a tunneling architecture that uses DTLS or TLS to send packets to the cloud-based system **100**. Because of this, the cloud tunnel **500** is capable of sending traffic from all ports and protocols.

Thus, the cloud tunnel **500** provides complete protection for a single user **102**, via the application **350**, as well as for multiple users at remote locations **118**, including multiple security functions such as cloud firewall, cloud IPS, etc. The cloud tunnel **500** includes user-level granularity of the traffic, enabling different users **102** on the same cloud tunnel **500** for the enforcement nodes **150** to provide user-based granular policy and visibility. In addition to user-level granularity, the cloud tunnel **500** can provide application-level granularity, such as by mapping mobile applications (e.g., Facebook, Gmail, etc.) to traffic, allowing for app-based granular policies.

FIGS. **9** and **10** illustrate the two communication channels, namely a control channel **530** and a data channel **540**, between the client **510** and the server **520**. Together, these two communication channels **530**, **540** form the cloud tunnel **500**. In an embodiment, the control channel **530** can be an encrypted TLS connection or SSL connection, and the control channel **530** is used for device and/or user authentication and other control messages. In an embodiment, the data channel **540** can be an encrypted DTLS or TLS connection, i.e., the data channel can be one or more DTLS or TLS connections for the transmit and receive of user IP packets. There can be multiple data channels **540** associated with the same control channel **530**. The data channel **540** can be authenticated using a Session Identifier (ID) from the control channel **530**.

Of note, the control channel **530** always uses TLS because some locations (e.g., the remote location **118A**, the branch office **1188**, other enterprises, hotspots, etc.) can block UDP port 443, preventing DTLS. Whereas TLS is widely used and not typically blocked. The data channel **540** preferably uses DTLS, if it is available, i.e., not blocked on the client **510**. If it is blocked, the data channel **540** can use TLS instead. For example, DTLS is the primary protocol for the data channel **540** with TLS used as a fallback over TCP port 443 if DTLS is unavailable, namely if UDP port 443 is blocked at the client **510**.

In FIG. **9**, the control channel **530** is illustrated with exchanges between the client **510** and the server **520**. Again, the control channel **530** includes TLS encryption, which is established through a setup or handshake between the client **510** and the server **520** (step **550-1**). The client **510** can send its version of tunnel **500** to the server **520** (step **550-2**) to which the server **520** can acknowledge (step **550-3**). For example, the version of the tunnel can include a simple version number or other indication, as well as an indication of whether the client **510** supports DTLS for the data channel **540**. Again, the control channel **530** is fixed with TLS or SSL, but the data channel **540** can be either DTLS or TLS.

The client **510** can perform device authentication (step **550-4**), and the server **520** can acknowledge the device authentication (step **550-5**). The client **510** can perform user authentication (step **550-6**), and the server **520** can acknowledge the user authentication (step **550-7**). Note, the device authentication includes authenticating the user device **300**, such as via the application **350**, the VNF **502**, the edge connector **150A**, etc. The user authentication includes authenticating the users **102** associated with the user devices **300**. Note, in an embodiment, the client **510** is the sole device **300**, and here the user authentication can be for the

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user 102 associated with the client 510, and the device authentication can be for the user device 300 with the application 350. In another embodiment, the client 510 can have multiple user devices 300 and corresponding users 102 associated with it. Here, the device authentication can be for the VNF 502, the edge connector 150A, etc., and the user authentication can be for each user device 300 and corresponding user 102, and the client 510 and the server 520 can have a unique identifier for each user device 300, for user-level identification.

The device authentication acknowledgment can include a session identifier (ID) that is used to bind the control channel 530 with one or more data channels 540. The user authentication can be based on a user identifier (ID) that is unique to each user 102. The client 510 can periodically provide keep alive packets (step 550-8), and the server 520 can respond with keep alive acknowledgment packets (step 550-9). The client 510 and the server 520 can use the keep alive packets or messages to maintain the control channel 530. Also, the client 510 and the server 520 can exchange other relevant data over the control channel 530, such as metadata, which identifies an application for a user 102, location information for a user device 300, etc.

In FIG. 10, similar to FIG. 9, the data channel 540 is illustrated with exchanges between the client 510 and the server 520. Again, the data channel 540 includes TLS or DTLS encryption, which is established through a setup or handshake between the client 510 and the server 520 (step 560-1). An example of a handshake is illustrated in FIG. 11. Note, the determination of whether to use TLS or DTLS is based on the session ID, which is part of the device authentication acknowledgment, and which is provided over the data channel 540 (steps 560-2, 560-3). Here, the client 510 has told the server 520 its capabilities, and the session ID reflects what the server 520 has chosen, namely TLS or DTLS, based on the client's 510 capabilities. In an embodiment, the server 520 chooses DTLS if the client 510 supports it, i.e., if UDP port 443 is not blocked, otherwise the server 520 chooses TLS. Accordingly, the control channel 530 is established before the data channel 540. The data channel 540 can be authenticated based on the session ID from the control channel 530.

The data channel 540 includes the exchange of data packets between the client 510 and the server 520 (step 560-4). The data packets include an identifier such as the session ID and a user ID for the associated user 102. Additionally, the data channel 540 can include "keep alive" packets between the client 510 and the server 520 (steps 560-5, 560-6).

The cloud tunnel 500 can support load balancing functionality between the client 510 and the server 520. The server 520 can be in a cluster, i.e., multiple servers 200. For example, the server 520 can be an enforcement node 150 cluster in the cloud-based system 100. Because there can be multiple data channels 540 for a single control channel 530, it is possible to have the multiple data channels 540, in a single cloud tunnel 500, connected to different physical servers 200 in a cluster. Thus, the cloud-based system 100 can include load balancing functionality to spread the cloud tunnels 500 from a single source IP address, i.e., the client 510.

Also, the use of DTLS for the data channels 540 allows the user devices 300 to switch networks without potentially impacting the traffic going through tunnel 500. For example, a large file download could continue uninterrupted when a user device 300 moves from Wi-Fi to mobile, etc. Here, the application 350 can add some proprietary data to the DTLS

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client-hello server name extension. That proprietary data helps a load balancer balance the new DTLS connection to the same server 200 in a cluster where the connection prior to network change was being processed. So, a newly established DTLS connection with a different IP address (due to network change) can be used to tunnel packets of the large file download that was started before the network change. Also, some mobile carriers use different IP addresses for TCP/TLS (control channel) and UDP/DTLS (data channel) flows. The data in DTLS client-hello helps the load balancer balance the control and data connection to the same server 200 in the cluster.

§ 8.0 CLOUD CONNECTIVITY

FIG. 11 is a diagram illustrating various techniques to forward traffic to the cloud-based system 100. These include, for example, use of the application 350 as a client connector for forwarding traffic, use of the connector 400 app, use of the VNF 502 or some other device, use of the edge connector 150A, and use of an eSIM/iSIM/SIM-card 600. The application 350 can be referred to as a client connector and it is via a native application executed on the user device 300 as well as being user ID-based. The connector 400 can be referred to as an app connector. The edge connector 150A can be referred to as a private service edge.

There is a requirement to get any customer traffic to/from the cloud-based system 100. However, there is a gap on some devices. The current approach, e.g., with the application 350, the connector 400, etc. there is a reliance on the device, namely installation of a forwarding app, a reliance on an operating system, namely virtual interfaces, and a reliance on forwarding gateways, namely the edge connector 150A. However, these may not be available with other types of devices such as IoT devices and the like. As described herein, the present disclosure utilizes the term client device to include, without limitations IoT devices (e.g., smart scooters, etc.), Operational Technology (OT) platforms (e.g., Supervisory Control and Data Acquisition (SCADA) systems, Industrial Control Systems (ICS), etc.), medical equipment (e.g., CAT and MRI scanners, etc.), connected vehicles, and practically any device that has a Subscriber Identification Module (SIM) in the form of a card, an eSIM, or an iSIM. Those skilled in the art will recognize that a client device differs from the user device 300 as it may not have the ability to implement the application 350, not support a user ID for identifying the user 102, etc.

The present disclosure includes two additional techniques for cloud connectivity for IoT devices including an eSIM/iSIM/SIM-card 600 based approach and a cloud/branch/thing connector 604. The eSIM/iSIM/SIM-card 600 based approach can be referred to as a device connector. The eSIM/iSIM/SIM-card 600 based approach is used for forwarding traffic from any SIM-based device (e.g., 2G to 5G and beyond). The key here is identity is based on the eSIM/iSIM/SIM-card 600, namely the International Mobile Equipment Identity (IMEI), as opposed to a user ID. There is no need for a SDK implemented by the third-party manufacturers, thereby bypassing development and patching processes. The eSIM/iSIM/SIM-card 600 approach provides its own network path and connectivity to the cloud-based system 100 as illustrated in FIG. 12; no gateway is needed, and it is a plug and play approach.

The eSIM/iSIM/SIM-card 600 approach leverages the fact that eSIM/iSIM/SIM-card 600 can have compute capabilities and the ability to implement functions including encryption. A TLS tunnel or the like is established from the

eSIM/iSIM/SIM-card **600** to the cloud-based system **100**. For example, this can include development via JavaCard which is a software technology that allows Java-based applications (applets) to be run securely on smart cards and similar small memory footprint devices, such as the eSIM/iSIM/SIM-card **600** which has low power and memory. Advantageously, this approach requires no reliance on the device and the forwarding is from the eSIM/iSIM/SIM-card **600**. Thus, this approach can work across various platforms, namely any device that uses the eSIM/iSIM/SIM-card **600**. The eSIM/iSIM/SIM-card **600** approach can also be implemented with a global Mobile Virtual Network Operator (MVNO)/Roaming agreement.

The cloud/branch/thing connector **604** can use the VNF **502** as well and includes forwarding of server traffic and is implemented on a hardware device connected to a network. The cloud/branch/thing connector **604** can be used in an Application-aware Networking (APN) approach with a mobile network **650** as illustrated in FIG. **13**. This approach leverages an APN as a gateway for traffic to forward to the cloud-based system **100**. This, similar to the eSIM/iSIM/SIM-card **600** approach, can be used on any Mobile Network Operator (MNO) network.

§ 9.0 APN

FIG. **14** is a network diagram of an APN network **700** for traffic forwarding of IoT devices **702** to the cloud-based system **100**, such as for secure access to the Internet **104**, to cloud services **106**, etc. Of note, the APN network **700** is illustrated with the client devices **702**, but it could work with any type of user device **300**. The client devices **702** are provisioned to operate on the APN network **700** with the eSIM/iSIM/SIM-card **600** as the ID. Traffic is passed from the local radio network to the carrier core, transparent to end client device **702**.

The APN **700** is illustrated with three example Mobile Network Operators (MNOs) **704A**, **704B**, **704C**, e.g., AT&T, Verizon, T-Mobile, etc. As is known in the art, MNOs **704** include radios **706** for wireless connectivity and servers **708** for processing. The MNOs **704** provide radio infrastructure, can include roaming agreements, and contract agreements with a Mobile Virtual Network Operator (MVNO) **720**.

The APN network **700** includes the MVNO **720** which is a wireless communications services provider that does not own the wireless network infrastructure over which it provides services to its customers. The MVNO **720** enters into a business agreement with the MNOs **704** to obtain bulk access to network services at wholesale rates, then sets retail prices independently. The MVNO **720** may use its own customer service, billing support systems, marketing, and sales personnel, or it could employ the services of a Mobile Virtual Network Enabler (MVNE). The present disclosure utilizes the known concept of the MVNO **720** to forward traffic to the cloud-based system **100**. The MVNO **720** can include servers **722** and the cloud/branch/thing connector **604** for connectivity to the cloud-based system **100**.

FIG. **15** is a flow diagram of communication in the APN network **700**. FIG. **15** includes interactions between the client device **702**, the MNO **704**, the MVNO **720**, a 3rd Generation Partnership Project (3GPP) **730** service, a Security Assertion Markup Language (SAML) **732** service, the cloud/branch/thing connector **604**, and the cloud-based system **100**. The device **702** connects via radio signals to the MNO **704** (step **S1**) and presents its configuration in the APN (step **S2**). The MNO **704** provides the APN association and forwarding to the MVNO **720** (step **S3**) and the device

702 is associated with the APN (step **S4**). The device **702**'s ESIM/iSIM/SIM-card **600** is authenticated using the 3GPP **730** services (step **S5**) which uses a proxy authentication of the ESIM/iSIM/SIM-card **600** to the SAML **732** service (step **S6**) which provides validation (step **S7**) and submission of a valid authentication token to the cloud connector **604** (step **S8**).

The SAML service **732** can provide accounting of access/logging/billing information to the MVNO **720** (step **S9**) and enable a path to the MVNO **720** (step **S10**). The device **702** has geolocation enabled as well through the 3GPP **730** services (step **S11**) and an IP path is enabled to the MVNO **720** (step **S12**). The device **702** now has access to the cloud edge via the MVNO **720** (step **S13**) and the cloud connector **604** forwards traffic to the cloud and applications (step **S14**).

Advantageously, the APN network **700** enables cloud connectivity for any ESIM/iSIM/SIM-card **600** connected device, including the client devices **702**, the user devices **300**, etc. All traffic from any device that is enrolled in the APN network **700** can be forwarded to the cloud-based system **100**, with traffic forward from the MNO **704** to the MVNO **720** to the cloud connector **604** to the cloud-based system **100**. The APN network **700** also removes the need for a client such as the application **350** or an SDK, as well as removing the need for physical network connections or gateways, just ride the APN network **700** to the cloud-based system **100**. This removes the barrier to entry—no software to manage at the client level, no patching, etc. Existing providers connect devices through the APN network **700** to a firewall and then on to the Internet. The firewall is used to isolate the devices on APN—no security layer. In an embodiment, this allows the client devices **702** to obtain security services from the cloud-based system **100**.

§ 10.0 ESIM/iSIM/SIM-CARD

FIG. **16** is a network diagram of a network **800** for traffic forwarding of any device having an ESIM/iSIM/SIM-card **600** to the cloud-based system **100**, such as for secure access to the Internet **104**, to cloud services **106**, etc. Of note, this approach can work with any device **300**, **702** having a ESIM/iSIM/SIM-card **600**. The devices **300**, **702** are provisioned to operate on the MNO **700** with the ESIM/iSIM/SIM-card **600** as the ID. This approach leverages the fact that ESIM/iSIM/SIM-cards **600** have the capability to execute code, such as using JavaCard, so that a tunnel is created between the ESIM/iSIM/SIM-card **600** and the cloud-based system **100**.

The ESIM/iSIM/SIM-card **600** is provisioned in advance with the functionality to support the cloud-based system **100**. The ESIM/iSIM/SIM-card **600** can launch TCP/IP functionality including a tunnel to the cloud-based system **100**, e.g., the cloud tunnel **500**, a TLS tunnel, or any other type of tunnel with encryption. The MNO **704** validates the device **300**, **702** and this can include SIM exceptions/authentication for known ESIM/iSIM/SIM-cards **600** supporting the tunnel to the cloud-based system **100**. This can include agreements between the MNO **704** and the cloud-based system **100**. Here, the cloud-based system **100** can provide details of allowable ESIM/iSIM/SIM-cards **600** to the MNO **704**. The cloud-based system **100** can provide an ingress point for the tunnel from the ESIM/iSIM/SIM-card **600** and validate the ID based on the ESIM/iSIM/SIM-card **600** ID (IMEI). In this approach, the ESIM/iSIM/SIM-card **600** is the client **510** and the enforcement node **150** can be the server **520** in the tunnel **500**.

FIG. 17 is a flow diagram of communication in the ESIM/iSIM/SIM-card 600 forwarding approach. FIG. 17 includes similar components as in FIG. 15 except the connector 604 and the MVNO 720. Specifically, FIG. 17 includes the device 300, 702, the ESIM/iSIM/SIM-card 600 in the device 300, 702, the MNO 704, the 3GPP 730, the SAML 732, and the cloud-based system 100.

The ESIM/iSIM/SIM-card 600 allows the device 300, 702 to connect via radio signals to the MNO 704 (step T1), the device 300, 702 requests access via the ESIM/iSIM/SIM-card 600 (step T2), and the ESIM/iSIM/SIM-card 600 is authenticated through the 3GPP 730 (step T3). The 3GPP 730 can perform proxy authentication of the ESIM/iSIM/SIM-card 600 via the SAML 732 (step T4) which can validate (step T5) and provide a valid authentication token to the cloud-based system (step T6).

The ESIM/iSIM/SIM-card 600 is configured to establish a TCP/IP connection to the MNO 704 (step T7) and the ESIM/iSIM/SIM-card 600 launches a tunnel (step T8). Using the 3GPP 730 can provide logging and billing information to the MNO 704 (step T9) so that the cloud-based system 100 can be charged and the 3GPP 730 can determine the geo location of the device 300, 702 (step T10). Once complete, the device 300, 702 has access to the cloud-based system 100 based on a tunnel between the ESIM/iSIM/SIM-card 600 and the cloud-based system 100 (step T11).

Advantageously, the ESIM/iSIM/SIM-card 600 based approach supports secure forwarding from the ESIM/iSIM/SIM-card 600, including physical ESIM/iSIM/SIM-cards, embedded ESIM/iSIM/SIM-cards (ESIM) and IP Multimedia Services Identity Module (ISIM). The ESIM/iSIM/SIM-card 600 is pre-enrolled and all traffic from such devices 300, 702 having these ESIM/iSIM/SIM-cards 600 is forwarded to the cloud-based system 100. This also removes the need for the application 350 or an SDK, as well as does not require physical network connections, i.e., the MVNO 720. The identity of the device 300, 702 is based on the ESIM/iSIM/SIM-card 600. This ESIM/iSIM/SIM-card 600 based approach is ideal for IoT devices having a tunnel running in a small form factor.

§ 11.0 DEVICE FORWARDING PROCESS

FIG. 18 is a flowchart of a process 800 for forwarding traffic to the cloud-based system using a ESIM/iSIM/SIM-card. The process 800 includes, responsive to a client device having a Subscriber Identity Module (SIM) card therein connecting to a mobile network from a mobile network operator, receiving authentication of the client device based on the ESIM/iSIM/SIM-card (step 802); receiving forwarded traffic from the client device (step 804); and processing the forwarded traffic according to policy, wherein the policy is determined based on one of a user of the client device and a type of the client device, each being determined based on the ESIM/iSIM/SIM-card (step 806).

The client device can be an Internet of Things (IoT) device. An identity of the user can be determined based on an International Mobile Equipment Identity (IMEI) of the ESIM/iSIM/SIM-card. The ESIM/iSIM/SIM-card can be preprogrammed for access to the cloud-based system. The ESIM/iSIM/SIM-card can be configured to implement a secure tunnel from the ESIM/iSIM/SIM-card to the cloud-based system. The ESIM/iSIM/SIM-card can execute JavaCard code for implementation of the secure tunnel. The secure tunnel can utilize any of Transport Layer Security (TLS), Secure Sockets Layer (SSL), and Datagram TLS (DTLS).

The forwarded traffic can be forwarded over an Application-aware Networking (APN) network.

§ 12.0 CLOUD-BASED SYSTEM FUNCTIONS

FIG. 19 is a block diagram illustrating functions of the cloud-based system 100, for example, cloud-based services of Internet Access (IA) and Private application Access (PA). Of course, the cloud-based system 100 can offer any cloud service, and IA and PA are shown for illustration purposes. Logically, the cloud-based system 100 can have forwarding functions, enforcement functions, and application steering functions. The users 102 can be web-based, connecting to the cloud-based system via a PAC file or the like. The users 102 can be tunneled to the cloud-based system 100 via a network edge, or connect to the cloud-based system 100 via the network edge through the VNF 502 or the edge connector 150A. Finally, the users 102 can be authenticated and connected via the application 350. The enforcement functions are performed via the nodes 150, including virtual service edges, physical service edges, etc. The nodes 150 can be customer-hosted (tenant specific) or cloud-hosted (multi-tenant).

§ 13.0 NETWORK EDGE OPTIONS

In an embodiment, the present disclosure includes hosting a node 150 in a service provider's MEC, and this node 150 can be referred to as a virtual service edge. Note, the node 150 itself can be physical devices and/or virtual devices. The term "virtual service edge" is used to note this node is part of the cloud-based system 100 but located in the MEC of the service provider's network. In this approach, the virtual service edge is configured to provide edge hosted enforcement. This includes policy enforcement for all service provider customers as well as individual policy enforcement for authenticated customers. That is, the service provider can provide the cloud-based security to all of its customers, as well as to enterprise users who already have defined policies.

The Radio Access Network (RAN) sends client traffic direct to the Virtual Service Edge (VSE) which is hosted in the MEC, and the VSE processes traffic locally in the MEC. A Multi-Access Edge Compute (MEC), also known as Mobile edge computing (MEC), is an ETSI-defined network architecture that defines cloud computing capabilities at the edge of any network, i.e., a cellular network. The MEC is designed to be implemented as cellular based stations or other edge nodes.

In another embodiment, the present disclosure includes hosting a connector application, called a cloud connector, such as the VNF 502, the edge connector 150A, etc., in the MEC. The cloud connector is configured to forward traffic to the cloud-based system 100. The RAN sends client traffic directly to the cloud connector, the cloud connector is hosted in the MEC and considered a gateway for all traffic on the RAN.

FIG. 20 is a block diagram of a MEC 900 located at a base station 902 of a service provider's network. FIG. 20 is a logical diagram of functions associated with the MEC 900. The MEC 900 is configured to enable user devices 300 to wirelessly connect to the Internet 104 via the base station 902. The MEC 900 includes physical resources, i.e., compute infrastructure, such as the servers 200. The MEC 900 can include virtual resources, such as Virtual Machines (VM), containers, bare metal, etc. On top of the resources are edge services, such as APIs, cloud connections, etc. The MEC 900 includes routing functions based on various

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protocols, traffic steering, DNS, Network Address Translation (NAT), etc. Next, the MEC 900 includes a hosted service layer and service gateways for various services—authentication, account, standards, etc. Finally, the MEC 900 includes mobile edge management.

FIG. 21 is a block diagram of the MEC 900 and the conventional approach with the cloud-based system 100. Conventionally, the MEC 900 forwards user traffic via the Internet 104 to the cloud-based system 100. Here, the MEC 900 is simply a pipe forwarding traffic, and the service provider is not offering the services of the cloud-based system 100.

FIG. 22 is a block diagram of the MEC 900 with a cloud connector 904 located in the MEC edge services. Here, the cloud connector 904 is hosted in the MEC 900 and configured to forward traffic to a node 150 in the cloud-based system 100. For example, the node 150 can be a public node or a private node. In an embodiment, the node 150 can be hosted in the MEC 900 as well. MEC-based nodes 150 can be only available to Users on the Mobile network.

FIG. 23 is a block diagram of the MEC 900 with nodes 150 located in the MEC hosted service layer. Here, there are private nodes 150 configured within the MEC 900, in the hosted service layer.

FIG. 24 is a block diagram of the MEC 900 illustrating a traditional security service from a service provider. Here, the service provider has to reroute traffic from the MEC 900 to a separate service.

FIG. 25 is a block diagram of the MEC 900 illustrating integration of cloud-security services in the MEC 900. Here, the functionality of the nodes 150 are provided within the MEC routing functions. Advantageously, the approach pro-

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vides improved latency, offers additional services for the service provider, and the like.

§ 14.0 INTELLIGENT EDGE STEERING

FIG. 26 is a network diagram of common edges and sizes encountered when accessing workloads from UE 300. The desire to deploy compute edges closer and closer to the end user (or service) requires smaller and smaller form factors to be deployed. In this example, there are various MECs 900A-900E. Within a controlled data center (e.g., the MECs 900D, 900E) or edge hosting facility (e.g., the MEC 900C), it is possible to run fully sized 19" and racked equipment for hosting edge solution components. Whereas an edge location may be as large as a data center or as small as a box of cards in an uncontrolled, harsh landscape such as a shipping container or farm vehicle, e.g., the MECs 900A, 900B.

The smaller the physical device is, the less powerful the compute functions are available to run workloads. Edge compute, such as with the multi-access edge compute MEC for 5G solutions will, if overwhelmed, act as a bottleneck for traffic passing from a wireless tower to the core, via the MECs. As edge compute evolves, there will be no “one size fits all” edge. Nor will all services be served from an edge. There will be requirements to connect and protect all sizes of workloads at many edge sites of varied compute/storage/network capabilities.

§ 14.1 Proposed Solution with the Cloud-Based System

The goal is to provide a solution that actively makes decisions about “how” and “where” to send traffic based on the topology of the data paths through the various Edge compute resources involved in a given customer use case. This solution needs to be broken down into three separate segments as shown in Table 1.

TABLE 1

Overview of the needed functions for dynamic steering		
Identification of traffic	Steering mechanism	Reception and processing
Governance and Tracking, to ensure all traffic is delivered correctly. Make sure traffic is not only steered, but monitors which path it follows, which edge nodes are available and how to use them	Understand the source traffic, its traffic type and destination location and/or type (e.g., known as basic website or low latency services AR/VR).	Reception and processing of traffic should be executed only if forwarded from a trusted forward node. Not an open gateway
Mechanism to actively identify traffic type & map this to possible steering actions. An example would be:	Steering must follow an intelligent path defined by the governance process. Traffic must be steered securely, against interception or unauthorized inspection, including protection from quantum and other emerging decryption attacks. e.g., decision to encrypt previously unencrypted traffic	Controls are applied as per security policy. Traffic is enforced and sent to destination. Any return content is returned via the state session established between reception/reception node-then “returned” to UE via the established tunnel
Source traffic from local UE in EMEA Is SSL encrypted Going to a destination in USA Forward to SSL decryption edge in smart Colo in USA This mechanism would require active input from the steering and processing functions Transparent to UE device- there should be little to zero awareness from the UE that this steering is happening	Steering should have fail-over options should the initial path be interrupted. Pinned flows for the UE over any edge. Should a device move from RAN to RAN or to new MEC, the traffic flow should “follow the UE device”	

The present disclosure includes an active set of processes that will assess all parts of a UE-to-workload traffic flow. This assessment will deliver a decision that instructs the forwarding platform on how to send this traffic. The platform decision will be made based on, but no limited to: i) Security Policy of the client, ii) ML knowledge of services, iii) The type of traffic, and iv) The location of the destination service.

By making these intelligent decisions on traffic flow, a cloud service can actively send traffic to the best and/or most effective processing node. Thus, optimizing the use of the “best compute” for that resource. e.g., UE is in EMEA, Primary Edge is in EMEA, Destination workload is in USA, Policy defined to SSL decrypt. Decision is to decrypt, but not at the local Edge—as it would be more effective to decrypt at an edge that is closer to the destination, thus saving edge compute close to the UE.

§ 14.2 Value

Beyond the direct value of steering the right user to the correct destination, the information collected in real time of workload components sourced from different technology providers (cloud, telco, etc.), will be useful to determine which offered data flow paths reduce Total Cost of Ownership (TCO). Thus, offering the customer data that is useful to determine which technology provider provides the most cost-effective location and/or path for end-to-end access to workloads has value.

The goal is to provide a solution that makes decisions actively about “how” and “where” to send traffic based on the total cost (as defined by latency, bandwidth costs, path pricing, and other customer-defined value measures). For example, additional values to the decision could be (but are not limited to):

- Cost of compute at edge vs destination—for more cost-effective compute

- Latency budget/bandwidth budget compared against actual network traversal costs

Bypass decisions that could include a default function for unknown traffic flow (e.g., block/allow) and other metrics defined by the customer.

This also allows workload providers the ability to select exactly where a workload may appear for a user and enforce local connectivity when appropriate (e.g., deliver a local version of a service rather than remove or route to a cloud service).

§ 14.3 Technical Details

FIG. 27 is a diagram of multi-workload access using the cloud-based system 100 providing edge and cloud solutions. FIG. 28 is a flow diagram of selective edge flow between the UE 300 and a workload 1000 with the cloud-based system 100 in-between for selective dynamic steering. Specifically, the cloud-based system 100 provides functions including a decision front end 1002 between the UE and the first MEC 900, and cloud intelligence between the end MECs 900D, 900E and the workload 1000, which can be on the Internet 104 or accessible therethrough. This intelligent traffic path steering allows for traffic to pass through an edge securely and uninterrupted or manipulated by other services, owners, or solutions. In addition, it allows for policy driven decisions as to which path to take to enable the workload in question.

To deliver the depicted flow in FIG. 27, there needs to be various steps of assessment, inquiry, and policy enforcement to deliver the necessary traffic flow for the UE-generated request. This is illustrated in FIG. 28. The flow is between the UE 300, a traffic type detection engine 1004, the decision

front end 1002, a machine learning engine 1006, a policy engine 1008, the intelligence in the cloud 100, and the workload 1000.

The cloud-based system 100 includes functionality integrated in a first MEC for the traffic type detection engine 1004 and the decision front end 1002, such as the MEC 900A. Of course, it could be in other MECs 900. The decision front end 1002 will need to intelligently identify which traffic should be steered and how. Delivering this requires active insights from a collection of data points, that are assessed and communicated at regular intervals.

In an example, the UE 300 requests access to a workload 1000 (step A1). The traffic type detection engine 1004 is configured to identify the traffic type (step A2) for the decision front end 1002 which queries the machine learning engine step 1006 (step A3). The traffic type detection engine 1004, the decision front end 1002, the machine learning engine step 1006, and the policy engine 1008 are functions and can be integrated in compute resources at one of the MECs 900. These engines 1002, 1004, 1006, 1008 can be separate or combined modules, that is this description is functional. The machine learning engine 1006 can perform a policy query for the identified traffic type (step A4). The decision front end 1002 receives the machine learning query and decides the path based thereon and informs the UE 300 (step A5).

The UE 300 again requests the workload 1000 (step A6), and the decision front end 1002 steers the traffic towards the workload 1000 (step A7). Periodically, the machine learning engine 1006 can receive intelligence from the cloud-based system 100 (step A8), such as types of workloads, steering, policy, etc.

§ 15.0 WORKLOAD ISOLATION

Edge compute, in the MEC 900, is a collection of software-driven services that will be hosted physically closer to end users and solutions than on-premises data centers or public/private clouds. The nature of this edge is to leverage the software-driven interconnects to deliver local and low-latency compute to UE 300—defined as end-user devices connecting to edge compute via wireless, or other network connectivity). Edge compute services will provide local connections from UE 300 to locally hosted edge services. Communication will use common network routable protocols and network functions. This will expose workloads to anyone or anything that shares this network context.

FIG. 29 is a diagram of intra edge connectivity in a MEC 900 between various services 1102, over a shared transport network 1104. To address this shared context, ideas for isolation have been proposed and rely on the virtualized network layer. An example often given is to share a network but to firewall-off access between each zone, between workloads, or some combination of the two. By utilizing network-based security services, all connectivity and protection is reliant on network security and its context, e.g., users and services must have a shared network context to be able to access services.

§ 15.1 Workload Isolation Solution

The present disclosure proposes using the foundational components of zero trust to control application access paths. Zero trust outlines that no service is accessible until validating the access conditions of the requestor. Before validation, there is no access or even visibility to any service or workload, rendering it “hidden” to the network. Workloads will only ever have a context to their hosted cloud service.

The cloud-based system **100** is used to implement a zero-trust layer for edge hosted workloads—ensuring that no workload or service can speak-to or observe anything without first receiving validation. Workloads can exist within networks, subnets, functions, etc. but will not be able to communicate without validation. This validation of access must occur at the substantiation of each connection—and there must never be permanent access granted or re-use of connections. Each access path is unique to the specific session and then torn down and forgotten at session completion.

A cloud service via the cloud-based system **100** only allows connections from authorized users, which is established through:

- i) Outbound path for control/signaling
- ii) Outbound path for data channel establishment
- iii) Internal cloud service will, only after approval, create a connection between it and the destination workload. The workload will only ever see sessions from the cloud service and nothing else.

All traffic between the cloud service, including (but not limited to) signaling and workflows, must be over an encrypted tunnel ensuring zero capability for interception or manipulation of the service access or function within the Edge. Thus, the service is only available to those who are allowed to access it and nothing more.

FIG. **30** is a network diagram of a MEC **900** illustrating cloud service-based edge isolation. This secure traffic path steering allows for traffic to pass securely through an edge, uninterrupted or manipulated by other services, owners or solutions. In addition, it allows for policy driven decisions about which path to take to enable the workload in question.

As depicted in FIG. **30**, a set of cloud services **1200** allows for any application or workload within the Edge Compute to have traffic correctly and securely steered to it. Regardless of the transport or service interconnect layers, traffic passes securely to only the required workload and nothing else. This design is capable of being deployed within physical or virtual edge compute locations. For example, within locations provided by a hyperscaler, or on premises, or hosted on portable edge compute.

§ 16.0 SECURE EDGE STEERING

FIG. **31** is a diagram of UEs connected to edge hosted services without intelligent steering. Edge compute services will enable low-latency access from User Equipment (UE) to locally hosted edge services, such as the Multi-Access Edge Compute (MEC) **900**. This connectivity will use common network routable protocols that will ensure traffic is passed between UE **300** and workloads via TCP/IP based controls (similar to the physical- or hypervisor-based routing mechanisms). Like traditional physical or logical controls, security is often overlooked or network centric. This includes security solutions that do not limit lateral motion of malicious software infecting UE or MEC workloads.

Connectivity within these edge hosted orchestration platforms, for example a RAN-connected smart device (UE) connected to a MEC-hosted mapping service, rely on a shared TCP/IP network context. The smart device needs to have a direct routing path to the edge mapping service. This implies that any UE and any RAN-connected service will need network level connectivity to the MEC-hosted services. The nature of a shared network context exposes applications/services to unwanted use or even attack and ultimately penetration. Any edge hosted service that can receive a packet from the UE opens itself up to risk.

§ 16.1 Edge Compute Architecture Overview

FIG. **32** is a diagram of possible edge compute **1300** components. Edge compute **1300** includes multiple form factors depending on use case. Within the Cellular/Mobile network, it is an ETSI-Defined architecture used to bring compute and control closer to the end subscriber or user equipment (UE). Running these compute and control services within a near-real-time environment allows for better user experience for current and future services.

The edge compute **1300** ecosystem does rely on virtualized and containerized application services that would be controlled through open standard hypervisors and/or Application Programming Interfaces (APIs). The virtualization of compute will allow for dynamic service chaining of all forms of applications to be consumed at the edge.

The edge compute **1300** will also provide gateway interconnects to cloud services, telecommunications cores, augmented and virtual reality providers, etc. This enables edge compute providers to move workloads from edge to edge, end to on-premises data center, and edge to cloud depending on the end user/workload use case.

The UE **300** will send traffic to the ultimate consumption destination, through the demonstrated edge transport of the edge compute platform. The edge will either direct traffic directly through and egress the edge without applying any control, or it will direct the traffic to an internally hosted service. This decision will generally be controlled by the edge management of the hypervisor layer of an edge. The actions will then be represented through traffic routing decisions in the edge transport layer/virtual switch. The action will be to send traffic either external or internal within the edge compute engine. The traffic passing through this switch or transport layer is exposed to the same network-based risks of any open/shared TCP/IP network—e.g., interception, manipulation, abuse, etc.

§ 16.2 Intelligent Steering Solution

Using an intelligent control service within the edge compute **1300** system, the cloud-based system **100** can affect three key outcomes to traffic traversing the Edge:

- 1) Overlay all controls and abstract the path steering to the overlay, on top of the network.
- 2) Steer the path locally, within the edge (or externally of the edge) directly to the required service
- 3) Ensure all traffic passing through the service, on to its destination (in the edge or externally) is end-to-end encrypted with zero exposure of content to neighboring networks, services, administrators, user etc.

FIG. **33** is a diagram of cloud edge hosted steering deployment for any type of wireless or other means of transport between UE **300** and the edge compute **1300**. This steering of the traffic path allows for the traffic not only to pass through an edge (securely and uninterrupted or manipulated by other services, owners or solutions) but more importantly deciding which path to take to the workload in question.

As depicted in FIG. **33**, cloud services allow for any application or workload within the Edge Compute **300** to have traffic correctly and securely steered to it. Regardless of the transport (wireless or wired) or service interconnect layers, traffic passes securely to only the required workload and nothing else. This design is capable of being deployed within physical or virtual edge compute locations. For example, within hyperscaler provided locations or on-premises or hosted on portable edge compute.

§ 16.3 Secure Edge Workload Steering and Access

Edge Workload deployment and orchestration is fundamentally empowering the consumption of new, low latency

experiences for users and services connected to that edge. This connectivity to edge services, hosted on any type of hosting environment (e.g., bare metal, virtualized, containerized, telecom operator-hosted multiaccess edge compute, hyperscaler-hosted edge compute, any other type of Edge device) needs to be effective, direct, and ultimately isolated. Delivery of connectivity to workloads within the edge is reliant on the user endpoints and workloads sharing a traditional IP network link. Thus, requiring workloads to expose a listening service to any and all devices sharing that network.

Securing these connections, to ensure protection of the workload, should be executed through unique, dynamic & encrypted paths on a per-app, per-connection basis. It should also not be reliant on any underlying transport protocol over any part of the path from source to Edge workload (e.g. S1, SGI, F1, N3 or other) or IP-based network controls for security. The correct workload path must be decided using values that best serve the endpoints, be it policy controls, geo specifics (in country hosted for in country content), specifically defined network paths, load management (e.g. traffic sent to various edges or burst new edge services depending on service load) and other uses.

Edge services, such as the Multi-Access Edge Compute (MEC) or other edge services (bare metal, virtualized, containerized, telecom operator-hosted multi-access edge compute, hyperscaler-hosted edge compute, any other type of Edge device) will simply leverage IP routes and paths to interconnect various access sources to the workloads including, but not limited to, 5G UPF to workloads in a MEC (and beyond to any arbitrary IP data network). Such connectivity will ensure that traffic is passed between source and workloads via TCP or UDP over IP. Similar to traditional physical or logical controls, security in the edge is often overlooked (or unspecified) or presumed to be network centric. This implies that any device on the ingress service will need network level connectivity to the MEC hosted RAN service.

FIG. 34 shows an endpoint device 300 connected to Edge Hosted Services over a traditional network path with service endpoints exposed and reachable by arbitrary endpoints. The nature of a shared network context exposes applications/services to unwanted use or even attack and ultimately penetration. Any connected network to an Edge hosted service can send and receive packet to/from the edge and thus may be impacted by the oft known risk of network attack surfaces, e.g., services exposed to untrusted network sources. To restrict such movement, edge services look to apply network layer controls such as Firewalls, to restrict access into various segments at the network layer. These network layer controls will still be faced with the attack surface problem, that in order to enable access a network path must be exposed to untrusted users. These network level security controls must contend with the aforementioned challenges. So that each time connectivity from different user sets is required, either these new users are added to the existing untrusted network, or the edge network must be extended to these untrusted user networks, again exposing the risk of the attack surface.

The traffic passing through the network layer switching or transport will be exposed to the same network-based risks of any open/shared IP networks, e.g., interception, manipulation, abuse, etc. Traditional approaches towards protection would be to apply network level firewalling between the endpoints and workloads. This still leaves the network path of endpoint to workload open for abuse, either from malicious actors or misconfiguration at any part of the network path between endpoints and workloads. Network based

security controls are dramatically restricted in their protection of the edge, but the applications will still need to be exposed to the sets of UE to allow interaction. This lack of effective protections at the Edge is considered to be the new frontier for malicious network attacks by cybersecurity specialists.

§ 16.3 Secure Edge Workload Steering and Access Solution

The present disclosure implements a zero trust layer for edge hosted workloads. Ensuring that no source, or interconnected service speaks to, or observes anything within the edge without first receiving validation. Workloads can exist within networks, subnets, functions, etc. but will not be able to communicate outside of their defined function without validation. This validation of access must occur at the substantiation of each connection, meaning there must never be permanent access granted. Each access path is unique to the specific connection and then torn down and forgotten at completion of that connection.

The disclosed service only allows connections from authorized sources, which are established through:

- Outbound path for control & signaling

- Outbound path for data channel establishment

The endpoint service will, only after approval, create a connection between it and the destination workload.

The workload will only ever see sessions from the endpoint services and nothing else

Using a secure traffic policy decision, the service can affect 3 key outcomes to traffic that pass the Edge:

- Overlay all controls & abstract the path steering to the overlay, on top of the network

- Connect the endpoint to the workload at an application layer, direct over the most effective path, either within the edge or externally of the edge

- Ensure all traffic passing through the service, on to its destination (in the edge or externally) is end-to-end encrypted with zero exposure of content to neighboring networks, services, administrators, users etc

The solution provides the following benefits being secure, agnostic, and transparent. Authorized traffic is always sent to the correct destination, securely and isolated between the endpoint and the workload not exposing any part of the application, connection, or session to anyone besides the authorized user. Edge controls are applied only to endpoint sources that are connected to MECs. Should the source be connected to another UPF, RAN or even IP network, the security steering will ensure that access is allowed or blocked, regardless of the underlying network. Consumers of the service will leverage the platform from the source location, thus passing through to the Edge layer of the infrastructure. At which point the service will steer traffic, based on policy, to the correct workload in that edge. The traffic path steering allows for the traffic not only to pass through an edge, securely and uninterrupted or manipulated by other services, owners, or solutions, but more importantly deciding which path to take to the workload in question.

FIG. 35 is a flow diagram showing the edge hosted steering allowing endpoint access only to authorized workloads. The set of services 1200 allows for any application or workload within the edge compute to have traffic, correctly and securely steered to it. Regardless of the transport or service interconnect layers, traffic passes securely to only the required workload and nothing else. This design is capable of being deployed within physical or virtual edge compute locations regardless of type of workload. For example, workloads can run in virtual machines, be containerized and within hyperscale-provided locations, on premises, or hosted on portable edge compute.

§ 17.0 MULTI-LAYER INTELLIGENT STEERING

Ensuring that endpoint traffic passes securely and directly to a chosen workload is key to the success of any connectivity deployment. Moreso, in the 5G drive for ultra-low latency, if the workload & its controls are located on the other side of the world, any speed advantage offered to the endpoint by the 5G network is lost. Moving workloads to the edge is only one step, as not all UE consumed workloads will exist on a local edge, the consumption and connectivity to workloads needs to be addressed.

§ 17.1 Multi-Layer Intelligent Steering Solution

The goal is to provide a solution that actively decides “how” and “where” to send endpoint-initiated traffic based on the topology of the data paths through the various compute resources involved in a given customer use case. This decision will need to be made at various granular levels that will be decided by the policyholder. There will need to be multiple layers of deployed control, from manual control (and override) to dynamically applied decisions using various mechanisms including but not limited to Artificial Intelligence (AI) and/or Machine Learning (ML), and/or user/device behavior analysis, and/or detection of anomalies in user/device behavior based on datasets that will ultimately influence how traffic gets from point A to point B. The solution is broken down into the three separate segments shown in Table 1.

The present solution provides two functions to enable the traffic steering, the first being path identification. A dynamic set of data identification services will assess all parts of an “endpoint to workload” traffic flow. This assessment will draw upon calculated data by the AI-engine for known sets of workload/protocol/latency/speed/endpoint combinations (e.g., “known protocol”, “known destination”, or “needed latency”). This will include data signals identified during the processing of traffic (e.g., content of traffic contains regional content that is better served in a regional egress) and will deliver an outcome to the steering platform that instructs it on how and where to send the traffic. The decision is concluded based on, but not limited to the high-level category definitions outlined in FIG. 36.

The second function of the two functions to enable the traffic steering is steering decision. Through preemptively collecting and calculating values based on the available information, across the connection path, plus enriching these values based on additional knowledge obtained through day-to-day operations of the global engine and additional external insights, decisions are made on how to steer each part of the endpoint traffic flow to achieve desired performance goals or desired cost goals (given that lower latency compute power is often charged at higher rates than higher latency compute power). This enables actively sending traffic to the best and most effective processing node or edge based on sets of rules defined by the network, the edge compute location, and the customer. Thus, optimizing the use of the “best choice compute” for that resource.

A high-level example of the two functions operating together would be e.g.,

Function 1:

- Endpoint is in EMEA
- Primary edge is in EMEA
- Required workload is located in the USA

Function 2:

- Policy defines to Secure Sockets Layer (SSL) decrypt

Outcome:

“Decision is to decrypt, but not at local edge, as it would be more effective to decrypt at an edge that is closer to the destination, thus saving edge compute close to the endpoint for additional functions”

It is noted that the engine exists as a core function of the Internet Access platform & can be queried for insights. The AI engine is a new function that will enrich traffic decision outcomes. This approach will enable steering and delivery of workload access from any endpoint to any workload, regardless of the integrity, risk and/or trustworthiness of the underlying network, due to misconfiguration, immaturity of network components, untrusted operators, equipment providers, and unavailability of underlying routing mechanisms, e.g., DNS.

§ 17.2 Multi-Layer Intelligent Steering Value

Directly steering the correct endpoint to the correct workload, through an informed and real-time decision, based on components sourced from different technology providers, delivers two key outcomes:

- Endpoint traffic will be optimized, to ensure the best-in-class path to deliver low latency and performant connections.

- Communication will have confirmed integrity and security, meaning content will be protected and steered over existing networks.

Delivering these functions requires building a service that actively steers traffic based on informed decisions about “how” and “where” to send traffic. Decisions should include, but not be limited to, values collected at:

- “Time” cost of compute at edge vs. destination, for more cost-effective compute

- “Dollar” cost of compute at various edges and steering to the best location

- Bypass decisions

- Policy decisions and overrides

- A default function for unknown traffic flow, e.g., block/allow

- The intelligence collected from the provider

- Partner intelligence

The outcome of the decision is represented in a mechanism/token that can be consumed to execute on the decision and steer traffic as defined. This also allows workload providers the ability to select exactly where a workload may appear for a user and enforce “only” local placement and connectivity, e.g., deliver local version of service rather than remove or conversely move a workload to an external location (through the various workload orchestration solutions).

FIG. 37 is a flow diagram representation of multi workload access using the service 1200 to steer to the correct enforcement point. The intelligent traffic path steering allows for traffic to pass through an edge securely, uninterrupted, or manipulated by other services, owners, or solutions. In addition, it allows for policy driven decisions, determining which path to take to enable the workload in question.

FIG. 38 is a flow diagram showing the selective edge flow. In order to deliver the depicted flow in FIG. 38, there needs to be various steps of assessment, inquiry, and ultimately policy enforcement to deliver the necessary traffic flow for the endpoint generated request. The decision “point” intelligently identifies which traffic should be steered and how. Delivering this requires active insights from a collection of data points, that are assessed and communicated at regular intervals.

§ 18.0 5G UDM IDENTITY FUNCTION

The present disclosure enables 3rd party, non-core, services to consume subscriber data from core data stores such

as, but not limited to, the Unified Data Management (UDM) and/or the User Data Repository (UDR) in 5G.

To correctly apply zero trust security controls, a true identity of the source of the connection needs to be identified. Within the Data Network (DN) of a 5G environment, a source identity cannot be limited solely to an IP address or source network, as this does not offer enough context of the source of a connection. A true zero trust control must be able to not only identify the initiator of the connection, but also differentiate it between various sets of device types. When running software on the initiator, it is very possible to extract an accurate identity of the device. However, when not running software on the initiating device there are limited mechanisms to provide identity of the device.

In 5G deployments, the information collected through the enrolment and enablement of a subscriber is stored in a set of services, tasked with managing all authorization, registration, network, and session definitions, amongst other data sets. This information, when collated together, allows the 5G provider to provide a unique identity for each and every subscribed UE. It is this information set that the present disclosure proposes to leverage for identity.

§ 18.1 5G UDM Identity Function Solution

FIG. 39 is a flow diagram showing identity values and stores in the 5G landscape. 5G UDM/UDRs 1400 provide unique sets of data, when consumed by a trusted mechanism to a trusted party, that allow for the direct identity of UE 300 to be categorized for User Plane Function (UPF) consumption and control. Within the context of zero trust access, this identity is used to differentiate the controls for traffic passing to and beyond the UPF and Data Networks (DN).

The present disclosure provides establishing a trusted communication path between UDM/UDR and a service to consume the UE values in assessing the source identity of a subscriber and thus any subsequent transactions. The service can take the form of a 5G Application Function (AF) that is authenticated by the 5G control plane and then granted access to UDM/UDR and/or other 5G core components that store and serve needed data. The service could also exist outside of the 5G core as an AF or other service function that can be authenticated and trusted by the 5G network using other mechanisms agreed to between the service provider and the 5G core owner.

In 5G environments, all events and factors related to the onboarding of devices, SIMS, and connection mechanisms (RAN, Geo, etc.) are paired with the subscriber identifier, e.g., account number, name, user identity, etc. This allows core operators to build unique identifiers for each connected device. 5G UDM/UDR values are updated based on signaling from RAN and 5G Core services. The present disclosure proposes to include continuous inclusion of updates and thus verification of session identity. This includes having AI/ML models to detect anomalies and terminate or force re-auth of a session, and periodically checking if any session or device parameters changed and re-apply policies which may be tied to session initiation.

For 5G networks, Data stored within a UDM, to be assessed as part of this query would include, but not be limited to, the following details:

- Subscriber Information (UE details, PDU, etc.)
- Session Management Data (QoS, SMF, PDU types, etc)
- Access and Mobility Management Function (Session response, tunnels, IP address, etc.)
- Policy (changes, implementations, controls)
- Among many other values
- Consuming 5G UDM/UDR values as an identity control from an UPF connected control function will give the

service provider (Zscaler) the ability to correctly assign identity to traffic flows. Which, in turn, will allow for the assignment of correct policy and steering decisions to be applied on the traffic initiating from the UE. These identifiers should be consumed, similar to how Security Assertion Markup Language (SAML) is used to consume the output of an Identity Provider (IDP). Note that the service could also signal the results of its analyses to 5G core components including (but not limited to) the Authentication Management Function (AMF), or the Session Management Function (SMF), to enable the 5G core to dynamically update how UE traffic is managed.

This set of information can then be used as a mechanism to form individual identity across multiple services. However, the service provider will consume these identifiers to then:

- Assign the traffic to the correct “owner”
- Assign the correct policy
- Steer the traffic over the best path
- Signal 5G core traffic management components the updates

FIG. 40 is a flow diagram of the edge process to query 5G UDM through the App Function (AF) 1500. The solution is for the process to directly query the UDM 1400 stored values. This is through a direct path request to the SMF 1502 or pass directly through to the UDM. Querying for values with the UDM 1400 directly allows the process to actively check the validity of multiple key values of the UE 300 connectivity. The full extent of the values stored in a UDM is outlined in the ETSI specification (ETSI TS 129 503).

By default, the UDM is not exposed to the UPF, or any UPF function. It is provided to enable trusted UPF/Network Function (NF) services to build a trusted connection to the UDM and provide API level query. As stated earlier, this API query will be delivered by an AF that can be located inside (or outside) the 5G core. This provided AF service would allow the service provider to receive information from the SMD, AMF, UDM, or potentially any other 5G core NF (based on permissions granted by the 5G core owner to the AF) to enable the process to apply policy.

FIG. 41 is a flow diagram of the proposed service flow including the Application Function (AF) 1500. By building an AF 1500 within the 5G core, the service provider can be actively informed of changes to the UE 300 state and values that could vary the identity of the UE 300 in the policy decision.

§ 19.0 5G UDM TO IDP FEDERATION

The present disclosure enables 3rd party, non-core, services to consume subscriber data from core data stores such as, but not limited to Unified Data Management (UDM) and/or the User Data Repository (UDR) in 5G cores through an Identity Federation of data to external Identity Provider (IDP). Additional data to enrich decision-making can be obtained by querying third party systems that include, but are not limited to, information on device classification and/or devices to be allowed or denied access.

Correctly integrating values of mobile subscribers' devices, sessions, authorizations, and other connectivity data as part of an identity value is solely restricted to internal-core functions of a 5G network. Implementing true zero trust controls must be able to not only identify the initiator of the subscriber connections, but also differentiate it between various sets of device types. In 5G deployments, the information collected through the enrolment and enablement of a subscriber is stored in a set of services, tasked with man-

aging all authorization, registration, network, and session definitions, amongst other data sets. This information, when collated together, allows the 5G provider to provide a unique identity for each and every subscribed UE **300** (FIG. **39**). It is this information set that is proposed to leverage for identity. For example, UDM acting like IDP.

§ 19.1 5G UDM to IDP Federation Solution

5G UDM/UDRs provide unique sets of data that, if consumed through a trusted mechanism to a trusted party, will allow for direct identity of UE to be categorized. Taken within the context of zero trust access, this identity will be used to differentiate the controls for traffic passing to and beyond the User Plane Function (UPF). The values stored within a UDM, when properly categorized for identity consumption, allow for very granular and specific insights into the User Equipment, their rights, responsibilities, roles, access criteria, allowed quality of service (QoS), and data flows within a telecommunications network. Enabling this data to be consumed as part of a standardized identity solution or broker would allow for non-core services and solutions to consume valuable attributes to deliver and apply controls to UE generated data, not only within the context of the 5G UPF, but also for external services who trust the IDP. In 5G environments, all events and factors related to the onboarding of devices, SIMS, and connection mechanisms (RAN, Geo, etc.) are paired with the subscriber identifier, e.g., account number, name, user identity, etc. This allows core operators to build unique identifiers for each connected device.

For 5G networks, data stored within a UDM, to be assessed as part of this query includes, but not be limited to, the following details:

- Subscriber information (UE details, PDU, etc.)

- Session management data (QoS, SMF, PDU types, etc.)

- Access and mobility management function (Session response, tunnels, IP address, etc.)

- Policy (changes, implementations, controls)

- Among many other values

Providing a mechanism that would translate the UDM/UDR values into attributable states within an external, non-core, non-5G identity service provides any consuming services of the Identity to granularly apply controls. These attributes are consumed from the IDP as simple Security Assertion Markup Language (SAML).

FIG. **42** is a flow diagram of the process to query IDP broker in 5G. The present disclosure provides an identity broker layer in which information from a UDM/UDR **1400** can be queried and/or collected/updated in near-live time. This information within the ID broker **1504** is contextualized from a 5G context to that of common Identity values. To deliver this broker layer, focus is placed on the following functions:

- Obtain information from the 5G UDM/UDR

- Contextualize the 5G data for IDP **1506**

- Broker the information to an external IDP **1506**

§ 19.2 UDM/UDR Function

The present disclosure provides a mechanism to query information from the UDM/UDR from outside the 5G control plane using one or more of the standard non-core Application Function (AF) approaches defined by the 5G specifications. These approaches allow the 5G core owner to validate and authenticate a non-core AF and enable it to communicate with core functions like UDM/UDR via long-lived encrypted connections. The UDM/UDR function contains all updated values from the UDM/UDR so that all updated values are signaled to the broker as being “fresh”. It is important to have the service provide continuous

inclusion of updates and thus verification of session identity. This means having AI/ML models to detect anomalies and terminate or force re-auth of a session and periodically check if any session or device parameters changed and re-apply policies which may be tied to session initiation.

The context function is as follows. Build a map of all 5G UDM/UDR data values that are mapped to values that would be consumed by an IDP. An example of the contextualization could be, but is not limited to:

TABLE 2

Contextualization example	
5G core context	IDP consumed context
AMF value change & modification of SMF with updated QoS and network criteria Subscriber Identity values	Network Criteria Change User Identity

The broker function is as follows. Provide an API-like subscription and consumption function that an IDP could consume and leverage for Identity validation and ultimately deploy for control. This is similar to that of a federation between IDPs. All traffic between the service, including but not limited to signaling and workflows, must be over an encrypted tunnel ensuring zero capability for interception or manipulation of the service access of function within the edge. Thus, the service is only available to those who are allowed to access it and nothing more.

§ 20.0 CONCLUSION

While the foregoing description refers to 5G, those skilled in the art will recognize the systems and methods could apply to any edge compute platform. It will be appreciated that some embodiments described herein may include one or more generic or specialized processors (“one or more processors”) such as microprocessors; Central Processing Units (CPUs); Digital Signal Processors (DSPs); customized processors such as Network Processors (NPs) or Network Processing Units (NPU), Graphics Processing Units (GPUs), or the like; Field Programmable Gate Arrays (FPGAs); and the like along with unique stored program instructions (including both software and firmware) for control thereof to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the methods and/or systems described herein. Alternatively, some or all functions may be implemented by a state machine that has no stored program instructions, or in one or more Application-Specific Integrated Circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic or circuitry. Of course, a combination of the aforementioned approaches may be used. For some of the embodiments described herein, a corresponding device in hardware and optionally with software, firmware, and a combination thereof can be referred to as “circuitry configured or adapted to,” “logic configured or adapted to,” etc. perform a set of operations, steps, methods, processes, algorithms, functions, techniques, etc. on digital and/or analog signals as described herein for the various embodiments.

Moreover, some embodiments may include a non-transitory computer-readable storage medium having computer-readable code stored thereon for programming a computer, server, appliance, device, processor, circuit, etc. each of which may include a processor to perform functions as described and claimed herein. Examples of such computer-

readable storage mediums include, but are not limited to, a hard disk, an optical storage device, a magnetic storage device, a Read-Only Memory (ROM), a Programmable Read-Only Memory (PROM), an Erasable Programmable Read-Only Memory (EPROM), an Electrically Erasable Programmable Read-Only Memory (EEPROM), Flash memory, and the like. When stored in the non-transitory computer-readable medium, software can include instructions executable by a processor or device (e.g., any type of programmable circuitry or logic) that, in response to such execution, cause a processor or the device to perform a set of operations, steps, methods, processes, algorithms, functions, techniques, etc. as described herein for the various embodiments.

The foregoing sections include headers for various embodiments and those skilled in the art will appreciate these various embodiments may be used in combination with one another as well as individually. Although the present disclosure has been illustrated and described herein with reference to preferred embodiments and specific examples thereof, it will be readily apparent to those of ordinary skill in the art that other embodiments and examples may perform similar functions and/or achieve like results. All such equivalent embodiments and examples are within the spirit and scope of the present disclosure, are contemplated thereby, and are intended to be covered by the following claims.

What is claimed is:

1. A method of identifying and categorizing identities of User Equipment (UE) in a 5G landscape comprising steps of:

receiving a plurality of 5G Unified Data Management (UDM) and/or User Data Repository (UDR) UE values;

assessing source identities of UE based on the UE values; assigning the source identities to traffic flows associated with the UE based on the assessing; and

forwarding from a device connector the traffic flows to a cloud-based system and performing actions based on the source identities, the actions including applying a correct policy, steering traffic over a best path, and/or signaling updates to 5G core traffic management components, wherein the steps are performed by a service that takes a form of a 5G Application Function (AF) defined by one or more 5g specifications that is authenticated by a 5G control plane and is granted access to UDM/UDR and/or other 5G core components that store and serve needed data.

2. The method of claim 1, wherein the service could also exist outside of the 5G core as an AF or other service function that can be authenticated and trusted by a 5G network.

3. The method of claim 1, wherein the steps are performed continuously for identity verification of sessions, and wherein Artificial intelligence (AI)/Machine Learning (ML) models are utilized to detect anomalies and terminate or force re-authentication of a session.

4. The method of claim 3, wherein the steps further comprise periodically checking if any session or device parameters change based on the UE values and re-applying policies within the cloud-based system which may be tied to session initiation to continuously verify session identity.

5. The method of claim 1, wherein the steps further comprise:

contextualizing via one or more mapped values the 5G data received from the UDM/UDR to that of common identity values; and

brokering the contextualized 5G data to an external Identity Provider (IDP).

6. The method of claim 5, wherein a subscription and consumption function is provided via a broker function to the IDP for identity validation and deploy for control, wherein the consumption function is consumable by the IDP.

7. A non-transitory computer-readable medium comprising instructions that, when executed, cause one or more processors to perform steps of

receiving a plurality of 5G Unified Data Management (UDM) and/or User Data Repository (UDR) User Equipment (UE) values;

assessing source identities of UE based on the UE values; assigning the source identities to traffic flows associated with the UE based on the assessing; and

forwarding from a device connector the traffic flows to a cloud-based system and performing actions based on the source identities, the actions including applying a correct policy, steering traffic over a best path, and/or signaling updates to 5G core traffic management components, wherein the steps are performed by a service that takes a form of a 5G Application Function (AF) defined by one or more 5g specifications that is authenticated by a 5G control plane and is granted access to UDM/UDR and/or other 5G core components that store and serve needed data.

8. The non-transitory computer-readable medium of claim 7, wherein the service could also exist outside of the 5G core as an AF or other service function that can be authenticated and trusted by a 5G network.

9. The non-transitory computer-readable medium of claim 7, wherein the steps are performed continuously for identity verification of sessions, and wherein Artificial intelligence (AI)/Machine Learning (ML) models are utilized to detect anomalies and terminate or force re-authentication of a session.

10. The non-transitory computer-readable medium of claim 9, wherein the steps further comprise periodically checking if any session or device parameters change based on the UE values and re-applying policies within the cloud-based system which may be tied to session initiation.

11. The non-transitory computer-readable medium of claim 7, wherein the steps further comprise:

contextualizing the 5G data received from the UDM/UDR to that of common identity values; and

brokering the contextualized 5G data to an external Identity Provider (IDP).

12. The non-transitory computer-readable medium of claim 11, wherein a subscription and consumption function is provided to the IDP for identity validation and control.

13. An edge compute system configured for cloud-based 5G security via an endpoint service, the edge compute system comprising:

one or more processors and memory storing instructions that, when executed, cause the one or more processors to

receive a plurality of 5G Unified Data Management (UDM) and/or User Data Repository (UDR) User Equipment (UE) values;

assess source identities of the UE based on the UE values;

assign the source identities to traffic flows associated with the UE based on the assessing; and

forward from a device connector the traffic flows to a cloud-based system and perform actions based on the source identities, the actions including applying a correct policy, steering traffic over a best path, and/or

signaling updates to 5G core traffic management components, wherein the steps are performed by a service that takes a form of a 5G Application Function (AF) defined by one or more 5g specifications that is authenticated by a 5G control plane and is granted access to UDM/UDR and/or other 5G core components that store and serve needed data. 5

14. The edge compute system of claim 13, wherein the service could also exist outside of the 5G core as an AF or other service function that can be authenticated and trusted by a 5G network. 10

15. The edge compute system of claim 13, wherein the steps are performed continuously for identity verification of sessions, and wherein Artificial intelligence (AI)/Machine Learning (ML) models are utilized to detect anomalies and terminate or force re-authentication of a session. 15

16. The edge compute system of claim 15, wherein the steps further comprise periodically checking if any session or device parameters change based on the UE values and re-applying policies within the cloud-based system which may be tied to session initiation. 20

17. The edge compute system of claim 13, wherein the instructions that, when executed, cause the one or more processors to contextualize the 5G data received from the UDM/UDR to that of common identity values; and broker the contextualized 5G data to an external Identity Provider (IDP). 25

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